

European National Transfer Accounts Manual V.2.0.0

A description of data and methods

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European National Transfer Accounts (ENTA) provide age- and sex-specific data on income, transfers between age groups, and consumption for European countries. The data can be accessed through direct download from Zenodo (`https://doi.org/10.5281/zenodo.20360412`) or the data explorer (`https://www.ntaeurope.eu/dataexplorer`), where descriptive information for users can also be found. This manual provides a codebook and an overview of the methodology and results. It also serves as a starting point for those with a deeper interest in the methodology, who may wish to consult the code, which is publicly available at `https://github.com/bbinderhammer/enta`.

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0. Organisation of the ENTA manual

This manual refers to ENTA version V.2.0.0. The manual describes the data and methods used to generate European National Transfer Accounts (ENTA) for 26 European countries. The age- and sex-specific data cover the income reference years 2008 to 2023, Croatia from 2009 and Bulgaria to 2022; the aggregate data cover 1995 to 2024, Bulgaria again to 2022. ENTA are mainly based on Eurostat data and provide age- and sex-specific information about the economy: on primary income, the redistribution of income through government and private transfers, and the use of disposable income for consumption and saving.

The ENTA data can be used to answer questions such as:

- Is the income of the young declining over time?
- Which countries redistribute the most to older age groups?
- Which age groups provide the most transfers to children?

National Transfer Accounts (NTA), in general, take aggregate data from National Accounts (NA) and estimate their age- and sex-specific distribution using data from a range of sources, including microdata from surveys and administrative records. This manual provides an overview of how this is done for ENTA based on Eurostat data. Furthermore, it serves as a starting point for those with a deeper interest in methodology and data. All code is publicly available, and with access to Eurostat microdata, all results can be reproduced.

Using ENTA data: a quick start

This section is for users of the published data who do not want to work through the methodology first. It tells where to find what, and which warnings to read before comparing countries or years.

The data files

ENTA are published as four files (Zenodo, `https://doi.org/10.5281/zenodo.20360412`, and the ENTA data explorer):

- *ENTA_esa*: age- and sex-specific values of income, taxes, social benefits, private transfers, consumption and saving in the ENTA–ESA format, ages 0 to 80 (80 and older), 2008–2023 (Section 6.2). This format keeps a close link to the European System of Accounts in concepts and terminology.
- *ENTA_nta*: the same in the NTA format. The NTA format uses the concepts and terminology of the National Transfer Accounts project.

- *ENTA_ aggregates*: the national totals by sector from the European System of Accounts and related sources, 1995–2024, in million euro;
- *ENTA_ units*: the factors that convert the stored values into other units.

Every file is in long format: one value per row. The columns are:

Column	Content
CNT2, COUNTRY	Country code (AT) and name (Austria)
YEAR	Year
AGE	Age, 0 to 80; 80 stands for 80 and older
SEX	Men, Women or Total
IND, INDICATOR	Code of the indicator (YEM) and its number and label
VALUE	The value, in euro per person at current prices

ENTA_ aggregates has *SECTOR* in place of age and sex, and the value is *VALAGGR*, in million euro. No stored value is missing: a country-year without data has no rows, and in the aggregates an indicator that is not reported in a year has no row. Real values, purchasing power standards, shares of income or GDP and the NTA norms are calculated from the stored values with *ENTA_ units* (Appendix A).

Which format?

The *ENTA–ESA format* keeps the organisation and terminology of the national accounts and describes the household sector. The *NTA format* combines households and corporations into the private sector, adds the government sector, and uses the concepts of the global NTA project, so that results can be compared with NTA for countries outside Europe. Chapter 4 translates one into the other; the main difference in content is that the NTA format splits mixed income into two thirds labour income and one third asset income.

Three routes through the manual

The manual has three parts: the methodology (Part I), the codebook and project reference (Part II; Appendices A to C are also published separately as the ENTA codebook), and the information needed to interpret the results (Part III).

- **Using the data:** choose the file and the unit (above and Appendix A), look up the indicator in the codebook (Appendices B and C), and read the simple rules and the table of what each indicator can be used for (Section 6.1).
- **Comparing countries or years:** in addition, read the limitations of the quantities you use (Chapter 6, ordered like the codebook), the table of country-specific treatments (Table 6.2) and the country patterns and events (Appendix E).

- **Replicating or extending ENTA:** start with the organisation of code and data (Appendix D), then the methods (Chapters 2 and 3) and the quality assessment (Chapter 5).

Finding an indicator

Table 1 lists the indicators for common questions; the codebooks define every indicator and give its formula.

Table 1: Common questions and the indicators that answer them

Question	ENTA–ESA format	NTA format
Earnings from work by age	YEM (employment income)	YL (labour income)
Income from property and housing	YA (asset income)	YA, YAH (households)
Pensions received from government	BSPEN	TGICPEN
All cash benefits from government	BS	TGIC
Taxes and contributions paid	T	TGOT
Net transfers from government, incl. public services	—	TG
Private consumption	CH	CF
Public health or education services received	CGH, CGE	CGH, CGE
Transfers between household members	BPW (received), TPW (paid)	TFW (net), TFWI, TFWO
Transfers between households	BPB	TFB
Disposable income	YD (after transfers within households)	YDH
Saving	SH (do not use by age)	SH, SF (do not use by age)
Employment rate, hours worked	EMPLSTAT, EMPLFT, WHRUSUT	—

Y and YA are household quantities in the ENTA–ESA file and totals of all sectors in the NTA file; see the codebooks.

When a result looks odd

- **The level is odd** (a jump in the total, or in all ages at once): levels come from the national accounts, so the cause is in the aggregate source. See the country patterns and events (Appendix E) and the aggregate data (Chapter 2).
- **The age shape is odd:** shapes come from the surveys or from an assumed shape. See the limitations of the quantity (Chapter 6) and the flows with an assumed shape (Table 6.1).
- **The profile is a step function:** the source reports age groups rather than single ages, as for Germany in some years and the Labour Force Survey of four countries (Section 6.4). Totals are comparable, the curvature is not.
- **A value is zero:** it may be a structural zero, a value the source does not report, rather than a true zero (Section 6.1).

Short glossary

Aggregate A national total from the national accounts, for example all wages paid in a country and year. It fixes the level of every profile.

Age profile The per capita value of a quantity at each age (and by sex). Its population-weighted sum over ages equals the aggregate.

Scaling Multiplying a survey-based profile by one factor so that it adds up to the aggregate; the survey decides only the distribution over ages.

ESA European System of Accounts, the national accounts framework of the EU (ESA 2010); the source of the aggregates.

NTA National Transfer Accounts, a method and global project that breaks the national accounts down by age.

EU-SILC European Union Statistics on Income and Living Conditions, the survey behind the income, tax and benefit profiles.

HBS Household Budget Survey, the survey behind the consumption profiles (waves 2010 and 2015).

LFS Labour Force Survey, the source of the employment indicators.

EUROMOD The EU tax-benefit microsimulation model, used to estimate tax rates by type of income and age.

COFOG Classification of the Functions of Government; splits government expenditure into old age, family, health, education and other functions.

COICOP Classification of Individual Consumption by Purpose; splits household consumption into education, health, housing and other purposes.

ISCED International Standard Classification of Education, the levels of education from pre-primary (0) to doctoral (8).

NPISH Non-profit institutions serving households; in ENTA part of the household sector.

ESA codes The ESA transactions and balances named in this manual: D.21 taxes on products; D.29 other taxes and D.39 other subsidies on production; D.59 other current taxes; D.74 current international cooperation; D.75 miscellaneous current transfers; D.76 VAT- and GNI-based EU own resources; D.8 change in pension entitlements; B.2g gross operating surplus; B.3g gross mixed income; B.6n net disposable income; B.8n net saving.

Life-cycle deficit Consumption minus labour income at a given age; positive for children and older people.

Structural zero A value that is zero because the source does not report it, not because the quantity is zero.

1. National Transfer Accounts: Analysing the Economy by Age and Sex

1.1 What is ENTA?

European National Transfer Accounts (ENTA) provide age- and sex-specific data on primary income, its redistribution through public and private transfers, and its use for consumption and saving, for 26 European countries over the period 2008–2023. ENTA are built mainly from Eurostat microdata (EU-SILC, HBS, EUROMOD) and the European System of Accounts (ESA), and are the European implementation of the global National Transfer Accounts (NTA) project.

The age profiles cover the income years 2008 to 2023; Croatia starts in 2009, the first year of its EU-SILC, and Bulgaria ends in 2022, the last year of its sector accounts. The aggregates cover 1995 to 2024. Malta is not included, because its EU-SILC files report age only in five-year groups.

Three features of the dataset open lines of analysis that single-country or single-year NTA cannot support. The time dimension shows how economies absorb shocks that hit age groups unevenly, such as the recession of 2008–09 or the COVID-19 pandemic. The cross-country dimension shows the diversity of European welfare systems in one comparable accounting framework. And the sex dimension allows the analysis of gender gaps in income, transfers and consumption by age.

This chapter introduces the NTA concept and the global NTA project (Section 1.2), including how NTA integrate age and sex into National Accounts (Section 1.2.2), and then describes what is specific to ENTA and how they relate to the European System of Accounts (Sections 1.3 and 1.3.3).

1.2 What are National Transfer Accounts (NTA)?

National Transfer Accounts (NTA) constitute a framework to measure and analyse the economy from a demographic perspective, with a focus on age and sex. The fundamental idea of NTA is the use of microdata to integrate age and sex into National Accounts (NA).

Basic NTA measure primary income by age and sex; the redistribution of primary income between age groups and sexes through public and private transfers; and the age- and sex-specific use of disposable income for consumption and saving. NTA provide the data and tools for understanding the macroeconomic consequences of demographic change, as well as for analysing inequalities between demographic groups.

A major innovation of NTA is the inclusion of private transfers within households in their accounting framework. National Accounts take households as a unit and ignore transactions within households. Estimates of primary income and consumption by age reveal a large gap between consumption and

income during childhood and in old age. In the NTA context, this gap is called the *life-cycle deficit* (LCD). The LCD of older persons is largely financed by public transfers, which are captured in NA and microdata on income. By contrast, the LCD of children is covered mostly by transfers from their parents, which are usually not captured in economic data. Extending the accounting system by age and sex requires consideration of these transfers within households.

Figure 1.1 illustrates what ENTA data look like, using employment income and total consumption (private plus public) by age for France in 2023. Children and older persons consume much more than they earn, whereas persons of working age earn more than they consume. The gap between the two curves at young and old ages is the life-cycle deficit; the surplus at working ages is what finances it.

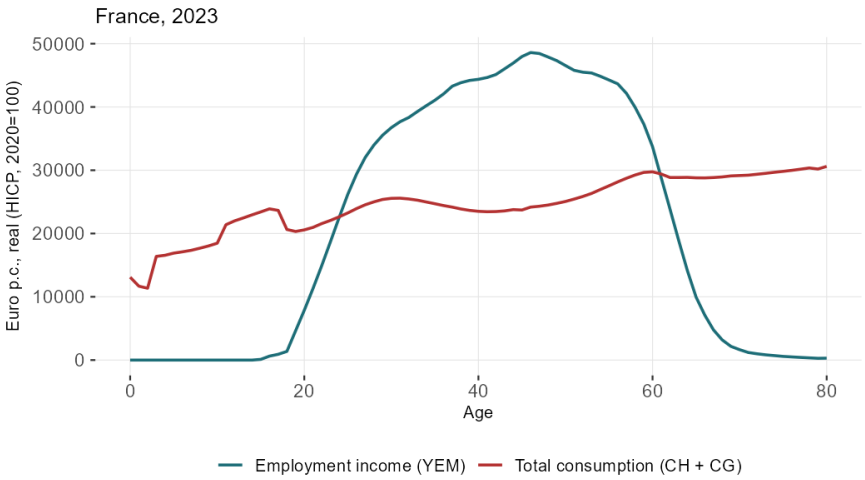


Figure 1.1: Employment income and total consumption by age, France, 2023. Total consumption is the sum of private consumption (CH) and public consumption (CG); euro per capita, real (HICP, 2020 = 100)

1.2.1 The global NTA project

NTA are developed and maintained by a network of research teams in more than 60 countries. More information about the NTA network, the framework, and NTA data can be found on the website of the NTA project at `https://www.ntaccounts.org`. The general NTA methodology is described in detail in a manual published by the United Nations (United Nations, 2013). d'Albis and Moosa (2015) provide a concise description of the concept, and Lee and Mason (2011) describe the methodology and NTA results for a wide range of countries. A useful reference is also the AGENTA manual (Istenič, Hammer, Šeme, Lotrič Dolinar, & Sambt, 2019), which describes an earlier version of European NTA data (reference year 2010) that are available on `https://www.wittgensteincentre.org/ntadata`.

NTA provide information on economic quantities by demographic characteristics and can be used for a range of purposes and research questions. Originally, NTA were designed to better understand the macroeconomic consequences of population ageing. See, for example, Mason, Lee, and members of the NTA network (2022) for a comprehensive overview. In particular, NTA have been used to assess the potential for a demographic dividend (e.g., Dramani & Oga, 2017; Ladusingh & Narayana, 2012).

Among the central NTA topics are also the consequences of demographic change for public transfer expenditures and revenues (e.g., Loichinger, Hammer, Prskawetz, Freiberger, & Sambt, 2017).

NTA have evolved over time and are currently used to study heterogeneities by age and other socioeconomic characteristics. Examples include differences between birth cohorts (Mérette & Navaux, 2025), between sexes (Hammer, Spitzer, Vargha, & Istenič, 2020), by education (Abio, Patxot, Souto, & Istenič, 2021), by income quintile (Hammer, Christl, & De Poli, 2023), by parental status (Vanhuysse, Medgyesi, & Gál, 2023), by race (Oosthuizen, 2019), and by religion or culture (Shraberman & Weinreb, 2024).

1.2.2 Integrating age and sex into National Accounts

The central feature of the NTA methodology is the combination of data and concepts from NA with distributional information from microdata. In doing so, NTA capture the advantages of both data sources: the accounting framework and the higher completeness and reliability of macro data in NA, and the distributional detail from microdata.

NTA use the accounting framework and data from NA. The NTA dataset is constructed and organised such that the central accounting identities of NA hold for each individual and age group: disposable income (YD), defined as primary income (Y), less transfers paid (T), plus all transfer benefits received (B), equals the sum of consumption (C) and saving (S).

$$Y - T + B = C + S = YD \tag{1.1}$$

This relation is also referred to as the NTA flow identity. NTA do not only use the concepts from NA, but also the data: the sum of age-specific means of income, public transfers, consumption, and saving over all individuals in all age groups equals exactly the values for the total economy as captured in NA.

However, the integration of demographic characteristics into NA and the focus on redistribution require two major adjustments. First, in NTA, institutional units are defined and aggregated differently than in NA. Second, quantities are measured at basic prices rather than at market prices.

1.2.3 Economic accounting with people

The most fundamental difference between NTA and NA is the definition and aggregation of institutional units. An institutional unit is defined as an *economic entity that is capable, in its own right, of owning assets, incurring liabilities, and engaging in economic activities and in transactions with other entities.*[1] In NA, an institutional unit could be a household, a corporation, or a government department. Institutional units in NA are grouped into sectors: households, corporations, government units, non-profit institutions serving households (NPISH), and institutional units abroad (grouped into the rest-of-the-world sector, ROW). NA measure the value of economic transactions in which institutional units engage and provide their total value by type of transaction and by sector.

[1] `https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Institutional_unit`

In contrast, in NTA the institutional units are individuals. This enables their aggregation by socioeconomic characteristics such as age and sex. Defining individuals as institutional units has an important consequence: it requires the measurement of transfers between household members. Transfers in NA are defined as *two-party transactions in which one party provides a good, service, or asset to the other but does not receive recompense in return*—a "something for nothing" transaction, or a transaction without a quid pro quo (United Nations, 2011, §3.58). The most important transfers are taxes and social contributions, social benefits, government consumption, and transfers from parents to children.

1.2.4 Measurement at basic prices

NA value transactions at market prices, whereas NTA value them at *basic prices*, i.e., excluding taxes on products and including subsidies on products (D21, D31).[2]

This is necessary for a consistent treatment of redistribution. At market prices, consumption includes taxes less subsidies on products. Disposable income would then have to cover these taxes, although it is meant to be income *after* taxes and benefits, and the taxes would never show up as a transfer to the government. NTA therefore treat taxes on products as a transfer from the private sector to the government in the redistribution accounts, and measure consumption at basic prices, net of these taxes. In ESA, by contrast, they are primary income of the government. Without this reclassification, redistribution through government transfers could not be integrated consistently into ENTA.

1.3 European National Transfer Accounts

1.3.1 Using harmonised Eurostat data

European NTA teams use the harmonised data that Eurostat provides for EU countries to generate NTA for all European countries jointly, which greatly improves cross-country comparisons. An earlier output was the European NTA data of the EU-funded AGENTA project (Section 1.2.1), which cover only the year 2010. ENTA, constructed mainly as part of the SUSTAINWELL and the Generational Satellite Accounts projects, build on this work (`https://www.ub.edu/sustainwell-eu-project`).

1.3.2 Differences between ENTA and global NTA

The NTA manual (United Nations, 2013) organises the quantities in a life-cycle account, which measures the age-specific difference between consumption and labour income, and in age reallocation accounts that measure how the life-cycle deficit is financed: public transfers, private transfers, and asset-based reallocations.

ENTA instead keep the organisation of ESA (Section 1.3.3), with a primary income account, redistribution accounts, and a use of disposable income account. The direct link between ENTA and ESA

[2] The basic price is the amount receivable by the producer from the purchaser for a unit of a good or service produced as output, minus any tax payable and plus any subsidy receivable by the producer as a consequence of its production or sale. Source: `https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Basic_price`

quantities makes it easier to control the quality of the data for 26 countries over 16 years, and the organisation is more intuitive for users familiar with NA. For comparison with the data of the global NTA project, ENTA are also provided in the NTA format (Chapter 4).

1.3.3 ENTA and the European System of Accounts (ESA)

ESA is the system of national and regional accounts used by the members of the European Union. It is EU-specific but consistent with the international guidelines for the compilation of national accounts (United Nations, 2011) and is described in detail in EUROSTAT (2013). The data are publicly available through the Eurostat database. ENTA use annual sector accounts as their central dataset and complement them with additional data, such as data on taxation trends and detailed information on public and private consumption expenditure.

In general, the ENTA aggregates are calculated to be as close as possible to the central quantities in ESA: the measure of total primary income in ENTA corresponds largely to the *balance of primary incomes* in ESA; total income after public and private redistribution in ENTA corresponds to *disposable income*; and the ENTA quantities of *consumption* and *saving* correspond to their equivalents in ESA.

1.3.4 The sequence of European transfer accounts

NTA inherit accounting identities from NA. Primary income plus all transfers received less transfers paid equals disposable income, which is used for consumption and saving. ENTA organise the components in a sequence of accounts (Table 1.1): an income account with primary income by type (1); the transfer accounts with taxes and contributions (2), government benefits in cash (3), funded pensions (4) and private transfers (5), which lead to disposable income (6); and its use for consumption (7, 8) and saving (9). Government consumption is received as a transfer in kind and is at the same time consumption, so it does not change saving. Transfers within households net to zero in the aggregate but not by age.

Table 1.1: The sequence of European National Transfer Accounts

0. Production account (only at aggregate level, not age-specific)		
0	Gross Domestic Product	GDP
0.1	Taxes less subsidies on products	TPS
0.2	Consumption of fixed capital	CFC
0.3	Employment income, net from ROW	YEROW
0.4	Property income, net from ROW	YPROW
1	Primary income	Y
1. Primary income (primary distribution of income account in ESA)		
1	Primary income	Y

continued on next page

Table 1.1 (continued)

1.1	Employment income	YEM
1.1.1	Compensation of employees	YE
1.1.2	Mixed income from self-employment	YM
1.2	Asset income	YA
	2. Government transfer contributions (taxes & social contributions)	
2	Taxes, total	T
2.1	Taxes on employment income	TYEM
2.2	Taxes on assets and asset income	TYA
2.3	Taxes on income, non-employed	TB
2.3.1	Taxes on pay-as-you-go pensions	TBSPEN
2.3.2	Taxes on non-pension social benefits	TBSN
2.3.3	Taxes on benefits, funded pensions	TBPPEN
2.4	Taxes on consumption	TCON
2.5	Other current transfers (paid less received)	TX
2.6	Net transfers from ROW, government (received less paid)	TROW
	3. Government transfer benefits in cash (social benefits)	
3	Social benefits	BS
3.1	Social benefits, pensions	BSPEN
3.2	Social benefits, family	BSFAM
3.3	Social benefits, unemployment	BSU
3.4	Social benefits, other	BSX
	4. Funded pensions, mandatory	
4	Funded pensions (received less paid)	BPPEN
4.1	Contributions	TPSOCfc
4.2	Benefits	BPSOCfc
	5. Private transfers	
5.1	Private transfers between households	BPB
5.1.1	Social contributions, nonfin corp.	TPSOCnfc
5.1.2	Social benefits, nonfin corp.	BPSOCnfc
5.1.3	Private transfers from ROW (received less paid)	BROW
5.1.4	Other private	BPX
5.2	Intra-household transfers received	BPW
5.3	Intra-household transfers paid	TPW
	6. Disposable income	
6	Disposable income	YD

continued on next page

Table 1.1 (continued)

	7. Government consumption	
7	Public consumption	CG
7.1	Public consumption, education	CGE
7.2	Public consumption, health	CGH
7.3	Public consumption, other	CGX
	8. Private consumption	
8	Private consumption	CH
8.1	Private consumption, education	CHE
8.2	Private consumption, health	CHH
8.3	Private consumption, housing	CHR
8.4	Private consumption, other	CHX
	9. Saving	
9	Saving	SH

2. ENTA aggregate data

The sequence of transfer accounts

2.0 Production

The production account shows how primary income is derived from GDP. It is available only at the aggregate level: production cannot be assigned to individuals or age groups, only the income distributed to the owners of the factors of production can. ENTA include it to show the difference between the widely used GDP and primary income, the starting point of the age- and sex-specific ENTA; the gap varies considerably across countries and over time.

The difference between GDP and primary income, as defined in ENTA, consists of four components: taxes on products (TPS), consumption of fixed capital (CFC), property income received from the rest-of-the-world sector less the property income paid (YPROW), and compensation of employees received from the rest of the world less the compensation paid (YEROW). The rest-of-the-world sector (ROW) refers to all institutional units in other countries.

The first step is GDP at basic prices, i.e. less the taxes on products (Section 1.2.4); for the economy as a whole it is the sum of value added across all sectors. The second step subtracts the *consumption of fixed capital* (CFC), the expenditure needed to maintain the capital stock; the ESA 2010 manual also notes that income is more meaningful in net terms (EUROSTAT, 2013, §8.06). The third step moves from production to income and from a domestic to a national view: the *compensation of employees* and *net property income* received from other countries are added, and those paid to other countries subtracted.

ENTA primary income corresponds largely to the ESA *balance of primary incomes* at the sector level and to net national income for the total economy, with two small differences. Taxes less subsidies on products paid to the rest of the world are part of primary income in ENTA and appear as transfers to the ROW in the transfer accounts. And taxes on production are primary income of the private sectors in ENTA, not of the government; they are a transfer to the government later in the sequence. ENTA primary income of corporations is therefore higher than the ESA balance of primary incomes, and that of the government lower.

The difference between GDP and income varies significantly across European countries (Table 2.1). Income flows to other countries are particularly large in Ireland and Luxembourg: a large part of the production of multinational companies is reported there, and the income is paid as property income to owners in other countries; in Luxembourg, a large part of the compensation of employees is paid to non-residents as well. Income paid to the rest of the world amounted to 25% of GDP in Ireland in 2023 and 32% in Luxembourg.

Table 2.1: The production account: from GDP to primary income, 2023, Euro per capita

ESA quantity	Germany	Ireland	Luxembourg
0 GDP	50,660	99,247	123,218
0.1 - Taxes less subs. on products (TPS)	-4,387	-5,249	-10,735
0.2 - Consumption of fixed capital (CFC)	-10,300	-24,307	-16,729
0.3 + Employment income, net from ROW (YEROW)	68	-82	-21,539
0.4 + Property income, net from ROW (YPROW)	1,624	-25,142	-17,780
1 Net primary income, basic price (Y)	**37,666**	**44,465**	**56,434**

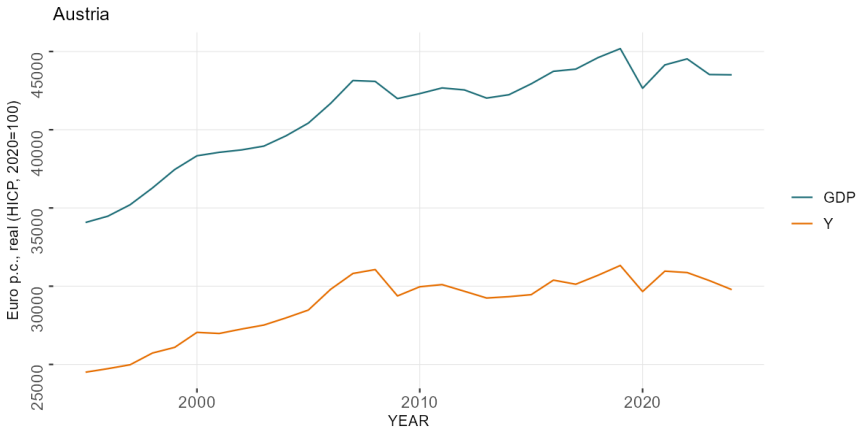


Figure 2.1: GDP and primary income (Y) over time, Austria

GDP and net primary income also develop differently over time. The consumption of fixed capital grows with the capital intensity of the economy, so net primary income increases less than GDP (Figure 2.1 for Austria).

2.1 Primary income

The ENTA income account summarises the information from the *allocation of primary income account* in ESA and provides details on the types and amounts of primary income received. *Primary income* refers to income earned directly from production, including wages, salaries, and income from ownership of assets — before any taxes or other transfers are paid. ESA distinguishes between *compensation of employees (YE)*, *mixed income (YM)*, and asset income (YA), which consists of *operating surplus (YO)* and net *property income (YP)*.

The components are defined in the codebook (Appendix B, Section 1). All are measured net of the consumption of fixed capital. Part of the taxes on production, less the subsidies, is classified in ENTA as compensation of employees: especially in Austria and Sweden, the taxes on production contain taxes on payroll and employment, which ENTA treat like employers' social contributions, as primary income of employees and later as a transfer from them to the government.

Table 2.2 shows the components of primary income; the compensation of employees is by far the largest. Components are shown for the household sector only: the tables of this chapter include only values for which ENTA have age-specific information.

Table 2.2: Primary income account with values for Germany 2023, Euro per capita

	Total economy	**Households**	**Corp.**	**Gov.**
1. Primary income (Y)	**37,666**	**35,028**	**2,789**	**-152**
1.1 Employment income (YEM)		**28,589**		
1.1.1 Compensation of employees (YE)		26,842		
1.1.2 Mixed income from self-empl. (YM)		1,747		
1.2 Asset income (YA)		**6,440**		
1.2.1 Property income (YP)		5,640		
1.2.2 Operating surplus (YO)		799		

Figure 2.2 shows real primary income per capita (in prices of 2020, with the exchange rate held at its 2020 value, Appendix A) in four countries: it grew strongly in Poland from a low base (+71% between 2008 and 2023), stagnated in France and fell in Italy after 2007.

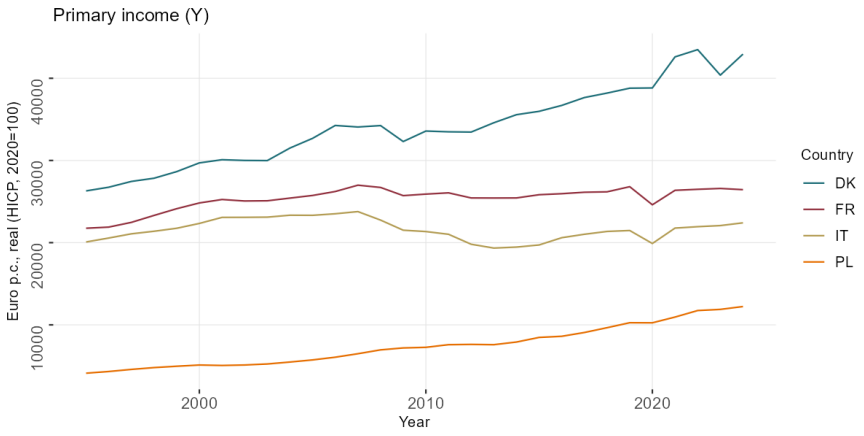


Figure 2.2: Primary income in selected countries and its change over time, prices of 2020

2.2 Government transfer contributions (taxes & social contributions)

Government transfer contributions in ENTA consist of all flows to the government sector in the form of taxes, social contributions, and other transfers. In ESA, these flows are captured in the secondary distribution of income accounts and consist primarily of *current taxes on income and wealth*, *social contributions*, and *other current transfers*. In ENTA, the public transfer account also captures public redistribution through *taxes on products*, with value added tax as the most important component, and other taxes on production. These indirect taxes are not part of the redistribution of income accounts in ESA.

The ENTA aggregates are based not only on ESA data but also on European data on taxation trends (European Commission, 2026) to distinguish taxes and social contributions by source or "economic function" (`https://webgate.ec.europa.eu/taxation_customs/redisstat/databrowser/view/tax_ec_func/default/table`). Based on the categorisation by economic function, the taxes and social contributions used in ENTA are classified into taxes on employment income (TYEM), taxes on assets and asset income (TYA), taxes on the income of non-employed persons (TB), taxes on consumption (TCON), and other transfers (TX).[1] The categorisation by economic function facilitates the allocation of tax payments by age, sex, and other demographic characteristics.

Additionally, EUROMOD is used to decompose the taxes paid by non-employed individuals into taxes on public pay-as-you-go pensions, taxes on non-pension social benefits, and taxes on mandatory funded pensions. Note that this decomposition is based on microdata; the values are estimates. In particular, small positive values may emerge even when the tax rates are effectively zero; for example, for taxes on non-pension social benefits.

In the tables of this chapter a payment of a sector carries a minus sign and a receipt a plus sign: taxes and contributions are negative in the household and corporation columns and positive in the government column, benefits the other way round. A payment shown positive is a net receipt: other current transfers of German households are received on balance in 2023.

The tax burden differs considerably between countries and changes over time (Figure 2.3): taxes relative to income declined in Sweden, from above 60% until 2007 to 54% in 2023, and increased in Italy, from less than 45% in 1995 to about 55% since 2012.

[1] Detailed lists of taxes and their assignment to functions for each country are available here: `https://taxation-customs.ec.europa.eu/taxation/economic-analysis/data-taxation-trends_en` (National tax lists)

Table 2.3: Public transfer contributions account, Germany 2023, Euro per capita

	Total economy	**Households**	**Corp.**	**Gov.**
2. Total taxes (T)		**-17,330**	**-1,351**	**18,681**
2.1 Taxes on employment income (TYEM)		-11,369		
2.2 Taxes on assets and asset income (TYA)		-536		
2.3 Taxes on income of non-employed (TB)		-1,415		
2.3.1 Taxes on pay-as-you-go pensions (TBSPEN)		-1,243		
2.3.2 Taxes on non-pension social benefits (TBSN)		-20		
2.3.3 Taxes on funded pensions (TBPPEN)		-152		
2.4 Taxes on consumption (TCON)		-4,387		
2.5 Other current transfers (TX)		377		
2.6 Net transfers from ROW, government (TROW)				-406

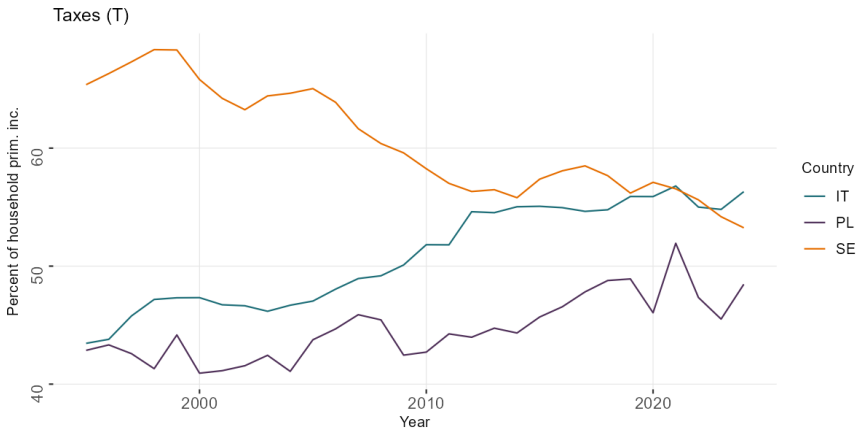


Figure 2.3: Taxes as a percentage of primary income and their change over time, selected countries, household sector

2.3 Government transfer benefits in cash

Government transfer benefits in cash consist of *social benefits* paid by governments to residents, with pensions as the largest component. The ENTA concept differs from the way government expenditure is usually captured in the data: government expenditure typically includes social benefits paid to residents of other countries, whereas ENTA focus on transfers to residents of the country itself. The ENTA expenditures by function of government are proportionally adjusted to the ESA estimates of social benefits received by the household sector from its own government.

ENTA distinguish social benefits by function, using ESA together with data on public expenditure classified by the functions of government (COFOG).[2] ENTA aggregate the detailed COFOG categories into fewer components to ensure consistency with the classification in microdata and to have enough observations to estimate age profiles. ENTA distinguish pensions, including the COFOG categories of old age, survivors, and sickness and disability (the last class does not separate sick pay from disability pensions); the category of family & children, which also includes the COFOG category of education; the category of unemployment; and other benefits, which include the cash benefits of the function health. Table 2.4 shows that pensions dominate all other categories of social benefits.

Eurostat's COFOG data do not cover Slovakia before 2010. For 2008 and 2009, social benefits and government consumption are split by function with the shares of 2010.

Table 2.4: Government transfer benefits in cash, Germany 2023, Euro per capita

	Total economy	**Households**	**Corp.**	**Gov.**
3. Social benefits, cash (BS)		**7,838**		**-7,838**
3.1 Pensions (BSPEN)		6,165		
3.2 Family & children (BSFAM)		490		
3.3 Unemployment (BSU)		688		
3.4 Other, incl. health (BSX)		495		

2.4 Funded pensions

ENTA distinguish between funded pensions and pay-as-you-go pensions. This requires approximations because funded pensions are treated very differently in ESA and in microdata. In ESA, (mandatory) funded pensions are captured as social contributions and benefits flowing between the household sector and financial corporations. In the microdata, they are included in public pensions together with benefits from pay-as-you-go pension systems.

At the aggregate level, the ENTA contributions to funded pensions consist of the social contributions paid by households to financial corporations. The benefits from funded pensions consist of the social benefits paid by financial corporations to households. The difference corresponds largely to the

[2] See `https://doi.org/10.2908/GOV_10A_EXP`

adjustment for the change in pension entitlements (D.8) in ESA, which ENTA do not use: ENTA saving plus D.8 equals ESA net saving (Appendix B, D.8). In Denmark, the Netherlands and Sweden, household saving in ENTA is therefore noticeably lower, and corporate saving higher, than ESA net saving.

While funded pension schemes exist in many European countries, they are of major importance only in Denmark, the Netherlands, and Sweden.

2.5 Private transfers

Private transfers in ENTA consist of transfers between individuals mediated by the private sector. NTA distinguishes between inter-household transfers, i.e. transfers between households, and intra-household transfers, i.e. transfers within households. Aggregate data are available only for transfers between households.

The most important types of private transfers between households in ESA are the *social contributions* and *social benefits* mediated by corporations, *non-life insurance premiums* and *claims*, as well as *miscellaneous current transfers*. The latter include transfers from households to the NPISH sector and transfers between households, such as remittances and alimony payments. The age-related redistribution through these transfers is small and difficult to measure. ENTA therefore focus on the net value, measured as the difference between transfers received and paid, and allocates it by a receipt profile, employment income plus pensions, which is zero below age 15 (see Chapter 3). Transfers between resident households cancel within the household sector and have no aggregate; ENTA add them from EU-SILC with received equal to paid, so they change the age profile of private transfers but not their aggregate.

ESA reports, for each sector, the total of miscellaneous current transfers (D.75) paid and received, but not who pays whom. ENTA need the flows between households, corporations, government and the rest of the world, and derives them from the sector totals with four proportional rules, applied in this order:

1. Transfers paid by the rest of the world go to households and corporations in proportion to what each receives, at most what the two receive together; any remainder goes to government.

2. Transfers received by the rest of the world come from households and corporations in proportion to what each pays, with the same limit; any remainder comes from government.

3. Government pays households and corporations in proportion to what they still have to receive, at most that amount; any remainder stays within government.

4. Households and corporations pay the remaining receipts of households and government in proportion to their remaining payments; the rest goes to corporations.

Government is not part of the first two rules, since its international transfers are recorded as D.74 and D.76. The rules reproduce the payments of every sector and the receipts of households, government

and the rest of the world exactly; the small difference between total D.75 paid and received in the source falls on the receipts of corporations. The rules determine only how household D.75 is split between government (`TX`), other households and corporations (`BPX`) and the rest of the world (`BROW`); the household total and household disposable income do not depend on them.

Table 2.5: Funded pensions and private transfers, Germany 2023, Euro per capita

	Total economy	**Households**	**Corp.**	**Gov.**
4. Funded pensions (received less paid, BPPEN)	**0**	**-752**	**752**	
4.1 Funded pens., contributions (TPSOCfc)		-1,578	1,578	
4.2 Funded pens., benefits (BPSOCfc)		826	-826	
5.1 Private transfers between households (BPB)	**-323**	**-172**	**-151**	

2.6 Disposable income

Accounting for all types of transfers results in disposable income. It represents the part of income "disposable" for the purpose of consumption and saving. Disposable income in ENTA corresponds to disposable income in ESA, with two differences that follow from the NTA concepts: it is measured at basic prices, i.e. after taxes less subsidies on products, and it is built from the ENTA definitions of primary income, taxes and social benefits described above. It therefore does not equal Eurostat's B.6n exactly; at market prices the household figure does.

Table 2.6: Disposable income and its use, Germany 2023, Euro per capita

	Total economy	**House-holds**	**Corp.**	**Gov.**
1 Primary income	37,666	35,028	2,789	-152
2 Government transfer contr. + net transfers from ROW	-406	-17,330	-1,351	18,275
3 Government transfer benefits in cash	0	7,838		-7,838
4 Funded pensions		-752	752	
5 Private transfers	-323	-172	-151	
6 Disposable income	**36,937**	**24,613**	**2,039**	**10,285**
7 Public consumption	**10,868**			**10,868**
7.1 Education				1,716
7.2 Health				4,106
7.3 Other				5,047
8 Private consumption	**22,250**	**22,250**		
8.1 Education		178		
8.2 Health		1,172		
8.3 Imputed rent		2,353		
8.4 Other		18,546		
9 Saving	**3,819**	**2,363**	**2,039**	**-583**

2.7 Government transfer benefits in kind—government consumption

Government consumption in ENTA comprises the *final consumption* of the government sector as defined in ESA. The distinction by function is based on data on government expenditure by function.[3]

All final consumption of the government sector in ENTA is classified as a transfer from the government to households. ENTA do not directly distinguish between individual consumption (social transfers in kind) and collective consumption. However, the distinction of the categories of education and health identifies the most important components of individual government consumption.

2.8 Consumption of households

Final consumption represents expenditure on goods and services that directly satisfy individual needs. Among the most important components are expenditure on housing (including energy), mobility, and food. ENTA consumption corresponds to the final consumption of the household sector in ESA, measured at basic prices.

To distinguish the strongly age-dependent components of education and health, as well as imputed rents, we use data on household final consumption expenditure by purpose.[4]

2.9 Saving

ENTA aggregate data include sector-specific saving, which corresponds to net saving in ESA less the adjustment for the change in pension entitlements (D.8; see the section on funded pensions). Household saving rates (saving relative to disposable income) differ considerably across countries. They are low or negative in Denmark, the Netherlands and Sweden, partly because the saving of funded pension schemes is not attributed to households in ENTA. Most countries show a peak in 2020–2021, when consumption was restricted during the pandemic.

[3] See `https://doi.org/10.2908/GOV_10A_EXP`
[4] See `https://doi.org/10.2908/NAMA_10_CO3_P3`

3. ENTA age-specific data

ENTA data consist of age profiles reporting age-specific per capita means for all ages from 0 to 80. The final age group (80) includes the population aged 80 and older. Age profiles are provided separately for men, women, and both sexes combined.

The age profiles derived from EU-SILC and from the consumption imputation are smoothed. Survey-based estimates of age-specific means exhibit considerable variation between single ages, reflecting the high uncertainty arising from the limited number of observations in each age group. ENTA smooth these profiles over age using Friedman's SuperSmoother, implemented via Stata's `supsmooth` command. It improves the age-specific estimates by taking into account the information from neighbouring age groups. The employment profiles from the Labour Force Survey are not smoothed, because the much larger sample makes it unnecessary.

Smoothing of the income, tax and benefit profiles starts at age 18; the values below age 18 are the observed means. The consumption profiles are smoothed over all ages. Negative income is set to zero at the individual level before the profiles are calculated. In the income and tax profiles, ages 15–17 are additionally capped so that they never exceed the value at the next older age (age 18 is the anchor), which prevents spikes just below the smoothed range, and all values below age 15 are set to zero. The benefit profiles apply neither the cap nor the zero below age 15.

The smoother errs most at the ends of the age range. The open top age group (80 and older; 73 and older for Germany 2014–2018) therefore stays in the smoothing but keeps its unsmoothed mean, in all smoothed income, tax and benefit profiles. It averages several hundred persons per sex over some twenty years of age and needs no smoothing, while a smoothed value there can differ markedly from the data. Unemployment benefits are smoothed differently, because they end abruptly at an age that differs by country, sex and year (see Section 3.3).

3.1 Primary income by age and sex

The age- and sex-specific estimates of primary income in ENTA are based on data from the European Statistics on Income and Living Conditions (EU-SILC). ENTA distinguish between employment income (YEM), consisting of income from employees (YE) and mixed income (YM), and asset income, which includes property income (YP) and operating surplus (YO).

To avoid age profiles being shaped by outliers, all income components are top-coded. Values that exceed ten times the mean of those who receive a certain type of income are replaced with ten times the mean; for property income, which is much more concentrated, the limit is six times the mean. The same top-coding at ten times the mean applies to taxes and social benefits.

3.1.1 Employment income

The ENTA estimates of employment income (YEM) are based on several EU-SILC variables. Income of employees is captured in the EU-SILC variables PY010G (wages), PY021G (company car), and PY030G (employers' social contributions) for each individual. A small part is captured in the income of persons below age 16 (HY110G), which is available only at household level and is split evenly among the household's 15-year-olds (if there is none, among the children aged 7–14, and failing that the 16-year-olds). These four EU-SILC variables together largely represent the same concept as the *compensation of employees* in ESA. The EU-SILC equivalent of *mixed income* in ESA is the income of self-employed persons (PY050G).

Although ENTA separate the income of employees (YE) and self-employed persons (YM), we recommend using the age profile of employment income (YEM), which includes both: the sample is larger and smoothing works much better for the combined profile.

The profiles of YEM peak at a much younger age in Eastern European countries than in Western European countries, where the peak lies between ages 50 and 60 (Figure 3.1).

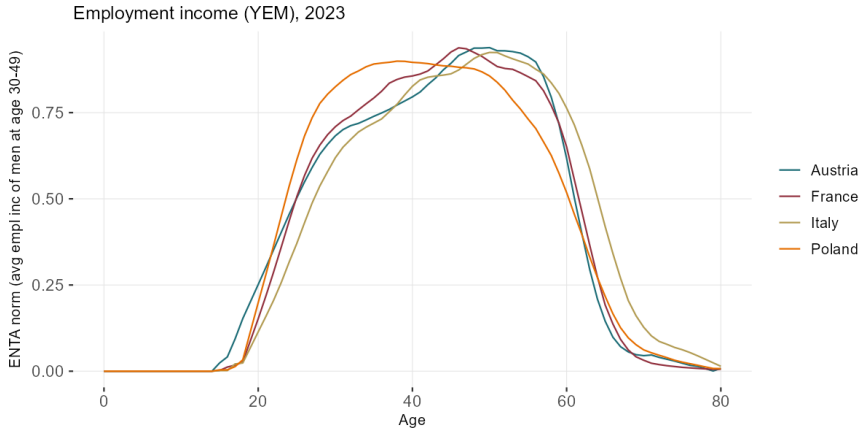
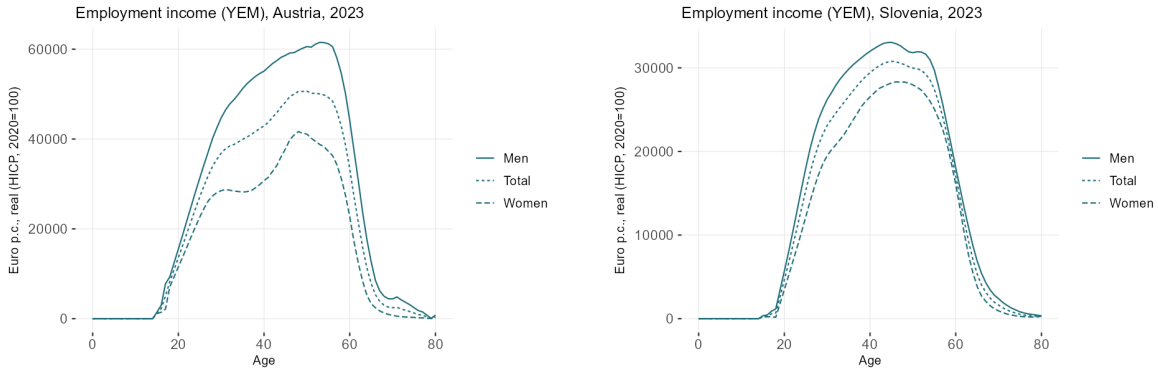


Figure 3.1: Employment income by age in AT, FR, IT, and PL, relative to the average of men aged 30–49

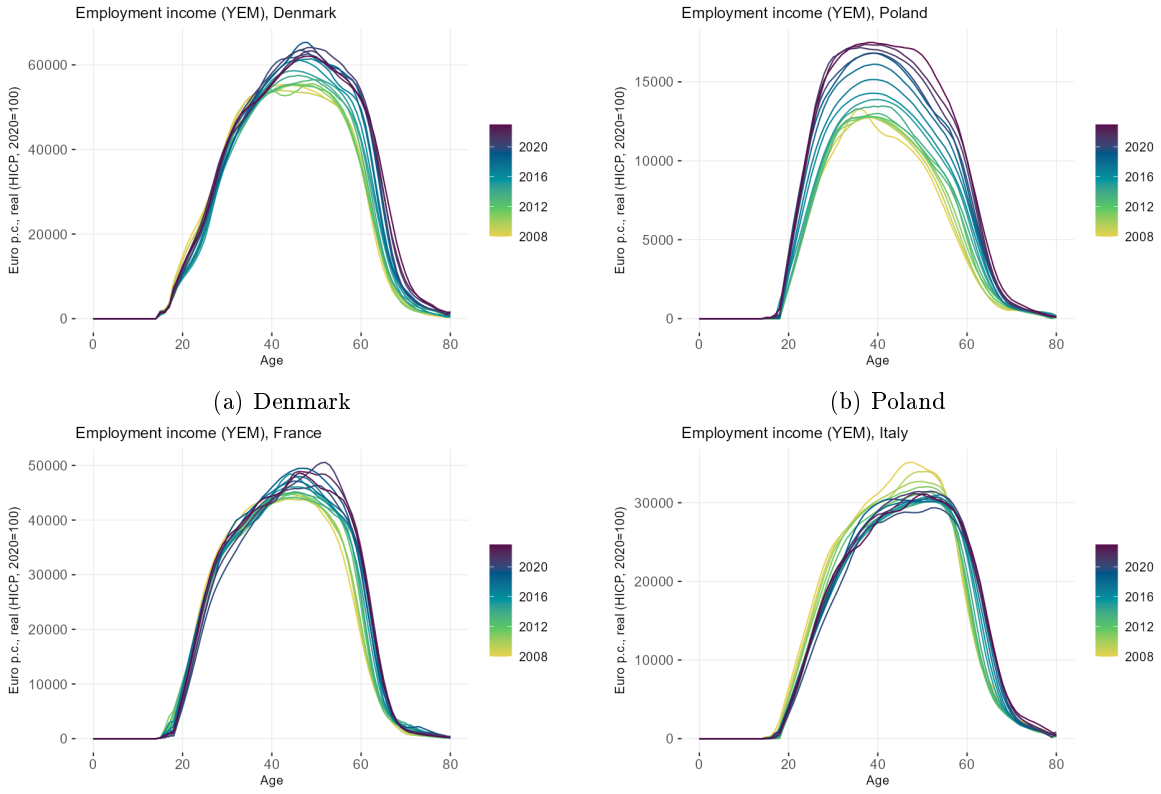
Employment income of men and women differs in all countries. Figure 3.2 shows the profiles by sex for Austria, where the difference is large, and Slovenia, where it is small.



(a) Austria (b) Slovenia

Figure 3.2: Employment income by sex, 2023

The data cover 2008 to 2023 and show changes over time (Figure 3.3): higher employment income at ages 55 to 64 in all countries, mostly because employment rates rose, and a large increase in Eastern European countries.



(a) Denmark

(b) Poland

(c) France

(d) Italy

Figure 3.3: Employment income over time, selected countries

3.1.2 Asset income

The age- and sex-specific estimates of ENTA asset income (YA) are based on the following EU-SILC variables: ESA *property income* (received) is distributed according to the combined EU-SILC categories *interest, dividends, and profit from investments in unincorporated businesses* (HY090G) and *rental of property or land* (HY040G). Property income paid by households is approximated by *interest repayments on the mortgage* (HY100G). ENTA provide the net values, defined as property income received minus property income paid. Income from individual private pension plans (PY080G) is added to property income received. The age profile of corporate income (YC), which ENTA allocate to households as the owners of corporations, uses property income received without these pensions: private pensions are paid out of pension assets and concentrate above age 60, and would move corporate income to older ages.

In Denmark, EU-SILC reports property income net of interest paid before 2019. For these years property income is approximated with rental income (HY040G) and imputed rent, and is not fully comparable with other countries.

All results on asset income based on EU-SILC should be taken with caution. Wealthy households are under-represented in the survey, and it records only a fraction of the capital income in ESA

(Section 5.2).

To allocate the *operating surplus* of households to individuals by age and sex, we use information on ownership status and size of dwelling in terms of the number of rooms. This is an approximation because imputed rent (HY030G) is not available for all years and countries: it is missing for Germany in every year, for every country in the income years 2020 and 2021, and for 17 countries in 2023. Operating surplus is assigned to all adult household members, except for adults who are below 40 years of age and live with their parents. In that case, it is assumed that the parents are the owners of the dwelling and the recipients of the imputed rent. Mortgage interest paid (HY100G) is assigned to the same persons. EU-SILC identified the persons responsible for the accommodation (HB080, HB090) only until the 2020 survey (income year 2019); ENTA do not use them, so the same rule applies in every year.

3.2 Taxes and social contributions

Public transfer contributions in ENTA include all transfers from private sectors (households & NPISH, corporations) to the government. The largest components are taxes on income, social contributions, and consumption taxes. ENTA do not distinguish between taxes and social contributions; the term "taxes" refers to both. ENTA distinguish different types of taxes. In particular, taxes on income are decomposed by type of income:

- Taxes and social contributions on employment income (TYEM)
- Taxes on asset income (TYA)
- Taxes paid by non-employed (TB)
- Taxes on government pensions (TBSPEN)
- Taxes on non-pension benefits, such as unemployment and family benefits (TBSN)
- Taxes on funded pensions (TBPPEN)
- Other taxes and transfers (TX)
- Consumption taxes (TCON)

The estimation of age- and sex-specific income taxes by type combines two data sources: the tax estimates from the tax-benefit microsimulation model EUROMOD and income data from EU-SILC.

The EUROMOD data are used to estimate tax rates by age and type of income. EUROMOD micro-data include the income tax paid by each individual (the variable *ils_tax*), the social contributions by type, and the income-tax base, i.e. the taxable income before deductions. The estimation proceeds in six steps; steps 5 and 6 concern the taxes on benefits (TB) and their split into taxes on pay-as-you-go pensions (TBSPEN), on funded pensions (TBPPEN) and on non-pension benefits (TBSN).

1. Taxability of each income component. The income components are employment income (including self-employment income), property income, pay-as-you-go pensions, funded pensions, and non-pension benefits by function (family, unemployment, sickness, other). For each country and year, the income-tax base is regressed on these components without a constant. The base is almost exactly a linear function of them: the coefficients are close to one for taxable and close to zero for tax-free income (pensions in Lithuania and Slovakia, for example), and the fit is close to perfect (R^2 of 0.97–1.00 in 2019). The coefficients are bounded to the range from zero to one: a euro of income cannot add more than a euro to the tax base unless the law says so. The only such case is the Czech "super-gross" wage (gross wage plus employer contributions), the tax base until 2020, for which the estimate is kept. EUROMOD offers a common income-tax base for 19 countries; Austria, Denmark, Greece, Finland, Italy and Slovenia use their own base variables, and the Netherlands, which has no single income-tax base, is given a fixed rule: all components are taxable except family benefits.

2. Taxes and contributions by income component. Each individual's income tax is split over the income components in proportion to their taxable part (taxability times amount). Tax-free income is thus assigned no tax, and child tax credits and other credits reduce the tax on all taxable income of the person alike. Social contributions are assigned to the income they are levied on: contributions on earnings to employment income; other contributions to all income of the person except family benefits, which carry no contributions, in proportion to the amounts; and the contributions that the state or social insurance pays on behalf of beneficiaries (credited contributions) to pensions and non-pension benefits, family benefits included, because the state pays them for parents on parental benefit as well as for the unemployed and pensioners. Income tax of people without any taxable income is not allocated; it is below 1% of the income tax in all countries except Greece (2.4% in 2019).

3. Funded and pay-as-you-go pensions. Funded pensions are distinguished from pay-as-you-go pensions where EUROMOD identifies them: occupational pensions in the Netherlands and Germany and labour-market pensions in Denmark. They go to higher incomes and are taxed more heavily — in the Netherlands in 2019 the income tax on funded pensions was 15% against 4.5% on the state pension. Elsewhere funded pensions are small (below 7% of all pensions, except in Sweden, Belgium, Cyprus and France) and take the pay-as-you-go rate.

4. Age-specific tax rates. Age-specific effective tax rates for each income type are the ratio of the allocated taxes to the corresponding income within age groups: 0–19, then ten-year groups up to 80 and older, so that the fall of the tax rate on pensions with age is kept. The tax rate on property income is computed once per country and year and applied to all ages, since property income is concentrated among few, mostly older, households. All rates are capped at 70%. A rate below 1% is set to zero; where this leaves no rate in the age groups that hold the income, the tax takes the age shape of that income (Table 6.2). These rates are not sex-specific; sex variation in the resulting tax estimates comes entirely from applying the same age-specific rate to sex-differentiated income in EU-SILC.

5. Composition of the taxes on benefits. The aggregate taxes on benefits (TB) are DG TAXUD's taxes on labour paid by the non-employed. DG TAXUD's national tax lists, which give every tax and social contribution of a country with the economic function it is assigned to, show that TB consists of two parts:

- items assigned directly to the non-employed — mostly social contributions: paid by pensioners and other non-employed persons (Austria, Belgium, Germany, France, Italy, Luxembourg, the Netherlands), paid for the unemployed (Spain, Hungary), or health insurance paid by the state for pensioners, children and other state-insured persons (the Czechia, Slovakia, Lithuania, Estonia) — and in single countries a tax: the income tax on social benefits in Poland, a tax on pension lump sums in Denmark, a levy on pensions in Greece;
- the part of personal income tax that the countries attribute to transfers (DG TAXUD's split of personal income tax into labour, self-employed, capital and transfers).

Each directly assigned item is classified by the income it is levied on: pay-as-you-go pensions only (the Greek levy, the contributions of retired Austrian civil servants), funded pensions only (the Danish tax on lump sums), pensions of both kinds, non-pension benefits, pensions and benefits together (contributions of the non-employed), state-paid contributions, or people without a benefit (students' fees, contributions of the uninsured). The two parts reproduce the published TB within 2% in every year since 2005, once the rows that DG TAXUD added to the lists without an ESA code (the split of an item over several economic functions, corrections for taxes unlikely to be collected) are counted with the item above them. The exception is Slovenia, where part of the social contributions is assigned to TB with a share that is not published; that remainder is treated as state-paid contributions. The composition differs widely: the directly assigned items are all of TB in Czechia, Slovakia and Poland, while in Italy, Sweden, Finland and Ireland TB is income tax only.

6. Splitting TB and building the age profiles. Each directly assigned item goes to the income it is levied on. The parts that cover several kinds of transfer — the income tax on transfers, the contributions of the non-employed and the state-paid contributions — are split between pay-as-you-go pensions, funded pensions and non-pension benefits with EUROMOD's income tax, other contributions and credited contributions per euro of each kind of transfer (steps 2 and 3), applied to the corresponding ESA amounts; where EUROMOD gives no such value, the ESA amounts alone are used. TBSPEN, TBPPEN and TBSN therefore add up to TB in every country and year. Their age profiles follow from the age-specific tax rates of step 4 applied to the EU-SILC pensions and non-pension benefits, and each profile is scaled to its own aggregate. Pay-as-you-go and funded pensions are not distinguished in EU-SILC, so both pension tax rates are applied to the pensions recorded there, and the two profiles differ only by the age pattern of their rates. Before EUROMOD data are available (2005–2007, depending on the country), only the total TB is published.

The tax rates are applied to the income data of EU-SILC because EUROMOD input data are not available for all years since 2008 and are published later than EU-SILC, while the EUROMOD models, and so the rates, exist for every year. Within an age group one average rate applies to everyone, so the approach cannot reproduce differences in tax allowances between persons of the same age;

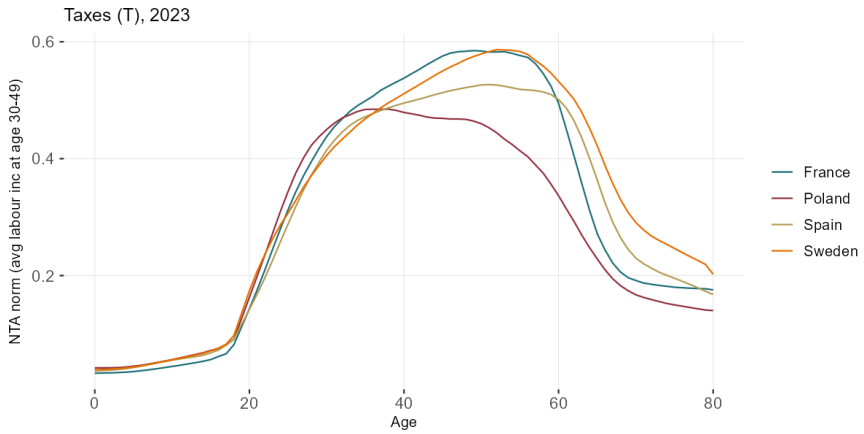


Figure 3.4: Taxes by age in FR, PL, ES, and SE, 2023, relative to the average labour income at ages 30–49

analyses by income or household type should use EUROMOD directly.

Consumption taxes (TCON) have the age profile of private consumption other than imputed rent. Other current transfers (TX) have a receipt profile, employment income plus pensions, for both components, the household part (TXH) and the part of non-profit institutions serving households (TXNPISH). These transfers are contained in household disposable income before the intra-household transfers are estimated, so they reach dependent household members through those transfers; allocating them to children as well would count them twice. The profile is therefore zero below age 15, and the same for both sexes.

Figure 3.4 shows total taxes by age relative to average labour income at ages 30–49. The shape largely follows the age profile of employment income.

3.3 Government transfer benefits in cash

The age- and sex-specific estimates of government transfer benefits in cash are based on EU-SILC. ENTA distinguish the following categories:

- Pensions (BSPEN)
- Family benefits (BSFAM)
- Unemployment benefits (BSU)
- Other social benefits (BSX), including health

Subsidies are part of the government transfer contributions, which are taxes paid less subsidies received. EU-SILC distinguishes social benefits by function in a similar way to COFOG, but the concepts are not fully consistent, and the smaller categories have too few observations for age profiles by sex. ENTA therefore combine several types of transfers into larger groups.

In ENTA, the social benefits (other than social transfers in kind) in the COFOG functions "old age", "survivors", and "disability" are combined into the category pensions (BSPEN). The age- and sex-specific estimates are based on the EU-SILC variables old-age benefits (PY100G), survivor benefits (PY110G), and disability benefits (PY130G). The ENTA category family and children (BSFAM) includes the COFOG categories "social protection family & children" and "education". Age-specific estimates are based on the EU-SILC variables family-related allowances (HY050G) and education-related allowances (PY140G). Family-related allowances are assigned to the parents of the children. Within couples, they are allocated inversely to the partners' shares of personal gross income (bounded between zero and one), so that the partner with the lower income receives the larger part. Where no child can be attributed, they are shared among all adults of the household, also inversely to their income shares. Where the income of the couple or of the household is not positive, or there is only one adult, the shares are equal. Unemployment (BSU) is based on PY090G. In Romania, Slovakia and Greece, where EU-SILC has too few receivers, the age shape of unemployment benefits is instead the share of persons receiving unemployment benefits or assistance in the Labour Force Survey (Section 6.4). In Denmark, the green check, a small transfer that almost every adult received and that Danish EU-SILC records in PY090G until 2018, is moved to other benefits (Section 6.4).

Unemployment benefits end within a year when the unemployed move into retirement or pre-retirement schemes, at an age that differs by country, sex and year. Smoothing across that age would spread benefits into ages at which almost nobody receives them. ENTA therefore determine, for each country, year and sex, the age at which BSU ends: the first age from 57 at which BSU stays at or below 15% of its mean at ages 50–56 for three ages in a row, replaced by the median over the previous, current and following year, or 62 where no such age is found. The profile is smoothed up to four years before that age, the last three years before it are a 1-2-1 moving average of the observed values, and at that age the observed value is kept. After that age few persons still receive unemployment benefits, and the observed means jump between zero and the amount of a single receiver. The ages from the year after it to 80+ are smoothed on their own, and the open age group 80+ may not exceed the value at 79. At ages 16 and 17, where the observed means also rest on few receivers, the value at 17 may not exceed that at 18, and the value at 16 not that at 17. Pensions rise steeply at the statutory retirement ages, within two or three years of age, and the smoother flattens this rise. From age 50 to the age at which the smoothed pension profile reaches its maximum (searched between 58 and 72), pensions are therefore not smoothed but fitted with a curve that may not fall (isotonic regression of the observed means), followed by a three-year moving average. The fit is bounded by the values of the smoothed profile at age 50 and at its maximum and joins it there, so that the parts meet without a jump; below 50 and from the maximum on the smoothed profile is kept; at older ages pensions per person may fall again, as in Belgium.

Where benefits in PY090G are in fact early-retirement pay, as with the Danish *efterløn*, pensions take over from them at the same age: their sum changes smoothly, while each part has a step. Keeping the step in unemployment benefits but smoothing it out of pensions would count both before that age and neither after it. From three years before the age at which BSU ends to two years after it, pensions are therefore the smoothed sum of unemployment benefits and pensions, minus the unemployment

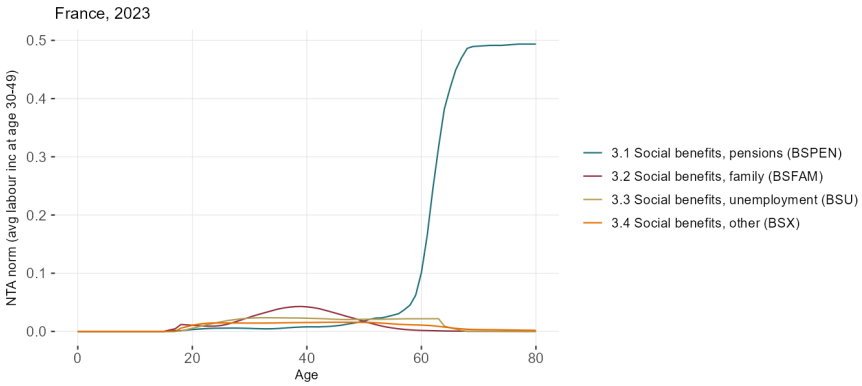


Figure 3.5: Social benefits by type and age, France 2023

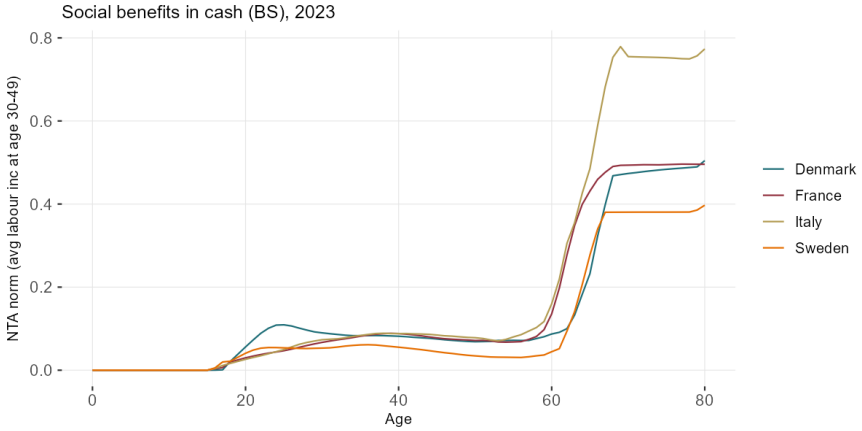


Figure 3.6: Social benefits in cash by age; DK, FR, IT, and SE in 2023

benefits. The adjustment matters mainly for Denmark, and to a lesser extent for Belgium, Finland and Ireland; elsewhere unemployment benefits are small at these ages.

The category other social benefits (BSX) combines social exclusion (HY060G) and housing allowances (HY070G), distributed to all adults in equal shares, because the amounts are small and there are simply too few observations to allow age- and sex-specific estimates. The survey's sickness benefits (PY120G) are not used: they are mainly sick pay from employers, while government sick pay is part of the pension aggregate (Appendix B, BSPEN).

Figure 3.5 shows the typical age patterns for France: cash benefits consist mainly of pensions, family benefits are concentrated at ages 30–45, and unemployment benefits are spread over the whole working age.

The differences between countries are dominated by the size of government pensions (Figure 3.6).

3.4 Funded pensions

Funded pensions in ENTA consist of contributions paid to mandatory funded pension systems and the benefits received from these systems. The aggregate values for contributions and benefits are approximated by social contributions paid by households to financial corporations and the benefits paid by financial corporations to households.

ENTA keep funded pensions apart from pay-as-you-go pensions, for consistency with ESA at the sector level and for research questions that distinguish the two. The microdata do not separate them, so ENTA assume that the age shape does not depend on the type: contributions have the age profile of taxes on employment income, benefits that of pension benefits including funded pensions (BSPPEN).

3.5 Private transfers

NTA distinguish between private transfers between households and private transfers within households.

3.5.1 Private transfers between households

On average, private transfers between households have little influence on the disposable income of households in EU countries. Private transfers between households comprise all transfers mediated by the household sector (e.g. remittances, alimony payments) and the corporation sector (non-life insurance premiums and claims). Note that social contributions and benefits paid to and received from financial corporations are not included here but under funded pensions.

EU-SILC records regular cash transfers received from other households (`HY080G`) and paid to other households (`HY130G`), such as alimony, support of relatives and remittances. Few persons in each age group are involved, the totals paid and received differ substantially (partly because transfers to and from abroad are included), and the concept does not match the national accounts, where transfers between resident households cancel. ENTA therefore use EU-SILC only for the transfers between resident households, with an age profile pooled over years, and rely on assumptions for the other private transfers.

Transfers between resident households

Transfers between resident households redistribute income between age groups — mostly from parents aged 50 to 74 to their adult children aged 18 to 39 — but have no ESA aggregate. ENTA estimate them from EU-SILC, assigning the household amounts received and paid to the adults of the household in equal parts:

- **Domestic volume.** In each country and year, the totals received and paid are both set to the smaller of the two. This is the domestic part, which nets to zero within the household

sector. The excess of receipts or payments is treated as cross-border and remains part of the ESA transfers from and to the rest of the world (BROW, see below).

- **Age profile.** Because single-year estimates are too noisy (small numbers of receivers and payers in several countries, large changes from year to year), the age and sex profile is pooled over all years: each person's amount is taken relative to the year's amount per person, averaged by age and sex over the years and smoothed. For each year, this profile is scaled to the year's domestic volume.
- **Consistency.** The same balanced amounts enter the individual income from which intra-household transfers are derived (see below). The published inter-household transfers received and paid are therefore included in BPB; as received equals paid in total, BPB keeps its ESA aggregate and only its age profile changes.

The transfers are of some weight (above 1% of household disposable income) in Bulgaria, Cyprus, Greece, Czechia, Latvia, the Netherlands, Hungary and Austria, and small elsewhere. Individuals use their actual EU-SILC amounts for the intra-household transfers, while the published profiles follow the pooled age shape; the difference cancels out over whole age groups and does not affect any total.

Transfers mediated by non-financial corporations

Social contributions and benefits mediated by non-financial corporations include, for example, the continued payment of wages during illness (Nelisse, 2021). EU-SILC does not separate them from government benefits and has too few observations for age profiles. ENTA therefore assume that contributions follow the age profile of taxes on employment income (TYEM), while benefits follow the age profile of employment income (YEM).

Other private transfers

For the remaining private transfers recorded in ESA (e.g. remittances to and from abroad, non-life insurance premiums and claims), ENTA treat the net values (received minus paid) as flows received from outside the household and distributes them by the age profile of employment income plus pensions (YEM + BSPPEN), taken for both sexes together rather than sex by sex, because the household member who receives such a transfer is as often a woman as a man.

3.5.2 Private transfers within households

The estimation of intra-household transfers and private consumption in ENTA goes hand in hand. Estimates of intra-household transfers require microdata that include income and consumption at the individual level. ENTA create such data by imputing consumption from HBS into EU-SILC (Section 3.8.5). A person's consumption is the sum of the four components of private consumption; for education, every person has the amount of his or her age and sex from the enrolment-based profile (Section 3.8.2). In the NTA framework, intra-household transfers are derived from the difference between consumption and income at the individual level: if an individual's income falls short of their

consumption, NTA assumes that the difference is covered through transfers from other household members.

Estimating intra-household transfers

Individual income before intra-household transfers comprises employment income, property income received less interest paid, public benefits and transfers received from other households less transfers paid to them (the balanced domestic part described above), less taxes on these incomes. It is scaled to the aggregate of household cash disposable income.

ENTA assume that couples share their income: the first type of intra-household transfers in ENTA are transfers between two partners to equalise their income. Since partners are usually of a similar age, this assumption has little effect on the net redistribution between age groups.

After redistributing between partners, based on the estimates of individual income and consumption, each person is classified as having a *surplus* (income exceeds consumption) or a *deficit* (consumption exceeds income). These individual surpluses and deficits are then aggregated at the household level to determine whether total household income is sufficient to cover total household consumption.

Case 1: Household surplus covers the deficit. If the combined surplus of all household members is large enough to cover the combined deficit, the estimation of intra-household transfers is straightforward. A household-specific "internal tax rate" is calculated as the ratio of the total deficit to the total surplus. Each member with a surplus is assumed to transfer this proportion of their surplus to those with a deficit, fully financing their consumption shortfall.

Case 2: Household surplus falls short of the deficit. If the total household surplus is insufficient to cover the total deficit, it is assumed that the available surplus of individual members is used to cover at least part of the deficit. Some members will remain with an uncovered deficit. This remaining deficit is assumed to be covered by the adult income earners in the household, with a contribution proportional to their income. Where the adults of the household have no positive income together, the adults cover the remaining deficit in equal parts, i.e. they dissave on behalf of the other members. Every household has at least one adult.

Household income falls short of consumption frequently in the microdata, because imputed consumption is only loosely related to the income of a single household. Such errors do not cancel out in the transfer profiles: a person's deficit is consumption less income where that is positive, and zero otherwise, so errors in either direction raise the gross transfers within households and can shift them between ages. The size of this effect has not been measured. The net transfers of an age group, and the consumption and income profiles themselves, are scaled to the aggregates and are less exposed than the gross flows `TFWI` and `TFWO`.

Figure 3.7 shows the result for two countries. Children and young adults receive the transfers, and adults of working age pay them. In Italy, where children leave the parental home late, the transfers

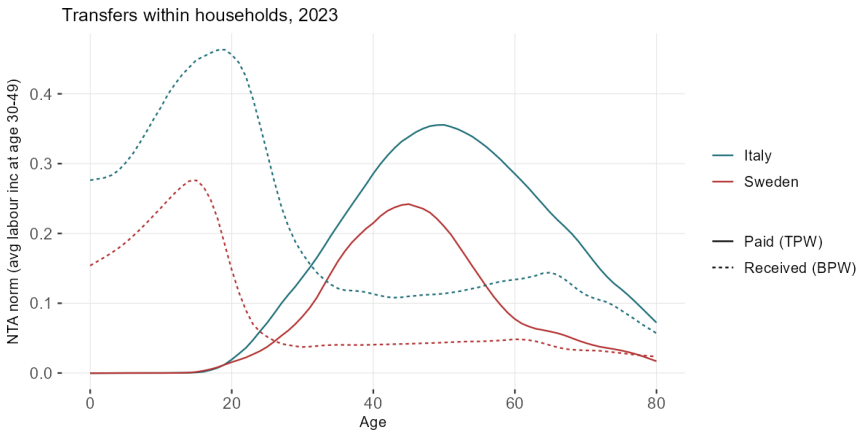


Figure 3.7: Intra-household transfers, paid and received, Sweden and Italy, 2023, relative to the average labour income at ages 30–49

received stay high up to the mid-twenties and the transfers paid peak around age 50; in Sweden both end about five years earlier.

3.6 Disposable income

Disposable income in ENTA reflects all types of primary income and all transfers, including transfers within households. It is calculated based on the age profiles of primary income and transfers:

$$YD = YEM + YA - T + BS + BPPEN + BPB + BPW - TPW \tag{3.1}$$

where YD is disposable income, YEM is employment income, YA is asset income, T is taxes and social contributions, BS is government transfer benefits in cash, $BPPEN$ is funded pensions (received less paid), BPB is private transfers between households, BPW is private transfers within households received, and TPW is private transfers within households paid.

The disposable income of children consists almost entirely of intra-household transfers received from their parents.

In most countries, NTA disposable income has a first peak in the late twenties or early thirties, before people have children (Figure 3.8). It is lower at ages 35–50, because part of income is transferred to children. Disposable income rises again in old age in most countries, as transfers to children decline and pensions replace employment income.

3.6.1 Alternative measures of disposable income

Asset income and private transfers vary considerably from year to year, because their age-specific estimates are uncertain. ENTA therefore provide three further measures that are better suited for the analysis of changes over time. The first ($YD1$) holds the components with the most reliable

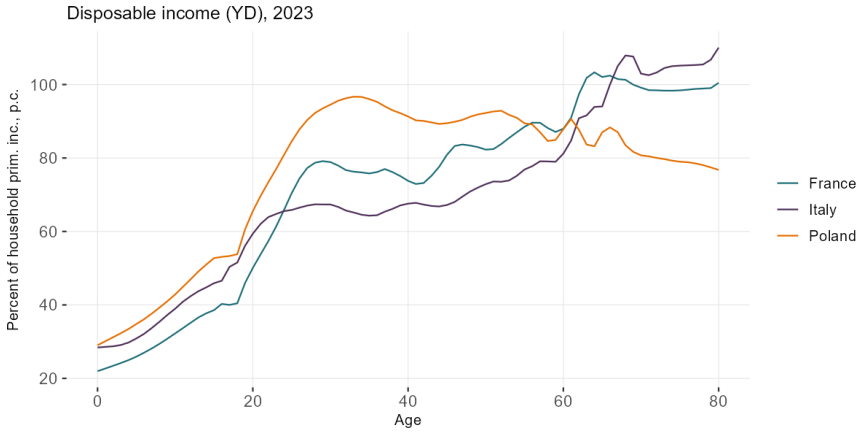


Figure 3.8: Disposable income by age in IT, FR, and PL, in percent of mean primary income of households, 2023

estimates: employment income, social benefits and funded pensions (received less paid), less the taxes on employment income, benefits and consumption. The second ($YD2$) adds the private transfers within households, and the third ($YD3$) the operating surplus of households, mainly imputed rent. None of them includes property income, the component most difficult to distribute by age and sex:

$$YD1 = YEM - TYEM + BS + BPPEN - TB - TCON \tag{3.2}$$

$$YD2 = YD1 + BPW - TPW \tag{3.3}$$

$$YD3 = YD2 + YO \tag{3.4}$$

3.7 Government consumption

Only expenditure in the functions of education and health is distributed to age groups. For the other consumption components, it is assumed that all residents consume the same amount.

3.7.1 Education

To distribute the consumption expenditure of the government sector to age groups, ENTA use information on enrolment by age, sex, and education level,[1] as well as information on total expenditure by education level from the data on government expenditure by COFOG. The expenditure data allow a distinction of only three educational groups: pre-primary & primary education (ISCED 0–1), secondary education including post-secondary non-tertiary education (ISCED 2–4), and tertiary education (ISCED 5–8). To each enrolled person, ENTA assign the average spending of the level. The part of education expenditure that COFOG does not assign to a level (administration, subsidiary services, research and development; between 5 and 20 per cent of the total in most countries) has

[1] `https://ec.europa.eu/eurostat/databrowser/view/educ_uoe_enra02/default/table` and `https://ec.europa.eu/eurostat/databrowser/view/EDUC_ENRL1TL/default/table`

no enrolment to follow and is spread equally over all ages. The enrolment data of Eurostat are partly reported in age groups and with gaps; how single ages are filled is described in Appendix D (Section D.3.4), and the resulting limitations in Section 6.5.6.

3.7.2 Health

Age profiles of public health care consumption are taken from a single source for all countries: age- and sex-specific per capita public health expenditure approximated from the 2018 Ageing Report of the European Commission and the Ageing Working Group (AWG) (European Commission, 2018). The Ageing Report publishes these profiles as charts only; ENTA extracted the curves with WebPlotDigitizer (Rohatgi, n.d.), and the extracted points, the script that cleans them and the resulting rates are published with the code (Section D.3.5). One profile per country is used for all years; in each year it is scaled to the government health expenditure aggregate, so the level follows the aggregate while the shape stays fixed.

Using one source for every country keeps the profiles comparable across countries. The 2021 Ageing Report is not used. Its profiles are very similar for most countries, but for some the old-age part is considerably steeper. This most likely reflects a difference between the two editions rather than a change over time (Section 6.5.6). The 2018 edition also lies closer to the middle of the period covered by ENTA.

3.8 Private consumption

ENTA include information on age- and sex-specific expenditure in four categories:

- Education services (CHE)
- Health (CHH)
- Housing (imputed rent of owner-occupiers) (CHR)
- Other consumption (CHX)

The estimates of private consumption in the categories of health and other consumption are based on the Household Budget Survey (HBS), housing consumption (imputed rent) is based on EU-SILC, and education consumption on the national accounts and enrolment.

3.8.1 Household Budget Survey

The Household Budget Survey (HBS) records household expenditure by purpose (Classification of Individual Consumption by Purpose, COICOP), together with household income and the age and sex of the household members. Two properties limit its use for age- and sex-specific estimates. First, it is available only for 2010 and 2015, for the Netherlands only 2015; the 2020 wave records only whether a household member is 60 or older and is not used. Second, expenditure is not observed over a whole year: frequent purchases are recorded over about two weeks and larger items retrospectively,

so the recorded expenditure of a single household is a noisy measure of its annual consumption. Reliable estimates can therefore be obtained only for large population groups and broad consumption categories.

3.8.2 Education

The HBS records education spending for the household and does not say for which member it is spent. ENTA therefore do not use it for education. Household consumption by purpose in the national accounts distinguishes education by level,[2] and each level is given to the persons enrolled at that level, with the enrolment data used for public education consumption (Appendix D, Section D.3.4): pre-primary and primary education (ISCED 0–1), secondary and post-secondary non-tertiary education (ISCED 2–4), and tertiary education (ISCED 5–8). Within a level every enrolled person has the same amount. The fifth class, education not definable by level (courses and adult education), has no enrolment to follow and is spread equally over all persons aged 20–64; it is 37 to 48 per cent of household education spending in Austria, Bulgaria, Czechia, Denmark, Estonia, Germany, Hungary, Slovakia and Spain, 61 per cent in Luxembourg, and 10 per cent or less in Belgium, Croatia, Cyprus, Ireland, Lithuania and the Netherlands.

Pre-primary and primary education are one class in the national accounts, so a child in pre-primary and a child in primary education have the same amount. Eurostat's statistics on education finance, which report the spending of households by level for 2012–2023,[3] show that households spend more on pre-primary education: of their spending on the two levels, pre-primary education takes more than half in nine of the 24 countries with data (more than 90 per cent in Finland and Sweden) and 40 to 50 per cent in seven more, although it covers fewer years of age than primary education; it is below 20 per cent only in Greece, Ireland and Luxembourg. In most countries ENTA therefore understate private education consumption at ages 3–5 and overstate it at the ages of primary school. These data are not used in this version.

The shares of the levels are one set per country, from the sum over the years 2008–2022, because the shares of single years jump in several countries (by 32 percentage points in Latvia and 24 in Romania from one year to the next), which points to changes in the classification. The age profile changes over time with enrolment and population only. Finland and Sweden report no split by level; their education consumption is spread equally over all persons aged 3–64.

3.8.3 Health

Age- and sex-specific estimates of private consumption expenditure on health are based on pooled HBS data for 2010 and 2015, with consumption measured relative to average consumption in each of the two years. The shape of the age profile is the same in all years: household spending on health is small and infrequent, and the data would not identify changes in the age pattern.

[2] `https://doi.org/10.2908/NAMA_10_CO3_P3`, COICOP classes 10.1 to 10.5.

[3] `https://doi.org/10.2908/EDUC_UOE_FINE01`; households are the non-educational private sector less the other private entities.

The estimation of age profiles from the individual microdata requires the allocation of household-level spending to individual household members. This is carried out using an iterative method, proceeding as follows:

1. In the initial step, health spending is distributed equally among all household members, and preliminary age profiles are estimated from these individual-level values.
2. In each subsequent iteration, the age profile from the previous step is used as a set of weights to redistribute household spending among its members. New age profiles are then re-estimated.
3. This process is repeated 15 times, by which point the age profiles have converged.

Unlike for education, where enrolment says who is in education, ENTA have no source that reports by age who uses the health services a household pays for. The iteration assumes no age pattern: it starts from an equal split, and the pattern comes from the comparison of households with members of different ages. The resulting profiles rise with age in every country; in 2023, private health consumption per person at ages 65–79 is 1.7 times that at ages 30–49 in the median country.

3.8.4 Imputed rent

The age profile of the housing consumption in terms of imputed rent (COICOP consumption category 4.2[4]) is allocated to individual household members according to the NTA equivalence scale (described below in Section 3.8.5). The imputed rent of a household is that of its owners, estimated from the number of rooms and assigned in equal parts to the adults of the household, except adults below 40 who live with their parents (see the operating surplus in the section on primary income); it is divided among all members of the household once. The part that the other members consume is an intra-household transfer from the owners to them.

3.8.5 Other consumption

Other consumption comprises all private consumption expenditure not classified as education, health, or imputed rent. Because consumption age profiles are needed also for years without an HBS survey, and because estimates of private intra-household transfers require individual-level consumption data in EU-SILC, other consumption is imputed into EU-SILC using a regression model. The model is estimated as a generalised linear model (gamma family, log link) rather than plain OLS, since OLS produced implausible negative predicted consumption values for some households; the log link guarantees positive predictions by construction.

The model uses HBS from 2010 and 2015 and relates standardised consumption expenditure per household member (from HBS, standardised means that consumption is measured relative to per capita consumption in the specific year) to a set of household characteristics available in both HBS and EU-SILC. These characteristics include the percentile of per capita household income, the number of persons in the household, the number of persons under age 20 (at most three), the number of persons aged 20–34, and the number of persons in the age groups 65–69, 70–74, 75–79, and 80+. The

[4] https://unstats.un.org/unsd/classifications/Econ/Download/COICOP_2018_pre_copy_edit_publication.pdf

age-count variables capture the tendency for per capita consumption to vary systematically with the household's age composition, in particular its decline in households with very old members. Where the HBS ends with an open age group below 80, as in Portugal, the EU-SILC members above it are counted in that group. In some countries, household income is not available in HBS. In that case, the income percentile is approximated using the percentiles of consumption per capita.

Matching and multiple imputation were tried as well; they carry the noise of the HBS into the age profiles, which are then hardly usable for comparisons between countries and over time.

The estimated household consumption is allocated to individual household members using the NTA equivalence scale. This scale assumes the following:

- Children aged 0–4 consume 40% of an adult's consumption.
- Adults aged 20 and older consume equal amounts.
- Between ages 4 and 20, the consumption weight increases linearly from 0.4 to 1.0.

For Germany from 2022 onward, the age-16 weight is set to 0.9 rather than following the linear ramp: EU-SILC's age variable for Germany in these years groups ages 16–19 together under the single value 16, so the age-16 weight has to stand in for the whole 16–19 range rather than age 16 alone.

3.9 Saving

Private saving is calculated as a residual, defined as the difference between disposable income and consumption:

$$SH_i = YD_i - CH_i \tag{3.5}$$

where SH_i denotes the saving of individual i, YD_i is disposable income, and CH_i is total private consumption (including education, health, imputed rent, and other consumption).

As a residual, saving inherits the measurement errors and assumptions of both the income and the consumption estimates: any error in either profile at a given age appears one for one in saving at that age. Figure 3.9 shows the age profile of saving for selected countries.

The profile does not follow the simple life-cycle pattern: saving is close to zero for children and positive in old age in all countries shown, where simple life-cycle models predict dissaving. The positive saving at old ages should be read with three qualifications. First, the surveys cover private households only (Section 6.2). Persons in care homes are therefore missing from the shapes, while the income and consumption aggregates include them. Where they pay for care partly from assets and so consume more than their income, the profiles at the oldest ages overstate saving. Second, consumption is allocated to the members of a household with an equivalence scale, not observed per person, and older persons living with younger adults receive the same share as these adults. Third, consumption is imputed from the HBS into EU-SILC. Section 5.3 shows that the imputation, if anything, overstates

consumption at ages 65 and over relative to working ages; correcting it would raise saving at old ages rather than lower it.

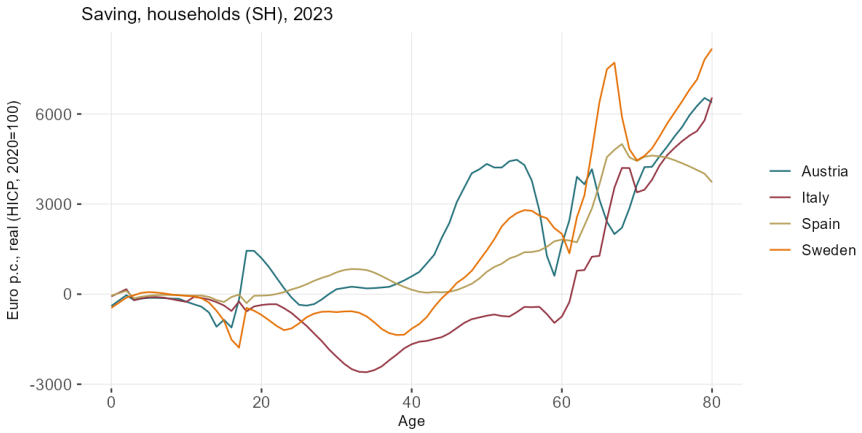


Figure 3.9: Saving by age, 2023, Austria, Italy, Spain and Sweden

4. Translating ESA/ENTA into NTA concepts and terminology

4.1 NTA variables

Although NTA are based on National Accounts, they use a different classification and terminology. ENTA are therefore provided in two formats: the *ENTA–ESA format*, which keeps a close link to ESA terminology and sector structure, and the *NTA format*, which follows the conventions of the global NTA project. This chapter describes the main differences between the two and how the NTA variables are derived from the ENTA–ESA data. Table C.1 in Appendix C lists all NTA variables with their correspondence in the ENTA–ESA format; the codebook gives the formula of each. The Stata programme *ENTA_compilation2_nta.do* does the translation.

The ESA-related format has two practical advantages. The microdata come from household surveys, which map more naturally to the ESA household sector than to the NTA private sector, which includes corporations; and the ESA sequence of accounts is more familiar to most users than the NTA life-cycle framework.

4.1.1 Treatment of mixed income

The largest conceptual difference concerns income. NTA distinguish between labour income (YL) and asset income (YA). Compensation of employees (YE) is labour income. Mixed income (YM) contains both a labour and a capital component. In the ENTA–ESA format it is one category within employment income. In the NTA format, following standard NTA practice, two-thirds are labour income and one-third is asset income of households:

$$YL = YE + \tfrac{2}{3}\, YM, \tag{4.1}$$

$$YAH = YA + \tfrac{1}{3}\, YM. \tag{4.2}$$

Household asset income (YAH) thus includes property income, the operating surplus (mainly imputed rents from owner-occupied housing), and the capital share of mixed income. Private-sector asset income combines households and corporations: $YAF = YAH + YAC$.

The two-thirds share is a convention, and its effect depends on the weight of mixed income. With an equal split instead, labour income in 2019 would be 1.6% lower in the median country, but 4.5 to 6.6% lower in Greece, Poland and Romania, where mixed income is 25 to 35% of employment income. The age shape of labour income hardly changes, but labour income at 65 and over falls more than at working ages, because many of the older persons still at work are self-employed or farmers. Comparisons of labour income and of asset income across countries with very different shares of self-employment should keep this convention in mind; employment income (`YEM`) of the ENTA–ESA

format does not depend on it.

Asset income of the government (YAG) is the primary income of the government sector. It is often negative, reflecting interest payments on public debt; the taxes on production that the government receives are not part of it, since ENTA record them as a transfer from the private sectors (Chapter 2).

4.1.2 Sectoral detail

The ENTA–ESA format reports quantities for the household sector (including NPISH). The NTA format combines households and corporations into private-sector totals. This applies to asset income (YAC), taxes ($TGOTC$), private transfers ($BPBC$), disposable income ($YDF = YDH + YDC$) and saving ($SF = SH + SC$). Corporations do not consume, so their disposable income equals their saving. The NTA format also includes income and saving of the government sector (YAG, SG), and net government transfers to and from the rest of the world ($TGNF$).

4.1.3 Terminology for public and private transfers

In NTA, all flows from private sectors to the public sector are *transfer outflows*; social benefits received from the government are *transfer inflows*. Total public transfer outflows (TGO) comprise all taxes plus the difference between non-tax government income and government saving, the public resources that do not stem from taxes; these are allocated by age with the profile of total taxes:

$$TGO = TGOT + YAG - SG.$$

Public cash transfer inflows are distinguished by function: pensions ($TGICPEN$), family ($TGICF$), unemployment ($TGICU$), and other ($TGICX$). Public consumption is an in-kind transfer inflow, split into education (CGE), health (CGH), and other (CGX), so that total government inflows are $TGI = TGIC + CG$. Adjusted disposable income of households adds it to disposable income: $YDHA = YDH + CG$.

Private transfers are divided into transfers *between* households (TFB) and *within* households (TFW). Between-household transfers include the flows of the corporation sector: $TFB = BPB + BPBC$.

5. Data quality assessment

ENTA combine two kinds of data that differ in quality. The aggregates come from the national accounts and are complete by construction; the age profiles come from surveys and inherit their sample sizes, their coverage and their changes over time. Every survey-based profile is scaled to its aggregate, so the surveys decide only how a total is distributed over ages and sexes, not how large it is. This chapter assesses that survey basis in three ways: by the number of observations behind each profile (Section 5.1), by how much of each national-accounts aggregate the survey sees (Section 5.2), and, for consumption, by how well its imputation into EU-SILC reproduces the age profile of the Household Budget Survey (Section 5.3). Comparisons with other NTA results, which test the age shapes themselves, are in Section 5.4; the checks run on the published files are described in Section D.4.

The figures in the first three sections are produced by scripts in the folder *ENTA_microdata_checks*: *cell_counts.do*, *coverage_silc.do* and, for the imputation, *imputation_check.do*. The first two cover the years for which ENTA publish age profiles: the income years 2008 to 2023 of EU-SILC (survey years 2009 to 2024); medians are taken over these years. The tables and the figure are generated from the output of the three scripts by *Manual_quality_tables.R* in the manual folder.

5.1 Number of observations

An age profile is a weighted mean by single year of age and sex, so its precision depends on the number of persons in each age and sex cell and, for a transfer or an income type that only some people receive, on the number of receivers among them. The profiles from EU-SILC are smoothed from age 18 (Chapter 3), which reduces the noise of single ages; the counts below refer to the unsmoothed cells. The number of observations is reported for EU-SILC, the source of the age shapes of income, taxes and benefits; the other two surveys are described at the end of the section.

5.1.1 EU-SILC

Table 5.1 gives, for each country, the persons in the sample per year and per single year of age at ages 20–79, and the effective sample size of such a cell. The effective sample size, $(\sum w)^2 / \sum w^2$ for the weights w of the persons in the cell, is the number of persons with equal weights that would give the same precision; it is smaller than the number of persons wherever the weights vary.

A single age at 20–79 holds a median of 136 persons in Luxembourg and 607 in Italy. The weights reduce this to an effective 70 persons in Luxembourg, about half of the persons. Split by sex, a cell holds half as many: in Luxembourg 24% of the single-age cells of one sex at ages 18–79 have fewer than 50 persons; in 13 countries none has. The open age group 80+ holds 232 persons a year in Luxembourg and up to 3,000 in Italy.

For income types and transfers that few people receive, the number of receivers matters more than the number of persons. Unemployment benefits are received by a median of 4 to 6 persons per single age at ages 25–59 in Slovakia, Croatia, Czechia and Luxembourg, by none in Romania, and by more than 100 only in Spain and Italy. In 14 countries every single-age cell of one sex at ages 25–59 has fewer than 20 receivers. The age profiles of unemployment benefits are therefore shapes drawn from few observations in most countries, and smoothing carries much of their form. In Romania EU-SILC records only 2 to 67 receivers a year from the income year 2011, in all ages together, and sees between 2 and 28% of the aggregate (Section 5.2); the age shape of Romanian unemployment benefits is therefore taken from the Labour Force Survey (Chapter 6). Property income is similarly thin in several countries: a median of 3 receivers per single age at ages 30–79 in Romania, 11 in Hungary and 19 in Poland.

Table 5.1: EU-SILC: observations per country, medians over the income years 2008–2023

		Per single age, 20–79				Receivers per single age			Cells
Country	Persons per year	Persons	Smallest cell	Effective sample	80+	BSU 25–59	BSFAM 25–44	YPr 30–79	< 50 (%)
AT	12,929	159	76	128	538	24	93	120	7
BE	14,468	180	84	145	536	28	118	106	6
BG	17,012	226	103	161	1,081	15	69	34	2
CY	11,098	140	66	102	486	10	55	32	10
CZ	18,954	240	124	199	913	6	47	37	0
DE	27,958	404	172	333	599	40	154	350	0
DK	13,318	160	77	116	506	24	81	136	18
EE	14,723	180	91	121	725	12	107	46	4
EL	22,626	311	164	243	2,239	13	55	55	0
ES	36,622	440	238	262	2,028	108	30	267	0
FI	24,348	278	118	179	696	52	180	258	0
FR	26,646	333	169	220	1,192	45	197	304	0
HR	19,071	227	123	170	1,094	5	42	25	0
HU	18,746	231	133	175	880	11	106	11	1
IE	11,978	143	64	81	510	28	90	30	20
IT	47,250	607	369	374	3,000	108	182	364	0
LT	12,000	149	70	83	724	12	60	26	23
LU	10,534	136	38	70	232	6	86	78	24
LV	13,894	173	104	136	850	16	95	24	6
NL	26,018	298	142	175	998	18	159	311	0
PL	37,960	480	242	305	1,502	8	152	19	0
PT	26,694	362	160	171	1,583	16	71	76	0
RO	17,404	225	148	148	948	0	79	3	0
SE	15,906	190	86	146	560	14	129	167	3
SI	25,996	340	144	225	1,002	18	165	120	0
SK	15,460	207	76	140	500	4	92	43	6

Persons: unweighted persons in the sample. Smallest cell: the smallest single age at 20–79 in the median year. Effective sample: $(\sum w)^2 / \sum w^2$ per single age. Receivers: persons with more than 1 euro of unemployment benefits (BSU), family benefits (BSFAM) or property income received (YPr). Cells < 50: share of the single-age cells of one sex at ages 18–79 with fewer than 50 persons. Source: *cell_counts.do*.

5.1.2 The 20-receiver rule

ENTA treat a survey-based age profile as unreliable on its own when it rests on fewer than 20 receivers per single age: the median number of receivers over the ages at which the profile matters (25–59 for unemployment and family benefits, 65–79 for pensions, 30–79 for property income received, 30–59 for property income paid, 30–64 for self-employment income) is below 20 in that country and year. Cells are not suppressed: every profile is smoothed and scaled to its aggregate, and a gap would break the accounts. The threshold is a guideline for where a profile takes help from elsewhere, not an automatic switch: where the estimated and smoothed profile shows a consistent pattern over the years, it is kept even if the number of receivers falls below 20 in single years or ages. Help is taken only where the own profile is not consistent over the years and the alternative is clearly better, and then in all years of the country, so that its method does not change between years:

- **Unemployment benefits** of Romania, Slovakia and Greece take their age shape from the share of persons receiving unemployment benefits or assistance in the Labour Force Survey, which has 270 to 4,900 such persons a year in these countries (Slovakia in five-year age groups). Poland, Croatia and Czechia keep EU-SILC: the Labour Force Survey counts registered unemployed persons and misses the payments to older workers, such as early retirement for labour-market reasons, that EU-SILC and the aggregate include (Section 6.4). Cyprus, Estonia and Luxembourg keep EU-SILC because their Labour Force Survey has no more receivers. Bulgaria, Hungary, Lithuania, Latvia, the Netherlands, Portugal, Sweden and Slovenia are below the threshold in the median year as well (Table 5.1); their smoothed profiles are consistent over the years and keep EU-SILC.

- **Property income paid** of Romania, Bulgaria, Croatia, Lithuania and Latvia, and **property income received** of Romania and Hungary, are mixed with the mean shape of the Central and Eastern European countries that meet the threshold in the same year and do not borrow the variable themselves: w times the own shape plus $1-w$ times the donor shape, with $w = n/(n+5)$ and n the country's typical number of receivers per single age (the median over its years). The weight is the same in all years of a country, so that the mix does not change when a single survey year has many more or fewer receivers. In a test that thinned out large samples of the region to 2–16 receivers of property income paid, this mixture had a smaller error of the age shape than the own shape alone and than the donor shape alone. Poland, with a median of 19 receivers of property income received, keeps its own shape, which is consistent over the years.

- **Self-employment income** is below the threshold in every year in Estonia, Luxembourg and Latvia and in some years in eight more countries, but takes no help from elsewhere: no source has more observations, and the age shape of self-employment differs between countries. Where it is thin, employment income (`YEM`), the sum of compensation of employees and mixed income, is the reliable profile; the split between the two should not be interpreted.

How often the profiles fall below the threshold in the 16 income years (15 in Bulgaria and Croatia) is summarised here by profile:

- **Unemployment benefits** are affected most: in 21 of the 26 countries in at least one year, and in every year in Cyprus, Czechia, Estonia, Croatia, Luxembourg, Poland, Romania and Slovakia. Only Austria, Spain, Finland, France and Italy have 20 receivers per single age in all years.
- **Property income received** is below the threshold in every year in Hungary, in 15 years in Romania, in 9 in Poland and in one to five years in Bulgaria, Cyprus, Croatia, Lithuania and Latvia.
- **Property income paid** is below it in every year in Romania and Bulgaria, in 14 years in Croatia, 12 in Lithuania, 9 in Latvia and 7 in Slovakia. In Germany, France and Hungary this is the case only in the single year without mortgage-interest data, which takes the average of the neighbouring years (Germany 2008: the profile of 2009).
- **Self-employment income** is below it in every year in Estonia, Luxembourg and Latvia, in 10 years in Belgium and Slovakia, and in one to seven years in Cyprus, Denmark, Ireland, Lithuania, Portugal and Sweden.
- **Family benefits** are below the threshold only in Lithuania, in two years, and **pensions** in no country.

Where no other source or donor is used, the profile is published as estimated; its age shape rests on few observations and should be interpreted with caution.

5.1.3 Household Budget Survey and Labour Force Survey

The number of observations is not reported for the other two surveys. The LFS, which gives the employment profiles, is larger than EU-SILC. In the HBS the central variable is private consumption other than education, health and imputed rent (`CHX`), which is available for most or all households. The persons of the HBS are recorded only in five-year age groups, and ENTA do not use them as cells: consumption is imputed into EU-SILC with a regression estimated on the pooled waves (Chapter 3), so the size of the HBS sample limits the precision of that regression rather than of single ages. Section 5.3 assesses the imputation.

5.2 Comparing microdata and National Accounts

Each EU-SILC profile is scaled to the household aggregate of the same code. The ratio of the weighted survey total to that aggregate, the *coverage*, shows how much of the national-accounts flow the survey sees. A coverage well below one means that the age shape is that of the part the survey captures; a coverage that changes abruptly between two years points to a change in the survey, which the age shape may share. Coverage is not a measure of error by itself: the survey and the national accounts define several of these flows differently, as listed below, and these differences are stable over time.

The survey totals are computed as the profiles use them: negative values set to 0 and outliers capped at 10 times the mean per receiver, 6 times for property income (Chapter 3). The pairs are:

- **YE**, compensation of employees: employee cash and non-cash income and the employers' social contributions (PY010G, PY021G, PY030G) against D.1 received.
- **YM**, mixed income: income from self-employment (PY050G) against the household share of operating surplus and mixed income.
- **YEM**, employment income: the sum of the two survey totals against the sum of the two aggregates. It is the profile to use where self-employment income rests on few observations (Section 5.1.2).
- **YPr**, property income received: rental income, income from investments (HY040G, HY090G) and private pension plans (PY080G) against D.4 received. The survey records no imputed interest and no investment income attributed to insurance policyholders.
- **YPp**, property income paid: mortgage interest (HY100G) against all interest paid by households (D.4 paid).
- **BSPPEN**, pensions: old-age, survivors' and disability benefits (PY100G, PY110G, PY130G) against pensions including those of funded schemes.
- **BSFAM**, family benefits: family and child allowances (HY050G) and education allowances (PY140G) against the corresponding COFOG benefits.
- **BSU**, unemployment benefits (PY090G) against COFOG unemployment benefits.

Table 5.2 gives the median coverage per country. Over the years, the median across countries lies between 0.87 and 0.96 for compensation of employees, 0.87 and 0.92 for pensions, 0.74 and 0.85 for family benefits and 0.85 and 1.01 for unemployment benefits. Mixed income is covered less, with a median between 0.60 and 0.91, and property income received least, 0.12 to 0.27. The main findings by item:

Employment income and pensions are well covered and stable. The exceptions are Romania (compensation of employees 0.55) and Hungary (0.68). Employment income as a whole (YEM) is covered at 0.89 in the median country and is lowest in Romania (0.44), Hungary (0.67) and Poland (0.70). Pensions lie between 0.75 (Lithuania) and 1.15 (Luxembourg). Compared with pensions without funded schemes, the survey totals exceed the aggregate in the Netherlands (1.60), Denmark and Cyprus, because occupational pensions are part of the survey's old-age benefits; the comparison with pensions including funded schemes is the appropriate one.

Mixed income is covered poorly in several countries, probably where small farms and informal self-employment are common: 0.13 in Estonia, 0.14 in Romania, 0.27 in Poland, 0.40 in Slovakia and 0.40 in Spain.

Property income received is covered at a small fraction of the aggregate in every country except France (0.84) and Denmark (0.94), and at or below 0.05 in Slovakia, Romania, Lithuania and Hungary. The age profile of property income is therefore the age pattern of the rents and investment income

the survey records, which may differ from that of interest and of income from insurance and pension reserves.

Property income paid compares mortgage interest with all interest paid, so its coverage is not meaningful as a level. It is unstable over time and zero or nearly zero in Bulgaria and Romania and in single years in Germany, France and Hungary.

Unemployment benefits vary most across countries: from 0.16 in Romania, 0.32 in Slovakia and 0.39 in Croatia to 1.68 in Sweden and 1.75 in Denmark, where the survey's benefits exceed the COFOG aggregate. In Romania coverage falls from 0.41–0.74 in 2008–2010 to between 0.02 and 0.10 in 2011–2016, while the aggregate stays at about 250 million euro: the break lies in the survey. This is why the Romanian age shape is taken from the LFS.

Family benefits lie between 0.41 (Cyprus) and 1.20 (Latvia), except for Germany (1.90), where the survey's family benefits are about twice the COFOG cash benefits.

Table 5.2: Coverage of the EU-SILC income components: survey total over the household aggregate, median over the income years 2008–2023

Country	YE	YM	YEM	YPr	YPp	BSPPEN	BSFAM	BSU
AT	0.89	1.09	0.91	0.18	0.86	0.94	0.95	0.96
BE	1.00	0.70	0.97	0.16	2.14	0.85	0.90	1.10
BG	0.91	0.92	0.89	0.14	0.00	0.99	0.59	1.16
CY	0.96	0.83	0.94	0.37	0.50	0.99	0.41	1.40
CZ	0.86	0.79	0.85	0.14	0.63	0.86	0.50	1.12
DE	0.81	0.77	0.80	0.12	0.83	0.91	1.90	0.59
DK	0.90	1.13	0.92	0.94	0.80	0.95	0.83	1.75
EE	1.05	0.13	0.96	0.15	0.56	0.89	0.86	1.10
EL	0.88	0.66	0.81	0.36	0.53	0.92	0.64	0.47
ES	0.94	0.40	0.85	0.50	0.87	0.90	0.46	0.92
FI	0.99	1.01	0.99	0.49	1.10	0.92	1.00	1.03
FR	0.89	0.82	0.88	0.84	0.87	0.91	0.79	0.97
HR	0.89	0.91	0.90	0.24	0.12	0.98	0.48	0.39
HU	0.68	0.55	0.67	0.05	0.54	0.88	0.96	0.70
IE	0.87	1.08	0.88	0.29	1.00	0.88	1.06	0.89
IT	0.94	1.06	0.97	0.16	0.76	0.94	0.64	1.41
LT	0.86	1.00	0.88	0.05	0.33	0.75	0.89	0.99
LU	0.91	1.06	0.92	0.42	0.74	1.15	0.78	1.16
LV	0.94	0.61	0.93	0.06	0.40	0.90	1.20	1.10
NL	1.00	0.91	0.99	0.12	2.91	0.97	0.79	0.86
PL	0.91	0.27	0.70	0.07	0.41	0.91	0.65	0.66
PT	0.89	0.50	0.83	0.19	0.68	0.90	0.52	0.87
RO	0.55	0.14	0.44	0.03	0.00	0.81	0.57	0.16
SE	0.78	0.62	0.78	0.29	0.88	0.84	1.05	1.68
SI	0.96	0.55	0.93	0.55	0.68	0.90	0.99	0.78
SK	0.83	0.40	0.75	0.02	0.85	0.78	0.47	0.32
Median	0.90	0.78	0.89	0.17	0.71	0.90	0.79	0.96

Survey totals with negative values set to 0 and outliers capped as in the profiles. Median: median of the country medians. Source: *coverage_ silc.do*.

5.3 Imputation of consumption

EU-SILC records no consumption, and the Household Budget Survey (HBS) is available only for 2010 and 2015. ENTA therefore impute consumption into EU-SILC (Chapter 3): a generalised linear model (gamma family, log link) of consumption per household member, relative to the national average, on the household's income percentile, its size, the number of members below 20 (at most three) and the numbers aged 20–34, 65–69, 70–74, 75–79 and 80 or older. It is fitted on the HBS households of both waves together and predicted for every EU-SILC household of every year. The consumption imputed this way excludes education, health and imputed rent, which are allocated separately. How well the imputation reproduces the age profile of consumption can be checked in the two HBS years, where the reported consumption is known.

The script *imputation_check.do* computes three profiles of consumption per person by five-year age group: the consumption reported in the HBS, the consumption the model predicts for the same HBS households, and the consumption imputed into EU-SILC for the same year, as ENTA use it. Household consumption is allocated to the members with the same equivalence scale in all three, 0.4 for children below 5 rising to 1 at age 20, on five-year age groups because the HBS records no single ages. The difference between the first two is the error of the model; the difference between the first and the third adds the difference between the two samples. As every profile is scaled to the consumption aggregate, only the age shape matters: each profile is divided by its mean, and the deviation is the mean absolute difference between two shapes over the age groups, weighted with the persons in the HBS.

Table 5.3 gives the result. The model reproduces the reported shape with a median deviation of 2.2% of mean consumption in 2010 and 2.5% in 2015; the imputation into EU-SILC deviates by 3.6% in both years, and by up to 7% in single countries. Part of the model error is probably noise in the HBS itself, which the regression smooths out.

At old ages the two shapes are close: relative to ages 25–59, consumption at 65 and over is 0.94 in the imputed profile and 0.92 in the reported one in the median country in 2015; it is higher in 12 countries and lower in 10. The model captures old age only through the number of members in four age groups and the income percentile, so single countries deviate in both directions.

Figure 5.1 shows the three profiles for Spain, where the imputation is close to the reported profile, and for Bulgaria, one of the countries where it deviates most.

Table 5.3: Imputation of consumption: deviation of the age shape from the HBS, in % of mean consumption, and consumption at 65 and over relative to ages 25–59

	Model (HBS)		EU-SILC imputed		65+ / 25–59, 2015	
Country	2010	2015	2010	2015	HBS	EU-SILC
AT	2.2	2.5	2.1	2.3	0.96	0.96
BE	1.5	2.2	3.1	4.5	1.00	0.96
BG	1.5	4.6	6.0	7.0	0.77	0.84
CY	4.0	3.2	7.2	3.7	0.88	0.82
CZ	0.7	0.9	4.2	4.2	0.90	0.88
DE	1.8	1.1	1.9	2.2	1.03	1.04
DK	2.4	3.7	3.5	5.0	1.01	1.02
EE	4.0	3.0	4.9	3.9	0.70	0.70
EL	3.6	2.9	3.8	3.5	0.80	0.85
ES	1.7	1.6	2.1	2.0	0.92	0.92
FI	3.4	2.5	3.6	3.0	0.77	0.76
FR	4.1	3.8	5.4	5.9	1.03	1.09
HR	1.3	1.5	1.7	2.1	0.92	0.89
HU	2.5	1.7	1.9	2.0	1.03	1.02
IE	2.8	2.3	4.4	2.6	0.94	0.94
IT	1.6	2.6	4.7	2.9	0.98	0.97
LT	2.3	3.6	5.3	3.4	0.73	0.75
LU	2.0	2.7	3.1	4.9	0.98	1.05
LV	4.2	2.4	3.6	3.0	0.74	0.70
NL	–	2.5	–	3.4	1.15	1.07
PL	1.6	1.8	2.8	2.3	0.96	0.97
PT	3.6	2.3	5.0	3.9	0.95	0.90
RO	1.8	2.1	4.1	5.1	0.89	0.97
SE	3.2	2.7	3.5	3.6	0.88	0.89
SI	2.2	2.6	2.5	4.3	0.85	0.95
SK	1.7	2.1	3.1	5.6	0.81	0.95
Median	2.2	2.5	3.6	3.6	0.92	0.94

Consumption other than education, health and imputed rent, per person, by five-year age group, allocated within the household with the same equivalence scale. Deviation: mean absolute difference between the shape of the reported HBS profile and the shape of the profile predicted for the HBS households (model) or imputed into EU-SILC, each shape divided by its mean, weighted with the persons in the HBS. Netherlands: no HBS 2010. Austria: Konsumerhebung 2010 and 2015. Open age group 80+, 75+ for Portugal. Source: *imputation_check.do*.

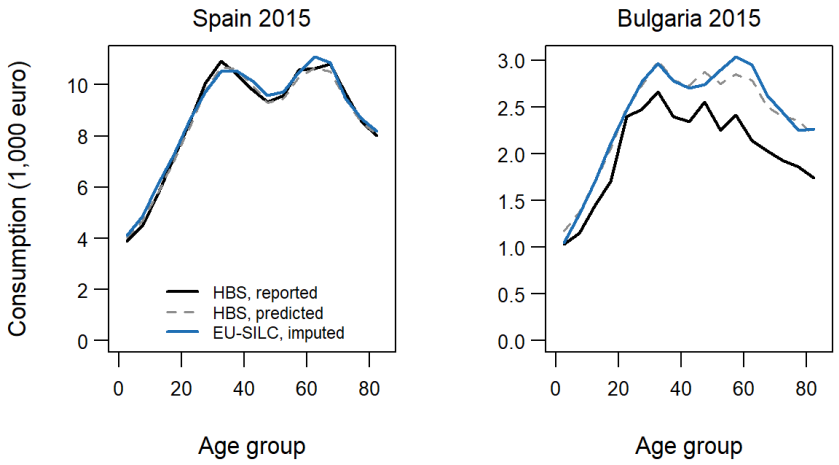


Figure 5.1: Consumption per person by age group in 2015: reported in the HBS, predicted by the imputation model for the same households, and imputed into EU-SILC, 1,000 euro

5.4 Comparison with AGENTA and national NTA results

The checks run on the published files (Section D.4) test the accounting of ENTA, not its age shapes, which meet their totals by construction. The comparisons in this section are the external test of the shapes: with the AGENTA data for 2010, which share most of their sources with ENTA, and with the results of national NTA teams, which are independent estimates.

The benchmark data are public. The AGENTA data were downloaded from the AGENTA data explorer of the Wittgenstein Centre (`https://www.wittgensteincentre.org/ntadata`): the whole dataset, all countries and all variables, the non-central variables included. The national NTA results were downloaded from the global NTA database (`https://www.ntaccounts.org/web/nta/show/Browse%20database`), all EU countries. The Austrian NTA for 2015 are part of that database and are also published as a dataset (Hammer, 2020). The downloaded files lie in *ENTA_output_control/benchmarks* (Appendix D).

5.4.1 Differences between ENTA and the AGENTA data for 2010

ENTA and the AGENTA data share the NTA concepts and most of the sources, EU-SILC 2011 (income year 2010) in particular, so the AGENTA results serve as a check of the ENTA profiles for 2010. The comparison covers the 24 countries in both datasets (the ENTA countries without Croatia and the Netherlands), both sexes together, at ages 0 to 80, where 80 is the open age group. Since ENTA are scaled to the current Eurostat release of the national accounts, the levels differ by construction; each profile is therefore divided by the mean labour income at ages 30–49 of the same source. Two measures summarise a pair of profiles: the correlation over age, and the mean absolute difference between the two normalised profiles, which is in units of prime-age labour income. Figure 5.2 shows six profiles as means over the 24 countries. The results by country and indicator are written by *4a_external_agenta.do* in the folder *ENTA_output_control*.

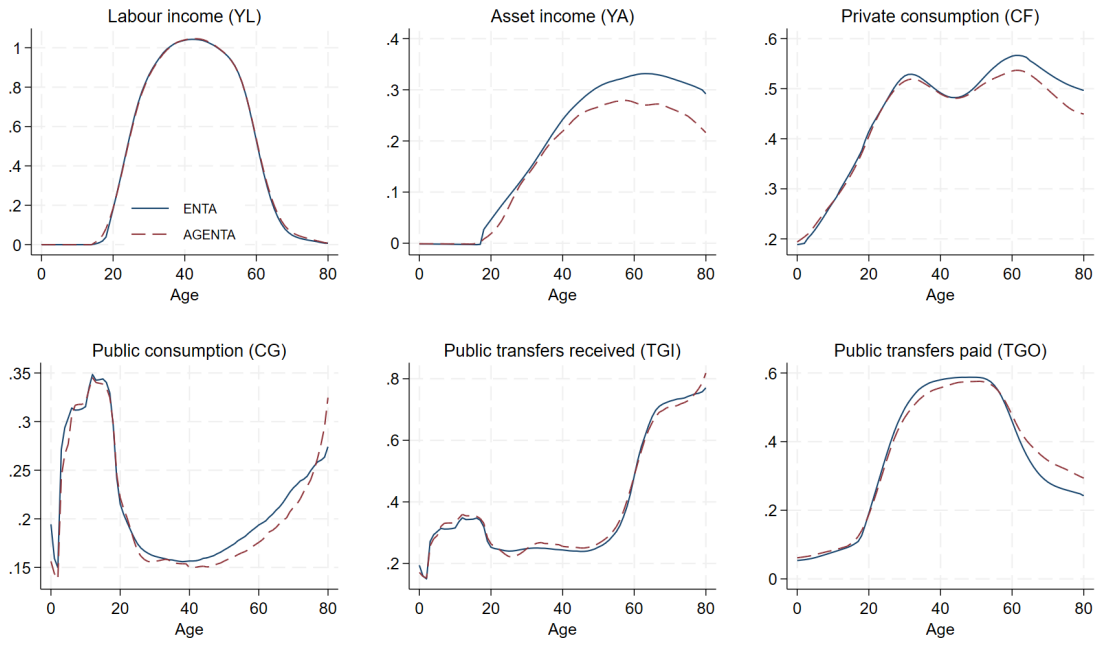


Figure 5.2: ENTA and AGENTA, 2010: age profiles per person relative to the mean labour income at ages 30–49 of the same source, mean of 24 countries

Where the two agree. Labour income is almost identical: the correlation over age is 1.00 in the median country and at least 0.995 in every country, and the mean difference is 0.01 of prime-age labour income. Public transfers received and paid, private transfers, private consumption and public education and health consumption have correlations of 0.98 to 0.99 in the median country and mean differences of 0.01 to 0.04. The life-cycle deficit, consumption less labour income, has a correlation of 1.00 and a mean difference of 0.04. In the median country the life-cycle surplus lasts from age 26 to 57 in ENTA and from 26 to 58 in AGENTA; it starts within one year of AGENTA's in 22 of the 24 countries.

Consumption at old ages. Private consumption per person at ages 65 and over is higher in ENTA in 22 of the 24 countries: 0.53 against 0.47 of prime-age labour income in the median country. AGENTA takes the age profile of private consumption from the HBS (Istenič et al., 2019). ENTA impute consumption into EU-SILC, which gives slightly more consumption at old ages than the HBS itself in the median country (Section 5.3), which accounts for a small part of the difference. Public consumption agrees less closely than public education and health consumption taken alone (correlation 0.95 in the median country, below 0.9 in six countries): it is somewhat higher in ENTA at ages 50 to 75 and lower in the open age group, 0.27 against 0.32 of prime-age labour income in the mean of the countries.

Public transfers paid at old ages. Public transfers paid per person at ages 65 and over are lower in ENTA in 23 of the 24 countries: 0.28 against 0.32 of prime-age labour income in the median

country, with the largest gaps in Portugal, Belgium, Cyprus and France. Which taxes are behind the difference was not examined.

Private consumption of education. AGENTA allocates the household's education spending in the HBS to the members enrolled in education, by level of education, and uses an average profile of the other countries where the HBS does not report enrolment (Germany, Spain, Finland, France, Italy, Romania, Sweden) (Istenič et al., 2019); the spending is therefore consumed mainly by pupils and students. ENTA do not use the HBS for education: household spending by level of education in the national accounts is given to the persons enrolled at that level (Section 3.8.2). The two differ in the part that the national accounts do not assign to a level, courses and adult education, which ENTA spread over the ages 20–64. The age shapes agree less closely than for the other consumption components: the correlation over age is 0.81 in the median country, between 0.39 (Slovenia) and 0.93 (Greece). The amounts are small: per person below age 20, 0.8% of prime-age labour income in ENTA and 0.9% in AGENTA in the median country.

Asset income. The total is similar (the level of ENTA's is 5% higher in the median country, correlation over age 0.98), but it is split differently: AGENTA credits corporations with their whole capital income, ENTA only with the income they keep, the dividends and interest they pay out appearing as property income of households. ENTA have more asset income at ages 65 and over in 20 of the 24 countries: in the median country 0.31 of prime-age labour income per person, against 0.22 in AGENTA. Two allocation rules account for about half of the gap. ENTA subtract mortgage interest (HY100G) at the ages of the adults who hold the dwelling, where AGENTA subtracts household interest with the shape of property income received, and ENTA assign imputed rent, estimated from the number of rooms, in equal parts to the adults who hold the dwelling, where AGENTA gives the reported imputed rent (HY030G) to the household member with the highest income. Switched to AGENTA's rules, the two lower the old-age level by 0.02 and 0.015 of prime-age labour income in the median country. The rest is not explained by the rules and is probably due to differences in data processing and smoothing. For Denmark, where EU-SILC reports property income net of interest paid, AGENTA uses the average asset income profile of the other countries, and ENTA rental income plus imputed rent (before 2019), so the two are not comparable.

Saving. Saving is a residual in both datasets, and its age shapes agree less well (correlation 0.82 in the median country). As explained in Chapter 6, age-specific saving should not be used for analysis.

5.4.2 Comparison with national NTA results

National NTA teams have published results for several EU countries in the global NTA database. Unlike AGENTA, they are independent estimates, made with other sources, methods and releases of the national accounts. Six of them are for years that ENTA cover: Austria 2010 and 2015, Germany 2008, France 2011, Italy 2008 and Slovenia 2010; Finland 2006, Hungary 2005 and Sweden 2006 can be compared with ENTA 2008. The age shapes are compared as for AGENTA, at ages 0–79.

In the six same-year comparisons, labour income, public and private transfers and the life-cycle deficit agree closely (median correlations over age of 0.99 to 1.00), and the life-cycle surplus starts and ends

at the same ages to within two years. Two systematic differences remain. First, consumption per person at ages 65 and over is higher in ENTA than in the national results in most countries (0.74 against 0.67 of prime-age labour income in Austria 2010, for example), while Germany agrees. The national files do not separate private and public consumption, so the source of the difference cannot be traced further here. Second, ENTA have more asset income at old ages in Italy and Austria 2010, which is the difference found against AGENTA (Section 5.4.1); in Austria 2015 it goes the other way. Asset-based reallocations, the difference between asset income and saving, agree least (median correlation 0.87), since they combine this difference with that of saving.

Where the national results are for a different year (Finland, Hungary, Sweden), the agreement is weaker, as expected: the differences include the change between the years as well as the method. In Sweden, ENTA's public transfers at old ages are lower than in the national results; one reason may be the treatment of the funded premium pension, which ENTA count among private pensions. In Hungary, the comparison spans the fiscal consolidation of 2006–2008.

Austria 2015: a benchmark from administrative data. The Austrian NTA for 2015 (Hammer, 2020) allocate employment and self-employment income, taxes on labour and cash benefits to age and sex with the integrated wage and income tax statistics, and public health consumption with hospital financing and social insurance data. For the survey-based parts of ENTA (EU-SILC, EUROMOD) these are independent benchmarks. Private consumption, consumption taxes and intra-household transfers are not: they rest on the same survey and the same method as in ENTA.

Income of employees, taxes on employment income, pensions (with disability and care benefits) and public health and education consumption agree closely: correlations over age of 0.99 to 1.00 and levels within 8%. The split between men and women agrees as well (male share of income of employees 0.65 against 0.64, of pensions 0.56 against 0.52). Three differences remain. Self-employment income starts at younger ages in ENTA (correlation 0.96). Unemployment and family benefits shift against each other at family-formation ages (correlations 0.96 and 0.98), but agree closely taken together (0.99). Private asset income agrees at old ages but not in detail (correlation 0.90): the Austrian NTA allocate it with net wealth in the Household Finance and Consumption Survey (HFCS), which gives negative values at ages 25–35 and a peak at 55–60 that the EU-SILC profile does not have.

6. Limitations

ENTA combine survey data with the national accounts. The level of every age profile comes from the national accounts; its distribution over ages and sexes comes from a survey or, where no survey reports the flow, from an assumption. This chapter states what the data can be used for, and lists the countries in which a profile is built with a different method than in the others. Changes that happened in the economy, and breaks that come from the sources, are in Appendix E.

6.1 Simple rules for analysing ENTA data

1. **The level is the national accounts, the age shape is the survey.** Every profile is scaled to its aggregate, so totals always agree with the national accounts. This says nothing about the age shape. The rules below concern the age shape.

2. **Employment income, taxes on labour income, pensions and family benefits are the most reliable.** Their age shapes can be compared across countries and over time. Use employment income (`YEM`) or labour income (`YL`), not its split into compensation of employees (`YE`) and mixed income (`YM`): few persons report self-employment income in Estonia, Luxembourg and Latvia in all years and in Belgium, Slovakia and Sweden in many (Section 5.1.2).

3. **Compare unemployment benefits and other benefits with care.** Few persons receive unemployment benefits (`BSU`) at each age, and what they include differs by country. Other benefits (`BSX`) are split equally among the adults of a household. Compare total benefits (`BS`) where possible.

4. **Do not compare asset income across countries or over time.** Wealthy households are under-represented in EU-SILC and the distribution is highly skewed. The profiles of property income (`YP`) and asset income (`YA`) show the basic age pattern only; their split by sex changes by up to 12 percentage points from one year to the next.

5. **Use private consumption and transfers within households to compare countries, not years.** Health and other consumption are estimated from the Household Budget Surveys of 2010 and 2015 only, and private education consumption follows enrolment, with the same amount for every person enrolled at a level. Transfers within households (`TFW`) are built from consumption and income.

6. **Do not use public consumption to analyse changes of the age shape over time.** Health consumption (`CGH`) has one age shape for all years, education (`CGE`) follows enrolment, and collective consumption (`CGX`) is equal per capita.

7. **Flows with an assumed age shape carry no information on age.** Funded pensions, other current transfers, transfers with the rest of the world and all flows of the corporate and

government sectors borrow their shape from another profile (Table 6.1). Transfers between households (`TFB`) have one shape for all years.

8. **Do not use saving by age.** Saving by age (`SH`, `SF`) is disposable income minus consumption and inherits the errors of both. Aggregate saving comes from the national accounts and is not affected.

9. **Check the age grid before comparing shapes.** Where the source reports age groups, the profile is a step function: Germany in some years, and the employment indicators of the Netherlands, Czechia, Ireland and Slovakia (Table 6.2). Totals are comparable, the curvature is not.

10. **A zero is not always a zero.** No stored value is missing (Chapter 0), but a zero can mean that the source did not report a value. Check a zero, or a sudden fall to zero, against the source before interpreting it.

6.2 What applies to every profile

Scaling. Suppose the national accounts report unemployment benefits of 10 billion euro and the survey records 8 billion. Scaling multiplies the survey value at every age and for both sexes by 1.25. If the survey misses part of the receivers at some ages, the published profile carries that error, although its total is right.

Persons outside private households. EU-SILC and the Household Budget Survey cover private households only. Persons in nursing homes and other institutions are not observed (0.5% to about 3% of the population of the EU countries in 2011, Törmälehto, 2019, Section 2), but the aggregates and the population figures include them. Scaling gives them the values of persons of the same age and sex in private households. The profiles at 80 and older are the ones most affected. Public health consumption is less affected, because its profile does not come from the surveys.

The oldest and the youngest ages. EU-SILC records age only up to 80, so 80 stands for everyone aged 80 and older; single ages above 80 cannot be estimated. Personal income is collected from age 16; the income and tax profiles are zero below age 15, and at 15–17 they are capped at the value of the next older age, because few persons of that age have income.

Smoothing. The profiles from EU-SILC are smoothed over age from age 18 (Chapter 3); the open group 80 and older keeps its unsmoothed mean. The employment indicators from the Labour Force Survey are not smoothed.

Non-profit institutions. The household sector in ENTA includes the non-profit institutions serving households (S.14 and S.15). Their consumption and their transfers have no age pattern of their own. Their consumption is 2% of the consumption of the combined sector in the median country and 6% in Luxembourg. They are net receivers of transfers from government, which is why net other current transfers paid (`TX`) are negative in most countries.

6.3 Flows with an assumed age shape

Where no microdata identify who pays or receives a flow, its age shape is taken from a related profile and scaled to the aggregate of the flow (Table 6.1).

Table 6.1: Flows with an assumed age shape

Flow	Age shape
Other current transfers (`TXH`, `TXNPISH`), other private transfers (`BPX`), transfers with the rest of the world (`BROW`)	Employment income plus pensions, the same for both sexes. These flows are part of household disposable income and reach children through the transfers within households.
Consumption taxes (`TCON`)	Private consumption other than imputed rent
Funded pensions	Contributions: taxes on employment income. Benefits: pensions. The microdata do not separate funded from pay-as-you-go pensions.
Social contributions and benefits via non-financial corporations	Taxes on employment income; employment income
Corporate sector: income, taxes, transfers, saving	Property income received, without private pensions
Government asset income, net transfers from abroad, government saving	Taxes paid
Collective government consumption (`CGX`)	Equal per capita
Public health consumption (`CGH`)	2018 Ageing Report, one shape for all years
Public education consumption (`CGE`)	Enrolment by level, age and sex, times spending per student by level
Private education consumption (`CHE`)	Enrolment by level, age and sex, times household spending per student by level; spending not definable by level: equal per person aged 20–64
Household operating surplus (imputed rent)	Adults of owner households, by number of rooms
Other social benefits (`BSX`)	Equal among the adults of the household
Transfers between households (`TFB`)	EU-SILC, pooled over all years

6.4 Countries with a different method

Table 6.2 lists the cases in which a profile of one country, or of a few countries, is built differently from the others, because a source is missing, is reported in age groups or is defined differently. Countries and years not listed follow the method of Chapter 3.

Table 6.2: Countries with a different method

Country	Years	What is different
All profiles from EU-SILC		
Germany	2014–2018, from 2022	EU-SILC reports age in groups: 73 and older in 2014–2018, five-year groups above 20 from 2022. Each age holds the mean of its group. No smoothing from 2022.
Employment		
Netherlands, Czechia, Ireland, Slovakia	all	The Labour Force Survey reports five-year age groups: the employment indicators are step functions.
Austria, Croatia, Italy	all	The Labour Force Survey ends with the group 75 and older: the employment indicators are constant from age 75.
Primary income		
Denmark	to 2018	Property income received is built from rental income and imputed rent, because interest and dividends are reported net of interest paid.
Romania, Bulgaria, Croatia, Lithuania, Latvia	all	Property income paid: the age shape is mixed with that of other countries of the region (fewer than 20 persons per age, Section 5.1.2).
Romania, Hungary	all	The same for property income received.
Hungary, France, Germany	2018, 2020, 2008	Mortgage interest is not reported: property income paid is the average of the neighbouring years; Germany 2008 takes the profile of 2009.
Ireland	2018	Private pensions in property income received are the average of 2017 and 2019 (four to five times as many receivers as in the other years).
Taxes		
Croatia	2009–2010	Tax rates by age are those of 2011, the first EUROMOD year.

Country	Years	What is different
Hungary	most years	Taxes on pensions have the age shape of pensions (one average tax rate), because the EUROMOD rates are below 1%.
Austria, Portugal, Germany; Greece, Croatia, Romania in many years	all	The same for taxes on other benefits.
Netherlands	all	The tax rate on pay-as-you-go pensions is zero below age 60.
Benefits		
All countries	all	Government sick pay is part of pensions (the COFOG class sickness and disability) and has the age shape of pensions. The survey's sickness benefits are not used: they are mainly sick pay from employers.
Romania, Slovakia, Greece	all	Unemployment benefits have the age shape of the share of receivers in the Labour Force Survey: the same benefit per receiver at every age.
Denmark	2010–2018	The green check, a small transfer to almost every adult, is moved from unemployment benefits to other benefits.
Denmark	to 2018	Unemployment benefits at ages 60–65 are mostly early-retirement pay. They are added to the shape of employment income plus pensions in Table 6.1.
Slovakia	2008–2009	Benefits and public consumption are split by function with the shares of 2010.
Ireland	from 2022	Primary and secondary education spending is split with the shares of 2021.
Private consumption		
Austria	all	National survey (Konsumerhebung 2010 and 2015) instead of the Eurostat HBS.
Netherlands	all	HBS 2015 only.
Italy; Luxembourg 2010	all	The HBS has no household income; consumption expenditure is used in its place.
Portugal	all	The HBS ends with the group 75 and older.

Country	Years	What is different
Finland, Sweden	all	The national accounts report no split of household education spending by level: private education consumption (CHE) is the same amount for every person aged 3–64.
Aggregates only		
Cyprus	2012–2013	Missing other current transfers (D.72, D.75) are derived from the reported total (D.7). Other subsidies on production (D.39) are zero in all years.
Denmark	to 2004	Social contributions are split by type with the shares of 2005.

6.5 Limitations by quantity

6.5.1 Primary income

Asset income is the least reliable part of primary income (rule 4). Three points add to it. Property income received includes income from renting out dwellings, which the national accounts record as mixed income or operating surplus; in Greece, Italy, Portugal and Spain rental income peaks 8 to 16 years later than interest and dividends, so the profile is somewhat older than the income it distributes. Net property income can be negative at ages 18 and 19, because the amounts received and paid are scaled separately. And corporate income, taxes and saving take their age shape from property income received and inherit its uncertainty.

6.5.2 Taxes and social contributions

Consumption taxes follow total private consumption, without differences in the consumption basket by age. Tax rates by type of income and age group come from EUROMOD, which does not cover every country-specific rule; rates are capped at 70% and rates below 1% are set to zero; where this leaves no rate in the age groups that hold the income, the tax takes the age shape of that income. The split of taxes on benefits into taxes on pensions and on other benefits starts with the first EUROMOD year of a country (2005 to 2007; Croatia 2011) and is missing before.

6.5.3 Government transfer benefits in cash

The age shape of each benefit comes from EU-SILC and the total of benefits from the national accounts. The split of the total into pensions, family benefits, unemployment benefits and other benefits comes from government expenditure by function (COFOG). It is the part that most often breaks between years, and it is missing in the early years of the aggregates in 13 countries. Government sick pay is part of the pension aggregate and has the age shape of pensions, although it is paid at

working ages (Table 6.2).

6.5.4 Funded pensions

The microdata do not separate funded from pay-as-you-go pensions, so both have the same age shape. This matters most where funded pensions are large: Denmark, the Netherlands and Sweden.

6.5.5 Private transfers

Transfers between households come from EU-SILC. Few households report them, the totals received and paid differ, and they are probably under-reported; the volume is a lower bound and the age shape is pooled over all years. Transfers within households are not observed. They are the difference between consumption and income of each person and depend on the assumed distribution of consumption within the household, which is the same for all countries and years.

6.5.6 Government consumption

The age shape of public health consumption is that of the 2018 Ageing Report in every year. For Spain, Slovakia and Ireland the 2021 edition gives a 28–42% higher ratio of spending at 65 and older to spending at 20–64, so the steepness at old ages is uncertain there. Public education consumption is distributed with enrolment by age. Eurostat reports enrolment partly in age groups and with gaps; these are distributed with a reference age profile or filled from neighbouring years. Spending at pre-school ages is not comparable over time in the Netherlands, Ireland and Poland, where pre-primary enrolment is reported only in part in 2008–2010.

6.5.7 Private consumption

Consumption is estimated from the HBS waves of 2010 and 2015, pooled. HBS expenditure is recorded over short periods and is a noisy measure of annual consumption. It is recorded for the household and is allocated to its members with the NTA equivalence scale (0.4 for children up to age 4, rising to 1 at age 20). The profiles of other years change only with household structure and income in EU-SILC. Private education consumption (`CHE`) is not from the HBS: spending by level of education in the national accounts is given to the persons enrolled at that level, the same amount each, and spending that is not definable by level to all persons aged 20–64 (Section 3.8.2). The age shape within a level and of the last part is therefore assumed, and the countries classify the spending differently. Pre-primary and primary education are one level, with the same amount per child, although households spend more on pre-primary education in most countries: consumption at ages 3–5 is understated and consumption at the ages of primary school overstated. As for public education, private education consumption at pre-school ages is not comparable over time in the Netherlands, Ireland and Poland, where pre-primary enrolment is reported only in part in 2008–2010.

6.5.8 Saving

Saving by age is a residual (rule 8). Aggregate ENTA saving is ESA net saving (B.8n) less the adjustment for the change in pension entitlements (D.8). Household saving is therefore lower, and corporate saving higher, than in ESA where funded pensions are large.

A. Units of measurement

Every ENTA dataset stores a single value in Euro at current prices: per capita in the age-specific files (*VALUE*), and as a national total in million Euro in the aggregates (*VALAGGR*). All other units are calculated from this value with the conversion factors in *ENTA_units*, one row per country and year (Section 1). Table A.1 gives the formula of each unit. In it, the *value per capita* is *VALUE* in the age-specific files and $VALAGGR \times 1{,}000{,}000/POPtot$ in the aggregates.

Per capita values divide by the population of the same age and sex, measured as the average of the population on 1 January of the current and of the following year (Eurostat table *demo_pjan*); in the last year of the data, only 1 January of that year is available and is used. The population by age and sex is published as the indicator POP, and its sum over all ages as POPtot. For countries outside the Euro area, values in national currency are converted into Euro with the exchange rate of the year.

Indicators that are counts or rates are not converted: population (POP) and the employment indicators (EMPLSTAT, EMPLSTATsilc, WHRUSUT, WHRACTT, EMPLFT).

Table A.1: Units of measurement: calculation from the stored value

Unit	Calculation	Available for
Euro per capita, nominal	Value per capita	both
Billion Euro, nominal	VALAGGR / 1,000	aggregates
Euro per capita, real (HICP, 2020=100)	Value per capita × EXR_NAC_EUR / (HICP/100) / EXR_2020	both
PPS-Euro per capita	Value per capita × PPP_EUR	both
PPS-Euro per capita, real (2020=100)	Value per capita × EXR_NAC_EUR / (HICP/100) / EXR_2020 × PPP_EUR_2020	both
Percent of GDP	VALAGGR / GDP × 100	aggregates
Percent of primary income	Value per capita / (Y × 1,000,000 / POPtot) × 100	both
Percent of household primary income	Value per capita / (YH × 1,000,000 / POPtot) × 100	both
NTA norm	Value per capita / NORMnta, the average labour income at ages 30–49	age profiles
ENTA norm	Value per capita / NORMenta, the average employment income of men at ages 30–49	age profiles

Real values. The value in Euro is converted into national currency with the exchange rate of the year (EXR_NAC_EUR), deflated with the Harmonised Index of Consumer Prices of the country (HICP, 2020 = 100), and converted back into Euro with the exchange rate of 2020 (EXR_2020). The deflation thus uses the consumer prices of the country, while the unit remains the Euro; in the Euro area both exchange rates are one. HICP is below 100 before 2020, so real values are higher than nominal ones before 2020 and lower after it: in Austria, 1,000 Euro of 2010 (HICP 83.1) are 1,203 Euro at 2020 prices. The base year does not affect growth rates or ratios between years.

Purchasing power standards (PPS). PPS values account for differences in price levels between countries: PPP_EUR is the purchasing power of one Euro in the country relative to the EU-27, at the prices of the year. The real PPS value combines the deflation described above with the purchasing power of 2020 (PPP_EUR_2020) and is comparable across both countries and years.

Shares of income or of GDP. Shares of primary income (net national income at basic prices) are generally preferable to shares of GDP for taxes, benefits, consumption and saving, because these flows are financed out of the income of the resident population rather than out of production. The share of GDP is misleading for Ireland and Luxembourg, where primary income was 45 to 51% of GDP on average in 2008–2023, against 63 to 75% in the other countries: in Ireland a large part of the value added of multinational firms accrues to foreign owners, and in Luxembourg a large part of wages goes to cross-border commuters. Shares of primary income are available for the aggregates and for the age profiles; the share of household primary income is the unit for analyses of the household sector.

The norms. The standard unit for comparing age profiles across countries is the average labour income of persons aged 30–49 (NTA norm). Averages over the whole population depend on the age structure and on the retirement age; the income of the prime working-age population does not. The ENTA norm uses the employment income of *men* aged 30–49 and therefore keeps the differences in employment and income between men and women, which a norm of both sexes would partly hide.

1 The ENTA_units dataset

ENTA_units holds the conversion factors, one row per country (CNT2) and year (YEAR). It combines the income age profiles (for the two norms), the ESA sector accounts (primary income and GDP), Eurostat exchange rates and purchasing power parities, and the Harmonised Index of Consumer Prices. Table A.2 lists every variable.

Table A.2: Variables in ENTA_units

Variable	Description
CNT2	Country code (ISO 3166-1 alpha-2)
YEAR	Reference year
NORMnta	NTA norm: average labour income (YL), population-weighted, ages 30–49; missing in years without an income age profile
NORMenta	ENTA norm: average male employment income (YEM), ages 30–49; missing in years without an income age profile
Y	Primary income, total economy — ESA aggregate, million EUR (not per capita)
YH	Primary income, household sector — ESA aggregate, million EUR (not per capita)
GDP	GDP, total economy — ESA aggregate, million EUR (not per capita)
CHXesa	Other private consumption (CHX), household sector — ESA aggregate, million EUR; used by the pipeline to standardise consumption across years, not in any unit calculation
POPtot	Total population, all ages
EXR_NAC_EUR	Exchange rate: national currency per Euro, year-specific
EXR_2020	Exchange rate: national currency per Euro, fixed at the country's 2020 value
HICP	Harmonised Index of Consumer Prices, 2020=100
PPP_EUR	Purchasing power of one Euro (PPS), year-specific
PPP_EUR_2020	Purchasing power of one Euro (PPS), fixed at the country's 2020 value
PPP_EU27_2020	Eurostat PPP conversion factor, national currency per PPS unit, relative to the EU-27 (Eurostat aggregate EU27_2020); intermediate input used to derive PPP_EUR, not itself used in any unit calculation

B. ENTA Codebook (ESA concept)

0 Production Account — Total Economy

The production account is part of the aggregate data only; it is not age-specific. It shows how primary income is derived from GDP. The age-specific analysis of national accounts in ENTA starts with primary income.

0.0 GDP — Gross domestic product

Gross domestic product at market prices, as defined in ESA 2010.

0.1 TPS — Taxes less subsidies on products

Taxes less subsidies on products (published in the aggregates file as TPSp) include, in particular, value added tax (VAT). In national accounts, taxes less subsidies on products appear twice: once in the final use of the private sector and once in the final use of the public sector. ENTA correct for this double counting to ensure a consistent measurement of redistribution: all quantities in ENTA are measured at basic prices, i.e. excluding taxes less subsidies on products (see Section 1.2.4).

0.2 CFC — Consumption of fixed capital

CFC captures the investment necessary to maintain productive capital due to wear and tear, obsolescence, and normal accidental damage.[1] ENTA measure income and saving net of CFC.

0.3 YEROW — Employment income received from rest of the world

YEROW refers to the compensation of employees received from abroad, less the compensation paid to non-residents. These flows of income are part of primary income. They are particularly important in Luxembourg, where a large part of employment income is paid to residents of Belgium, Germany, or France.

0.4 YPROW — Property income received from rest of the world

Property income received from other countries less property income paid to other countries. This is a particularly important component for Ireland, which records part of the production of multinational companies; however, the income derived from these activities is paid to owners abroad.

[1] https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Consumption_of_fixed_capital_(CFC)

0.5 Y — Primary income

Primary income is derived from GDP as follows:

$$Y = GDP - TPS - CFC + YEROW + YPROW.$$

1 Y — Primary Income

Primary income in ENTA captures the income that residents receive for contributing to production, either through their work or the use of their assets. It includes wages before paying taxes and social contributions, income from self-employment, as well as asset income such as dividends, interest, and rents.

ENTA primary income of each sector corresponds largely to the *balance of primary income* in ESA, measured at basic prices. The primary income of all sectors combined represents *national income* in ESA. The age profile is calculated as the sum of the subcomponents:

$$Y = YEM + YA.$$

1.0 POP — Population

Age- and sex-specific population count, the denominator underlying every per-capita age-profile value in ENTA. Sourced from Eurostat population data (the World Population Prospects are used to distribute the oldest ages where Eurostat reports them as one group; see *Population (Suppl_ population.do)* in Appendix D). Unlike the other indicators, population is a count, not a monetary value, so unit conversions such as real, PPS, or percent of income/GDP do not apply to it.

1.1 YEM — Income from employment

YEM includes the compensation of employees (YE) and income from self-employment (YM):

$$YEM = YE + YM.$$

We recommend analysing the two components together because the age-specific estimates are more reliable. In particular, the smoothing of the age profiles works much better for the combined profile.

1.1.1 YE — Compensation of employees

Compensation of employees in ENTA includes gross wages, employers' social contributions, all other taxes that employers pay on payroll or employment, and in-kind compensation, all before paying any taxes.

In ESA, YE corresponds to the *compensation of employees* plus taxes on production paid on payroll or employment. The age- and sex-specific estimates are based on the EU-SILC variables *employee cash*

or near cash income (PY010G), *employers' social contributions* (PY030G), *company car* (PY021G), and *income received by people aged under 16* (HY110G).

1.1.2 YM — Mixed income / self-employment income

YM captures the income of self-employed persons. Implicitly, it includes both the return to labour and the return to capital of self-employed persons. In ESA, it is therefore captured as *mixed income*. Age profiles are based on the EU-SILC variable *cash benefits or losses from self-employment* (PY050G).

1.2 YA — Asset income

Asset income of households consists of all types of income derived from assets, such as interest, dividends, and rent; it also includes imputed rents, which capture the value of housing services enjoyed by owner-occupiers.

In ESA, the components of household asset income are captured as *property income* (YP) and *operating surplus* (YO):

$$YA = YO + YP.$$

1.2.1 YO — Operating surplus

The operating surplus of households consists mainly of the value of housing for owner-occupiers (imputed rent). Many households have it, so its age profile is more reliable than that of property income. The age- and sex-specific estimates are based on an estimate of imputed rent from the number of rooms of the dwelling in EU-SILC (HH030), assigned in equal parts to the adults who hold the dwelling: all adults except those below 40 who live with their parents (Chapter 3). EU-SILC's own imputed rent (HY030G) is not used, because it is missing for several countries and years.

1.2.2 YP — Property income

Property income received less property income paid, $YP = YPr - YPp$. The two parts are internal quantities (not published). The age- and sex-specific estimates of property income received (YPr) are based on the EU-SILC variables *interest, dividends, profit from capital investments in unincorporated businesses* (HY090G), *income from rental of property or land* (HY040G) and *pensions from individual private plans* (PY080G); property income paid (YPp) is based on *interest repayments on mortgage* (HY100G).

1.3 Employment status (EMPLSTAT)

Employment status based on the Labour Force Survey. A person is counted as employed if they worked for pay in the reference week (variable WKSTAT) or were absent due to holidays, special working time arrangements, sick leave, or job-related training (variable ABSREAS).

The four indicators based on the Labour Force Survey (EMPLSTAT, EMPLFT, WHRUSUT, WHRACTT) are published for the full age range 0–80 in every country and year. The Labour Force

Survey covers persons aged 15 and over, so these indicators are zero at every age below 15.

1.4 Employment status, EU-SILC based (EMPLSTATsilc)

The second indicator on employment status is based on EU-SILC. A person is counted as employed if their self-reported economic status is employed or self-employed (variable PL030/PL031/PL032 in EU-SILC).

1.5 Usual working hours (WHRUSUT)

Usual working hours in the main and second job. Based on the variables HWUSUAL and HWUSU2J in LFS.

1.6 Usual working hours, EU-SILC (WHRUSUsilc) [not published]

Usual working hours in the main job and beyond. Based on the variables PL060 and PL100 in EU-SILC.

1.7 Actual working hours (WHRACTT)

Actual working hours for the main and second jobs during the reference week. Based on the variables HWACTUAL and HWACTU2J in the LFS.

1.8 Employment, full-time equivalents (EMPLFT)

Employment in full-time equivalents is based on the usual working hours (HWUSUAL and HWUSU2J in the LFS). If the number of working hours is 35 or more, employment is counted as one full-time equivalent. Fewer working hours are counted proportionally; e.g., an employment of 17.5 hours is counted as 0.5 full-time equivalents.

1.9 Employment, full-time equivalents, EU-SILC (EMPLFTsilc) [not published]

Based on the number of usual working hours in EU-SILC.

2 T — Taxes and Contributions

"Taxes" in ENTA include all transfers paid by residents to the government, less the subsidies received. In particular, they also include social contributions. ENTA distinguish taxes by economic function, i.e. whether they are paid on employment income (TYEM), assets or asset income (TYA), social benefits (TB), or consumption (TCON). They also include a component of *other current transfers* (TX), which includes, for example, fees, other charges, fines, and penalties.

$$T = TYEM + TYA + TB + TCON + TX.$$

The aggregate value for total taxes is based on ESA data; the aggregates for the subcategories are mainly based on data on tax revenues by economic function.[2] The age- and sex-specific values for total taxes are calculated as the sum of the subcomponents.

2.1 TYEM — Taxes and social contributions on employment income

TYEM includes wage taxes, income taxes paid by the self-employed, and social contributions paid by employees, employers, and self-employed persons.

The ENTA aggregates by type of tax are based on the dataset on taxes by economic function. TYEM corresponds largely to *taxes on income from employment* paid by employees and employers, plus *taxes on the income of the self-employed*.

The age- and sex-specific estimates of TYEM combine data from the tax-benefit model EUROMOD and EU-SILC. EUROMOD is used to estimate age-specific tax rates by type of income, including income from employment. These rates are then applied to the income variables in EU-SILC.

2.2 TYA — Taxes on asset income and assets

Taxes on asset income and assets of households. The aggregate values are based on *taxes on capital — income of households* in the data on tax revenues by economic function. Additionally, the aggregate includes *other taxes* in the ESA category *taxes on income and wealth*.

The age-specific estimates use tax rates by type of income calculated from EUROMOD data, which are applied to impute age-specific estimates in EU-SILC. In general, they follow the age profile of property income, based on the corresponding EU-SILC variable (HY090G).

2.3 TB — Taxes and social contributions on income of non-employed

TB captures taxes and social contributions paid on transfer benefits, such as pay-as-you-go pensions, benefits from funded pensions, or other social transfers. The aggregate of TB comes from the data on tax revenues by economic function. Its split into taxes on pay-as-you-go pensions, on funded pensions and on non-pension benefits combines DG TAXUD's national tax lists, which assign each item of TB to the income it is levied on, with EUROMOD's taxes per euro of each kind of transfer (Chapter 3, taxes and social contributions). The split is published from the first EUROMOD year of a country (2005 to 2007; Croatia 2011):

$$TB = TBSPEN + TBSN + TBPPEN.$$

2.3.1 TBSPEN — Taxes on pay-as-you-go pensions

Taxes and social contributions paid on pension benefits received from government pay-as-you-go pension systems.

[2] https://webgate.ec.europa.eu/taxation_customs/redisstat/databrowser/view/tax_ec_func/default/table

2.3.2 TBSN — Taxes on non-pension social benefits

Taxes and social contributions paid on non-pension social benefits, such as unemployment benefits or family allowances.

2.3.3 TBPPEN — Taxes on funded pension benefits

Taxes and social contributions paid on benefits received from funded pension systems.

2.4 TCON — Consumption taxes

TCON captures primarily the taxes paid when purchasing a specific good or service, with value added tax as the most important component. The corresponding quantity in ESA is *taxes less subsidies on products*. It includes the part received by the rest of the world — mainly EU customs duties, less EU subsidies on products (large only before the 2005 reform of the Common Agricultural Policy). Government collects that part with the other consumption taxes and passes it on to the rest of the world (TROW), as it does with the taxes on production paid to the EU. The age profile is that of private consumption other than imputed rent.

2.5 TX — Other current transfers

TX refers, in particular, to transfers from government to NPISH, as well as penalties or fines. The corresponding quantity in ESA is *other current transfers*. It is reported by recipient sector, TXH for households and TXNPISH for non-profit institutions serving households:

$$TX = TXH + TXNPISH.$$

Both components have the age profile of employment income plus pensions (YEM + BSPPEN), the same for both sexes and zero below age 15; the reason is given in Section 3.2. The split is available from 1995 for 24 of the 26 countries, from 2012 for Germany and from 2010 for Ireland; in the earlier years the whole of TX is in TXH, so the two components are not comparable across these breaks (Table 6.2).

2.6 TROW — Net transfers from the rest of the world, government

Published for the government sector only (aggregates). Current transfers the government receives from the rest of the world less those it pays to it, including the taxes on production and the consumption taxes it passes on to the European Union. It equals TGNF of the NTA format. In the accounts of chapter 2 it is the only transfer contribution of the total economy, since taxes net out between the domestic sectors. No age profile is published.

3 BS — Social Benefits

BS consists of the social benefits (in cash) paid by the government to residents. In ESA, these transfers are captured as *social benefits other than social transfers in kind*. The most important transfers in the ENTA category of social benefits are government pensions (BSPEN), followed by support for families (BSFAM) and unemployment benefits (BSU). All other benefits (BSX, including health) are of minor importance.

$$BS = BSPEN + BSFAM + BSU + BSX.$$

The distinction of social benefits by function is based on data on general government expenditure by function (COFOG), which is consistent with ESA 2010.[3]

3.1 BSPEN — Pensions

The ENTA category of pensions includes all types of pension-like benefits, including regular old-age pensions, survivor pensions, and disability pensions.

The aggregate values are derived from expenditure on social benefits in the COFOG categories of old age, survivors, and sickness and disability. The last class does not separate sickness cash benefits from disability pensions, so government sick pay is part of BSPEN and has the age profile of pensions; the survey's sickness benefits are not used (see BSX). There is no clear-cut distinction between these functions in the microdata, and although EU-SILC contains a variable for each, it is not possible to distinguish them in the microdata in the same way as in the aggregates. Consequently, the age- and sex-specific data is based on the sum of the EU-SILC variables PY100G (old age), PY110G (survivors), and PY130G (disability).

Pensions including the benefits of funded schemes, $BSPPEN = BSPEN + BPSOCfc$, are an internal quantity (not published) that serves as the receipt profile of several transfers and as the base of the taxes on pensions.

3.2 BSFAM — Family and children

BSFAM includes cash benefits for families, children, and their education. The aggregate values are based on social benefits, not in-kind, in the COFOG categories family and children (GF10.04) and education (GF09). The combination enables better harmonisation of microdata and aggregate data. The age- and sex-specific data are based on the EU-SILC variables HY050G (family/children allowances) and PY140G (education-related allowances).

3.3 BSU — Unemployment

BSU mainly captures unemployment benefits. In the data on government expenditure by COFOG, these transfers are captured in the category of social protection and unemployment. The age- and sex-specific estimates are based on the EU-SILC variable PY090G.

[3] https://doi.org/10.2908/GOV_10A_EXP

3.4 BSX — Other social benefits

Captures social benefits not classified elsewhere. At the microdata level, BSX is based on the EU-SILC variables for social exclusion (HY060G) and housing allowances (HY070G). The EU-SILC sickness benefits (PY120G) are not used in any profile: a comparison with the aggregates showed that they consist mainly of sick pay from employers, which is not a government benefit.

At the aggregate level BSX is the residual of the COFOG total of social benefits in cash after the categories named separately, so that the categories add up to the total. It contains housing, social exclusion and the other social protection categories, the cash benefits of the function health, and any cash benefit that a country reports under another function (of some size only in Germany and Belgium).

4 BPPEN — Funded Pensions

BPPEN captures the net transfer from funded pension systems, defined as benefits received less contributions paid:

$$BPPEN = BPSOCfc - TPSOCfc.$$

ENTA approximate contributions with social contributions paid by households to financial corporations and benefits with social benefits paid by financial corporations to households. As in ESA disposable income, both flows enter household disposable income gross. The difference between contributions and benefits is close to the ESA *adjustment for the change in pension entitlements* (D.8), which ENTA do not use in its accounts; see D.8 below.

In the microdata, it is not possible to distinguish between pay-as-you-go pensions and funded pensions. Both social contributions and benefits are therefore based on the same age profiles as pay-as-you-go pensions: contributions follow the age profile of taxes on employment income (TYEM), and benefits follow the age profile of total pension benefits (BSPPEN).

4.1 TPSOCfc — Funded pensions, contributions

Social contributions paid by households to financial corporations, which operate the funded pension schemes. The age profile follows that of taxes on employment income (TYEM).

4.2 BPSOCfc — Funded pensions, benefits

Social benefits paid by financial corporations to households from funded pension schemes. The age profile follows that of total pension benefits including funded pensions (BSPPEN).

4.3 D.8 — Adjustment for the change in pension entitlements (not published)

In ESA, the adjustment for the change in pension entitlements (D.8) is recorded after disposable income: it moves the accumulation of pension entitlements from the saving of the pension schemes to the saving of households. It is positive for households, negative for financial corporations, and close to the net contributions to funded pensions, $TPSOCfc - BPSOCfc = -BPPEN$.

ENTA do not use D.8 in any concept, for any sector, and do not publish it. It is the difference between ENTA saving and ESA saving:

$$SH + D.8 = \text{ESA net saving (B.8n)},$$

and the same for corporations, with SC and a negative D.8. Household saving in ENTA is therefore lower than ESA saving in countries with large funded pension schemes, in particular Denmark, the Netherlands and Sweden, and corporate saving is higher by the same amount. Household disposable income at market prices equals ESA disposable income (B.6n); the published YD is at basic prices, i.e. after consumption taxes.

5 BP — Private Transfers

Private transfers include all transfers between private sectors as well as transfers between private sectors and the rest of the world (ROW). Among them are social benefits and contributions that are financed and/or organised by corporations themselves, such as continued remuneration during sick leave. Total private transfers are defined as:

$$BP = BPB + BPW - TPW.$$

5.1 BPB — Private transfers between households

BPB captures inter-household private transfers, including transfers between the household sector and non-financial corporations (BPSOCnfc, TPSOCnfc), other private transfers between households (BPX), transfers received from the rest of the world less transfers paid (BROW), and transfers between resident households received (TFBHI) less paid (TFBHO):

$$BPB = BPSOCnfc - TPSOCnfc + BPX + BROW + TFBHI - TFBHO.$$

TFBHI and TFBHO come from EU-SILC (`HY080G`, `HY130G`); they are equal in total, so they do not change the aggregate of BPB, only its age profile (Section 6.5.5 and the chapter on private transfers).

5.1.1 TPSOCnfc — Social contributions, non-financial corporations

Social contributions paid to employers, including imputed social contributions for social benefit schemes that are not financed by explicit contributions. An important example is the continued remuneration of employees in the event of sickness. The age profile follows that of taxes on employment income (TYEM).

5.1.2 BPSOCnfc — Social benefits, non-financial corporations

Social benefits paid by employers, such as the continued remuneration of employees in the case of sickness. The age profile follows that of employment income (YEM).

5.1.3 BPX — Other private transfers

Includes transfers in the national accounts category *other current transfers*. These are flows received from outside the household, so the age profile follows a receipt shape: employment income plus pensions ($YEM + BSPPEN$), taken for both sexes together rather than sex by sex, because the household member who receives such a transfer is as often a woman as a man.

5.1.4 BROW — Transfers from rest of the world

Includes social contributions, social benefits, and other private transfers received from the ROW less those paid to the ROW. This encompasses, in particular, benefits paid to non-resident workers (such as cross-border commuters) and remittances. As for BPX, the age profile follows employment income plus pensions ($YEM + BSPPEN$) for both sexes together.

5.2 BPW — Private transfers within households, received

Intra-household transfers received, representing resources received by household members whose consumption exceeds their individual income (typically children and non-working adults). These are imputed as the difference between each household member's consumption and their individual income net of saving and inter-household transfers.

5.3 TPW — Private transfers within households, paid

Resources provided by household members whose individual income exceeds their consumption (typically prime-age earners).

6 YD — Disposable Income

ENTA provide several measures of disposable income that differ in the income components included. Each successive measure adds components, from the narrowest definition based on employment income and transfers (YD1) to the broadest measure that includes all income and transfer flows (YD).

6.1 YD — Disposable income

The broadest measure of disposable income, including all primary income, all public and private transfers, and intra-household transfers:

$$YD = YEM + YA - T + BS + BPPEN + BPB + BPW - TPW.$$

This is the income that is available for consumption and saving. The NTA accounting identity ensures:

$$YD = CH + SH.$$

The aggregate value in ENTA corresponds largely to the disposable income of the household sector in ESA. Age-specific values are the sum of all income and transfer age profiles.

6.2 YD1 — Disposable income from employment and transfers

A narrow measure of disposable income based on employment income and public transfers only:

$$YD1 = YEM - TYEM + BS + BPPEN - TB - TCON.$$

It excludes all asset income (property income and operating surplus), taxes on assets, other current transfers, private transfers between households, and intra-household transfers. YD1 captures the income available to individuals from their own work and the social transfer system. Estimates of these components are the most reliable and best suited for comparisons over time.

6.3 YD2 — Disposable income including intra-household transfers

Extends YD1 by adding private transfers within households:

$$YD2 = YD1 + BPW - TPW.$$

This measure captures the resources available from employment, public redistribution, and family support, but still excludes asset income and private transfers between households.

6.4 YD3 — Disposable income including operating surplus

Extends YD2 by including the operating surplus of households:

$$YD3 = YD2 + YO.$$

This measure includes imputed rent from owner-occupied housing but still excludes property income (dividends, interest, rent), taxes on assets and asset income, other current transfers, and private transfers between households.

6.5 EHI — Equivalised household income

Equivalised household income by age is based on total disposable household income (HY020) in EU-SILC and the OECD modified equivalence scale. EHI is not derived from the ENTA accounting framework but provided as a complementary measure of household welfare from the microdata.

7 CG — Government Consumption

Government consumption measures public consumption expenditure, distinguishing education, health, and other. In NTA, government consumption is categorised as an in-kind transfer. The aggregate values are based on general government expenditure by function (COFOG).[4]

$$CG = CGE + CGH + CGX.$$

7.1 CGE — Government consumption, education

Education expenditure of the government, covering all levels from pre-primary to tertiary education. Aggregates are based on COFOG, category education. The age profiles are based on government expenditure by education level (ISCED) and enrolment by education level, age, and sex.[5] Enrolment data from two Eurostat datasets are combined: *educ_uoe_enra02* (from 2013) and *educ_enrl1tl* (until 2012). Enrolment rates by ISCED level are multiplied by per-student expenditure at each level to obtain age-specific education consumption.

7.2 CGH — Government consumption, health

Government expenditure on health care services. Age profiles are based on the health expenditure profiles of the 2018 Ageing Report of the European Commission and the Ageing Working Group, the same source for all countries. One profile per country is used for all years and scaled to each year's aggregate.

7.3 CGX — Government consumption, other

All remaining government consumption not classified under education or health. In the ENTA framework, all expenditure of the government is treated as transfers. Expenditure that cannot be assigned to a particular age group — such as public administration or defence — is assumed to benefit all individuals equally, with everyone receiving the same per capita amount.

[4] https://ec.europa.eu/eurostat/databrowser/view/gov_10a_exp/default/table

[5] https://ec.europa.eu/eurostat/databrowser/view/educ_uoe_enra02/default/table and https://ec.europa.eu/eurostat/databrowser/view/EDUC_ENRL1TL/default/table

8 CH — Household Consumption

Household consumption expenditure at basic prices, with housing, food, and transport as the largest components:

$$CH = CHE + CHH + CHR + CHX.$$

In ESA, total household consumption is captured as *final consumption expenditure*. For the distinction by purpose, ENTA use data on household final consumption expenditure by purpose (COICOP 1999).[6]

The age- and sex-specific estimates of health and other consumption are based on the Household Budget Survey, imputed into EU-SILC; education follows enrolment, and owner-occupied housing the imputed rent in EU-SILC (Section 3.8). Household consumption in ENTA is measured at basic prices, i.e. excluding taxes less subsidies on products.

8.1 CHE — Household consumption, education

Household expenditure on educational services, including tuition fees, textbooks, and other education-related spending. The aggregate is the share of education in household consumption by purpose (COICOP 10). The age profile gives the spending at each level of education to the persons enrolled at that level, and the spending not definable by level to all persons aged 20–64 (Section 3.8.2); in Finland and Sweden, which report no split by level, it is the same amount for every person aged 3–64.

8.2 CHH — Household consumption, health

Household out-of-pocket expenditure on health, including co-payments, pharmaceuticals, and medical services not covered by public or private insurance.

8.3 CHR — Household consumption, owner-occupied housing

Imputed rental value of owner-occupied housing, representing the housing services consumed by owner-occupiers.

8.4 CHX — Household consumption, other

All remaining household consumption expenditure not classified under education, health, or owner-occupied housing, including food, clothing, transport, communication, and recreation.

9 SH — Saving, Households

Saving of the household sector, defined as disposable income less consumption expenditure:

$$SH = YD - CH.$$

[6] https://doi.org/10.2908/NAMA_10_CO3_P3

It represents the change in household net worth due to current transactions. ENTA saving excludes the adjustment for the change in pension entitlements; adding D.8, which is not published, gives ESA net saving (B.8n).

Household saving is published under the code `SH`. Earlier releases of the ENTA–ESA file used `S`; in the ENTA–NTA file `S` denotes total saving, so a script keyed on the old ESA code needs updating.

C Corporation Sector

Income, taxes, transfers and saving of non-financial and financial corporations. All corporate variables have the age profile of property income received by households, without income from private pension plans (`PY080G`), scaled to the corporate aggregate from ESA (Chapter 3, asset income). The suffix C distinguishes them from their household counterparts; the aggregates of the sector Corporations carry the same codes (YC, TC, BPPC with its parts TPSOCfcC and BPSOCfcC, BPBC, YDC, SC).

C.1 YC — Primary income, corporations

Primary income of corporations corresponds to their income after compensating the factors of production (compensation of employees, paying net property income such as interest and dividends) but before paying taxes. The ENTA aggregate of YC is the ESA *net value added* less *compensation of employees* less net *property income* paid. Age profiles are based on property income received in EU-SILC (`HY090G`, `HY040G`), without private pensions.

C.2 TC — Taxes, corporations

Taxes paid by corporations mainly include taxes on production and taxes on corporate income. The age profiles of TC are based on those for property income of households, using information on property income in EU-SILC.

C.4 BPPC — Funded pensions, corporations

Funded pensions from the perspective of financial corporations mirror the flows of the household sector: contributions to funded pension systems constitute an inflow for financial corporations, while benefits represent an outflow:

$$BPPC = -BPSOCfc + TPSOCfc.$$

C.5 BPBC — Private transfers, corporations

Private transfers of corporations include flows to and from the household sector, as well as some flows to and from the rest of the world. The age profile is based on property income of households.

C.9 SC — Saving, corporations

Saving of corporations largely represents retained profits, i.e. the profits that are not paid out to owners but are kept in the company for investment. Total corporate saving consists of saving excluding funded pensions (SCX) and saving from funded pensions (SCP):

$$SC = SCX + SCP.$$

As for households, corporate saving excludes the adjustment for the change in pension entitlements: SC plus the (negative) corporate D.8 gives ESA net saving of corporations.

C. Codebook ENTA (NTA concept)

0 Introduction and General Variables

ENTA are provided in two formats. The ENTA–ESA format describes the household sector and keeps a close link to the European System of Accounts (ESA). The NTA format, described here, uses the terminology and concepts of global NTA, so that the data can be compared with NTA data of other countries. Chapter 4 explains the translation.

0.1 Differences between ENTA in NTA format and global NTA

- **Order of the accounts.** ENTA keep the ESA sequence of accounts: how income is generated, redistributed, and used for consumption and saving. Global NTA show the life-cycle deficit (consumption less labour income) and how it is covered by transfers and asset-based reallocations.
- **Sectors.** Global NTA distinguish the private sector (suffix F, for "family") and the government (suffix G). ENTA split the private sector further into households (H) and corporations (C), for example $YAF = YAH + YAC$.
- **Mixed income.** As in global NTA, two thirds of self-employment income (mixed income) are labour income and one third is asset income: $YL = YE + \frac{2}{3}\,YM$ and $YAH = YA^{\mathrm{ESA}} + \frac{1}{3}\,YM$, where YA^{ESA} is household asset income in the ESA format (published there as YA). In the ESA format employment income includes the whole of mixed income.
- **Taxes on benefits.** Social benefits are gross in both formats ($TGIC = BS$). The ESA format shows the taxes on benefits (TB) separately; in the NTA format they are part of total taxes ($TGOT$).

0.2 Overview

Table C.1 shows the NTA variables and their relation with ENTA in ESA format.

Table C.1: ENTA variable definitions in NTA format

Number	Variable	Abbrev.	ENTA-ESA format
0	Population	POP	*POP*
	1. Income		
1	Primary income	Y	$YL + YA$
1.1	Labour income	YL	$YE + \frac{2}{3}\ YM$
1.2	Asset income, private	YAF	$YAH + YAC$
1.2.1	Asset income, households	YAH	$YA + \frac{1}{3}\ YM$
1.2.2	Asset income, corporations	YAC	*YC*
1.3	Asset income, government	YAG	
1.4	Asset income, total	YA	$YAF + YAG$
	2. Transfers to government (taxes and contributions)		
2	Taxes, total	TGOT	$T + TC$
2.1	Taxes, paid by households	TGOTH	*T*
2.2	Taxes, paid by corporations	TGOTC	*TC*
2.3	Public transfers, net from ROW	TGNF	
	3. Transfers from government (cash)		
3	Social benefits	TGIC	*BS*
3.1	Social benefits, pensions	TGICPEN	*BSPEN*
3.2	Social benefits, family	TGICF	*BSFAM*
3.3	Social benefits, unemployment	TGICU	*BSU*
3.4	Social benefits, other	TGICX	*BSX*
	4. Funded pensions		
4	Funded pensions (received less paid)	TPEN	*BPPEN*
	5. Private transfers		
5	Private transfers	TF	
5.1	Private transfers between households	TFB	$BPB + BPBC$
5.2	Private transfers within households	TFW	$BPW - TPW$
5.2.1	Intra-household transfers received	TFWI	*BPW*
5.2.2	Intra-household transfers paid	TFWO	*TPW*
	6. Disposable income		
6	Disposable income, private sector	YDF	
6.1	Disposable income, households	YDH	*YD*
6.2	Disposable income, corporations	YDC	

Table C.2: ENTA variable definitions in NTA format (continued)

Number	Variable	Abbrev.	ENTA-ESA format
	7. Government consumption		
7	Public consumption	CG	
7.1	Public cons., education	CGE	
7.2	Public cons., health	CGH	
7.3	Public cons., other	CGX	
7.4	Adjusted disp. income, households	YDHA	$YDH + CG$
	8. Private consumption		
8	Private consumption	CF	CH
8.1	Private consumption, education	CFE	CHE
8.2	Private consumption, health	CFH	CHH
8.3	Private consumption, housing	CFR	CHR
8.4	Private consumption, other	CFX	CHX
	9. Saving		
9	Saving, total	S	$SF + SG$
9.1	Saving, private	SF	$SH + SC$
9.1.1	Saving, households	SH	SH (same code in ESA format)
9.1.2	Saving, corporations	SC	SC
9.2	Saving, government	SG	
	NTA aggregates		
10.1	Net government transfers	TG	$TGI - TGO$
10.2	Total government inflows	TGI	$TGIC + CG$
10.3	Total government outflows	TGO	$TGOT + YAG - SG$
10.3.1	Government outflows, asset-based	TGOABR	$YAG - SG$
10.4	Total consumption	C	$CF + CG$

0.3 Basic variables

AGE — Age group

Single-year age groups for ages 0 to 79. The age group 80 includes all persons aged 80 and older.

Country-specific notes:

- **Germany:** In the years 2014–2018, the oldest available age group is 73. For EU-SILC–based variables, averages for the 73+ group are applied to all older age groups up to age 80. From 2022 onward, age is provided only in five-year age groups.

YEAR — Reference year

Calendar year to which the observation refers.

CNT2 — Country code

Two-letter country code as used by Eurostat (ISO 3166-1 alpha-2, except EL for Greece; e.g. DE for Germany, FR for France). *COUNTRY* holds the country name.

SEX — Sex

Men, Women or Total. The value for Total is the per capita value of both sexes together, not the sum of the two.

IND, INDICATOR — Indicator

IND is the code of the indicator as used in this codebook (YL, TG, . . .); *INDICATOR* is its number and label, for example "1.1 Labour inc (YL)". The same two columns identify the indicators of the ENTA–ESA file and of the aggregates.

VALUE — Value

The stored value, in euro per capita at current prices (Appendix A). In *ENTA_aggregates* the value is *VALAGGR*, in million euro, and the rows are identified by *SECTOR* instead of age and sex.

Missing values

A country-year without data (Croatia before 2009, Bulgaria 2023) has no rows, and in the aggregates an indicator that is not reported in a year has no row. In the profiles every indicator has a value at every age; a zero can be a structural zero, a value the source does not report (Section 6.1).

1 Y — Primary Income

Primary income in NTA is classified into labour income (YL) and asset income (YAF, YAG):

$$Y = YL + YA, \qquad YA = YAF + YAG.$$

It covers all sectors, and so differs from primary income in the ESA format, which is that of households only ($Y = YEM + YA$ there). The codes Y and YA therefore mean different quantities in the two formats.

1.1 YL — Labour income

Compensation of employees, including employers' social contributions and taxes on payroll, plus two-thirds of self-employment income (mixed income), the standard NTA convention (Section 4.1):

$$YL = YE + \tfrac{2}{3}\, YM.$$

1.2 YAF — Asset income, private sector

Asset income of the private sector, calculated as the sum of household and corporate-sector asset income:

$$YAF = YAH + YAC.$$

1.2.1 YAH — Asset income, households

Asset income of the household sector, including property income (interest, dividends, rent on land), imputed rents on owner-occupied housing (operating surplus), and one-third of self-employment income (the capital share of mixed income):

$$YAH = YA^{\mathrm{ESA}} + \tfrac{1}{3}\, YM,$$

where YA^{ESA} is household asset income in ENTA–ESA format (property income plus operating surplus, published there as YA) and YM is mixed income.

1.2.2 YAC — Asset income, corporations

Primary income of corporations, corresponding to the operating surplus of corporations less net property income paid to other sectors, before the payment of corporate taxes. Corresponds to YC in the ENTA–ESA format.

1.3 YAG — Asset income, government

Asset income of the government sector. This quantity is negative in most countries, largely reflecting interest payments on public debt.

1.4 YA — Asset income, total

Asset income of the private sector and the government, $YA = YAF + YAG$, as in NTA. It is not the YA of the ESA format, which is household asset income only.

2 TGOT — Public Transfer Outflows (Taxes and Contributions)

Government transfer outflows (taxes) comprise all transfers paid by the private sector to the government, net of subsidies received. The most important components are income taxes (wage and other income taxes), social contributions (employees' and employers' shares), and taxes on consumption (VAT and excise duties):

$$TGOT = TGOTH + TGOTC.$$

2.1 TGOTH — Taxes paid by households

All taxes and social contributions paid by the household sector to the government, including taxes on employment income, asset income, benefits, and consumption, as well as other current transfers. Corresponds to T in the ENTA–ESA format.

2.2 TGOTC — Taxes paid by corporations

Taxes paid by the corporation sector, mainly including taxes on production and taxes on corporate income. Corresponds to TC in the ENTA–ESA format.

2.3 TGNF — Public transfers, net from rest of the world

Net transfers between the domestic government sector and the rest of the world (ROW), including contributions to and receipts from international organisations and foreign governments. This component affects only the government sector and is included for accounting consistency with total government revenue and expenditure.

3 TGIC — Public Transfer Inflows in Cash

Social benefits paid in cash by the government to individuals. The categories are those of the ENTA–ESA format, where they are defined (Appendix B, BS); the prefix TGIC (transfers, government, inflows, cash) replaces BS:

$$TGIC = TGICPEN + TGICF + TGICU + TGICX.$$

3.1 TGICPEN — Public transfer inflows in cash, pensions

Old-age, survivor and disability pensions, including government sick pay. *BSPEN* of the ENTA–ESA format.

3.2 TGICF — Public transfer inflows in cash, family

Cash benefits for families, children and education. *BSFAM* of the ENTA–ESA format.

3.3 TGICU — Public transfer inflows in cash, unemployment

Unemployment benefits. *BSU* of the ENTA–ESA format.

3.4 TGICX — Public transfer inflows in cash, other

All remaining social benefits in cash, mainly housing and social exclusion. *BSX* of the ENTA–ESA format.

4 TPEN — Funded Pensions

Funded pension transfers (*TPEN*) capture benefits received from funded (i.e. capital-reserve–based) pension systems, net of household contributions to those systems. Corresponds to *BPPEN* in the ENTA–ESA format. In the aggregates file the two flows are also published separately for the household sector, as *TFPI* (benefits received, *BPSOCfc*) and *TFPO* (contributions paid, *TPSOCfc*), so that $TPEN = TFPI - TFPO$.

For the private sector as a whole, funded pensions net to zero, because financial corporations and households both belong to it; they reallocate resources between age groups, from workers who pay contributions to retirees who receive benefits. Depending on the purpose of the analysis, funded pensions can be treated as private transfers, as public transfers (participation is often mandatory) or as saving and dissaving of households.

5 TF — Private Transfers

Private transfers include all forms of redistribution that do not involve the government sector:

$$TF = TFB + TFW.$$

5.1 TFB — Private transfers between households

Inter-household transfers, including social contributions and benefits organised by non-financial corporations (such as continued remuneration during sick leave), remittances, regular cash support between households, and transfers to and from the rest of the world. In the NTA format, *TFB* combines the

household component (BPB) and the corporate component ($BPBC$) from the ENTA–ESA format:

$$TFB = BPB + BPBC.$$

Most of these transfers have no microdata by age: their net value has a receipt profile (employment income plus pensions), which says nothing about who pays or receives them. Transfers between resident households are taken from EU-SILC with an age profile pooled over years; received equals paid in total, so they move resources between ages, mainly from ages 50–74 to young adults, without changing the total.

5.2 TFW — Private transfers within households

Net intra-household transfers, representing the reallocation of resources among members of the same household:

$$TFW = TFWI - TFWO.$$

These are imputed as the difference between each household member's consumption and their individual income net of saving and inter-household transfers. Intra-household transfers are the mechanism through which children and non-working household members receive resources for consumption.

5.2.1 TFWI — Intra-household transfers, received

Received by household members whose consumption exceeds their income. BPW of the ENTA–ESA format.

5.2.2 TFWO — Intra-household transfers, paid

Paid by household members whose income exceeds their consumption. TPW of the ENTA–ESA format.

6 YDF — Disposable Income, Private Sector

Disposable income of the private sector is calculated as primary income less taxes plus social benefits, funded pensions, and private transfers:

$$YDF = YL + YAF + TFB + TFW - TGOT + TGIC.$$

The NTA accounting identity ensures that private disposable income equals private consumption plus private saving:

$$YDF = CF + SF.$$

YDF refers to the combined disposable income of the household and corporation sectors. It can be decomposed as:

$$YDF = YDH + YDC.$$

6.1 YDH — Disposable income, household sector

Disposable income of the household sector, calculated as:

$$YDH = YL + YAH + TFB - BPBC + TFW - TGOTH + TGIC + TPEN.$$

This corresponds to the NTA version of disposable income (YD) in the ENTA–ESA format. It includes all household primary income, less taxes paid by households, plus social benefits in cash, funded pensions (received less paid), and all private transfers (between and within households).

6.2 YDC — Disposable income, corporations

Disposable income of the corporation sector, calculated as corporate asset income less corporate taxes, plus the corporate component of private transfers, less funded pension transfers:

$$YDC = YAC - TGOTC + BPBC - TPEN.$$

Since corporations do not consume, corporate disposable income equals corporate saving: $YDC = SC$.

7 CG — Government Consumption

Government consumption represents the value of goods and services provided by the government to individuals or the community. In NTA, it is treated as an in-kind transfer from the government to individuals and is allocated by age:

$$CG = CGE + CGH + CGX.$$

7.1 CGE — Government consumption, education

Government expenditure on education, allocated by age with enrolment by level of education. The same indicator as in the ENTA–ESA format.

7.2 CGH — Government consumption, health

Government expenditure on health care, allocated by age with the profiles of the 2018 Ageing Report. The same indicator as in the ENTA–ESA format.

7.3 CGX — Government consumption, other

All other government consumption, the same amount per capita at every age. The same indicator as in the ENTA–ESA format.

7.4 YDHA — Adjusted disposable income, household sector

Disposable income of the household sector including government consumption allocated to individuals (i.e. in-kind transfers):

$$YDHA = YDH + CG.$$

In ESA, *adjusted disposable income* refers to disposable income plus individual (but not collective) government consumption. The ENTA definition includes all government consumption, including the collective component, which is allocated equally per capita.

8 CF — Private Consumption

Final consumption expenditure of households at basic prices, with housing (including energy), mobility, and food and non-alcoholic beverages as major components:

$$CF = CFE + CFH + CFR + CFX.$$

Aggregate values are based on ESA final consumption expenditure at basic prices. The breakdown by purpose follows the Classification of Individual Consumption According to Purpose (COICOP 1999).[1] The age- and sex-specific estimates are those of the ENTA–ESA format (Appendix B, CH).

8.1 CFE — Private consumption, education

Household expenditure on education. *CHE* of the ENTA–ESA format.

8.2 CFH — Private consumption, health

Household out-of-pocket expenditure on health. *CHH* of the ENTA–ESA format.

8.3 CFR — Private consumption, imputed rent

Imputed rent of owner-occupied housing. *CHR* of the ENTA–ESA format.

8.4 CFX — Private consumption, other

All other household consumption expenditure. *CHX* of the ENTA–ESA format.

9 S — Saving

Saving is defined as the residual of disposable income less consumption. It represents the change in net worth due to current transactions:

$$S = SF + SG.$$

[1] https://doi.org/10.2908/NAMA_10_CO3_P3

9.1 SF — Private saving

Saving of the private sector (households and corporations combined):

$$SF = SH + SC.$$

9.1.1 SH — Saving, households

Saving of the household sector, equal to household disposable income less household consumption expenditure:

$$SH = YDH - CF.$$

Published under the same code, *SH*, in the ENTA–ESA format.

9.1.2 SC — Saving, corporations

Saving of the corporation sector, comprising retained earnings (profits not distributed as dividends or other property income). It equals corporate disposable income (*YDC*).

9.2 SG — Saving, government

Saving of the government sector, equal to government revenue less government current expenditure (i.e. the government budget balance on current account).

10 NTA Aggregates

NTA is based on a flow identity that holds for each individual and each age group. For the total economy:

$$\underbrace{Y}_{\text{Income}} + \underbrace{TGI}_{\text{Gov. inflows}} - \underbrace{TGO}_{\text{Gov. outflows}} + \underbrace{TFB + TFW}_{\text{Private transfers}} = \underbrace{C}_{\text{Consumption}} + \underbrace{S}_{\text{Saving}}.$$

These aggregate variables are constructed in the code and included in the standard ENTA output, as indicators 10.1–10.4, plus 10.3.1 for the asset-based component of TGO (TGOABR). They can be used to verify the accounting identity.

10.1 TG — Net government transfers

Net public transfers received by an age group, i.e. total government transfer inflows less total government transfer outflows:

$$TG = TGI - TGO.$$

A positive value indicates an age group is, on net, a beneficiary of public redistribution; a negative value indicates it is, on net, a contributor.

10.2 TGI — Total government transfer inflows

Total resources received by individuals from the government, comprising cash transfers and in-kind transfers (government consumption):

$$TGI = TGIC + CG.$$

10.3 TGO — Total government transfer outflows

Total resources transferred from individuals to the government, calculated as:

$$TGO = TGOT + YAG - SG.$$

This expression reflects that government expenditure allocated to individuals is financed by tax revenue ($TGOT$), government asset income (YAG), and government dissaving ($-SG$).

10.3.1 TGOABR — Government outflows, asset-based

The part of total government transfer outflows financed by government asset income net of government saving, rather than by tax revenue:

$$TGOABR = YAG - SG.$$

It is the non-tax component of TGO: $TGO = TGOT + TGOABR$.

10.4 C — Total consumption

Total consumption of an age group, combining private and government consumption:

$$C = CF + CG.$$

D. Organisation of ENTA code and data

D.1 Code and data

The code of ENTA is written in Stata and is public (`https://github.com/bbinderhammer/enta`). The file *README.md* of the repository describes the folders, the order in which the files are run and what each file does; each do-file starts with a header that states what it reads and what it writes. The tables downloaded from Eurostat are part of the repository, so the aggregates can be rebuilt without access to microdata. The files with the prefix *Micro_* and the files in *ENTA_ageprofiles* need the microdata, which are not part of the repository, in compliance with the Eurostat rules for access to microdata.

The published data are four files, *ENTA_esa* and *ENTA_nta* (age profiles, euro per person, long format), *ENTA_aggregates* (million euro) and *ENTA_units* (conversion factors, Appendix A), each as CSV and as Parquet file. Values in other units are calculated from these files and are not stored.

D.2 Data versions

ENTA V.2.0.0 was built from the data versions in Tables D.1 and D.2. Eurostat revises its tables, so data downloaded later will not reproduce the published results exactly; by default the code uses the tables stored in the repository.

Table D.1: Eurostat tables and other public sources, with download date

Table	Content	Downloaded
nasa_10_nf_tr	Non-financial sector accounts	27.09.2026
gov_10a_exp	Government expenditure by function	27.09.2026
nama_10_co3_p3	Household consumption by COICOP	28.09.2026
demo_pjan	Population on 1 January by age and sex	27.09.2026
educ_enrl1tl	Students by ISCED level (to 2012)	28.09.2026
educ_uoe_enra02	Pupils and students by level (from 2013)	28.09.2026
prc_hicp_aind	HICP, annual data	27.09.2026
prc_ppp_ind	PPPs and price level indices	27.09.2026
tax_ec_func (DG TAXUD)	Tax revenue by economic function	02.09.2026
National tax lists (DG TAXUD)	Taxes by item and economic function, PIT split	29.09.2026
World Bank CPI	Consumer price index	09.05.2026
UN WPP	Population by age and sex	02.09.2026
Ageing Report 2018, 2021 (EC)	Health expenditure by age, digitised charts	17.06.2026

The code was run with Stata 19 SE and the Stata packages *eurostatuse*, *sumdist* and *supsmooth*; the Parquet files, the figures and the tables of Chapter 5 with R 4.6.1.

Table D.2: Microdata versions

Source	Version
EU-SILC cross-sectional UDB, survey years 2009–2024	2025 release, revision 1 (December 2025; DOI 10.2907/EUSILC2004-2024REV1)
EU-LFS yearly files, data up to 2024	2025 release (January 2026; DOI 10.2907/LFS1983-2024)
HBS	Waves 2010 and 2015
EUROMOD	J2.0+, with input data from 2024
Konsumerhebung (Austria only), Statistics Austria	2010 and 2015

D.3 Notes on the input data

D.3.1 Population (Suppl_population.do)

The population comes from Eurostat, *population on 1 January by age and sex* (`https://doi.org/10.2908/DEMO_PJAN`). ENTA use the average population of a year: the mean of 1 January of that year and 1 January of the next year.

The population by age of the World Population Prospects 2024[1] is used in three cases:

- **The oldest ages are reported as one group.** In some countries and years the open age group of Eurostat starts below 100. The group is distributed over its single ages in proportion to the UN population at these ages; the total population of Eurostat does not change. This occurs mostly before 2008; in the years of the age profiles it concerns Germany (2008–2010), Hungary (2008–2012), France (2009–2010), Croatia (2011–2012) and Luxembourg (2008).
- **Romania, 1995–2001.** Eurostat reports implausibly many persons aged 100 and older. Ages 90 and older are treated as one group and distributed in the same way.
- **Croatia, 1995–2000.** Eurostat reports only the total population, and the UN population is used.

D.3.2 Consumer prices (Suppl_prices.do)

The *Harmonised Index of Consumer Prices* (`https://doi.org/10.2908/PRC_HICP_AIND`) starts in 1996 (Bulgaria 1997, Croatia 1998). It is extended back to 1995 with the growth of the World Bank consumer price index.

D.3.3 Household consumption by COICOP (Suppl_consumption_hh_coicop.do)

Imputed rent is the level of COICOP 04.2; education and health are their shares in total COICOP consumption, applied to household consumption of the sector accounts. Where the COICOP data of

[1] `https://population.un.org/wpp/downloads?folder=Standard%20Projections&group=Population`

2023 or 2024 are not yet published, the value of the year before is carried forward with the growth of household consumption.

D.3.4 Enrolment in education (Suppl_enrollment_educ.do)

Eurostat reports the number of students partly by single year of age and partly in five-year or open age groups. Single years are used where all five ages of a group are reported. Otherwise the group is distributed to single ages in proportion to a reference age profile: the enrolment rate of the country by level, sex and age, averaged over up to three years around 2015–2017 with complete single ages, times the population. The same is done for open age groups, for sources that spread the students of a group evenly over its ages, and for students of unknown age. Where the age distribution of a year is not usable (the age groups add up to less than half of the total, or the students of some ages are missing), the ages concerned take the mean of the nearest earlier and later year. Where a total has no split by sex, the shares of the nearest year with a split are used. Greece has no data for 2009; it takes the mean of the enrolment rates of 2008 and 2010.

D.3.5 Public health expenditure (CGH_AWG.dta)

Health expenditure by age and sex is taken for all countries from the 2018 Ageing Report. The Ageing Reports publish these profiles only as charts, so the curves were extracted with WebPlotDigitizer (Rohatgi, n.d.), for men and women, from the 2018 and the 2021 edition. The extracted points, the R script that cleans them and the resulting rates are published with the code. The script rounds the ages to whole years, averages points that fall on the same age and fills missing ages by linear interpolation.

D.3.6 Household Budget Survey (Micro_HBS.do)

The survey weights of the members of a household differ, because they are adjusted to the population by age group. Values of the household (consumption, the income percentile) are therefore weighted with the mean weight of its members, in the HBS and in EU-SILC.

D.3.7 Labour Force Survey (Micro_LFS.do, ENTA_employment.do)

A full-time equivalent is the usual weekly working hours divided by 35, at most one. Where the usual hours vary, the hours worked in the reference week are used, or, if none are reported, the average usual hours of the same country, year and sex. The employment profiles are not smoothed.

D.4 Control of the output

The published files are checked after every change to the code, with the scripts in the folder *ENTA_output_control*, which need no microdata:

- *1_method_check.do* tests what must hold: every profile adds up to its aggregate to within 0.01 per cent, components add up to their totals, men and women add up to the total, quantities

that cannot be negative are not, and every profile has all ages. A failure is a defect.

- *2_aggregates_check.do* and *3_profiles_check.do* list the largest changes from one year to the next: in the aggregates, in the age shape of the profiles and in the share of men, and single-age spikes. These findings are not defects; most are real events in the data. The file *ledger.csv* records each examined pattern with its cause and the evidence for it; the patterns that matter for interpreting the data are described in Appendix E.
- *4a* to *4c* compare ENTA with independent NTA results (Section 5.4).

What these checks show is limited. The tests of the first script hold by construction: every profile is scaled to its aggregate, and the identities are built into the compilation. They show that nothing was dropped, double counted or mislabelled on the way to the published files, not that an age shape is right. The checks that need the microdata (the number of observations behind each profile, the coverage of the national accounts by EU-SILC, the test of the imputation of consumption) are in the folder *ENTA_microdata_checks* and produce the tables of Chapter 5.

E. Country-specific patterns and events

Some ENTA series contain level shifts, sign changes or single-year spikes that look like errors but are not. They come from events in the economy, from national policy, or from the way a country reports to Eurostat. This chapter lists the cases found so far.

Sections E.1 and E.2 cover changes in the economy and in national law. Section E.3 covers breaks that come from the data: a series shifts because the way it is measured, reported or built changed. Each entry gives the pattern in the data, the explanation with its source, and the consequence for analysis. Where an explanation is not supported by a document, the entry says so.

A series that changes its level permanently makes every year of one period look unusual against the average of the whole period. Before treating a value as an error, check whether the series has such a break, and compare the year with the years of its own period. The quick start (Chapter 0, "When a result looks odd") says where to look first.

E.1 Changes caused by the economy

E.1.1 The financial crisis, 2008–2010

Pattern. Between 2008 and 2009 social benefits (`BS`) rise as a share of household primary income in all 26 countries (median +7%), pensions in all 26 (+5%) and unemployment benefits in all 26 (+31%). Household disposable income falls as a share of household primary income in all 26 countries. National income falls in 24 of the 26 countries (median −5.7%), most in the Baltic countries. The primary income of the German corporate sector is negative in 2009, and its age profile is negative at every age.

Explanation. The recession that followed the financial crisis of 2008. Over the whole year 2009 the GDP of the euro area fell by 4.0% and that of the EU by 4.1%.[1] Benefits rose and incomes fell, as in every recession.

Consequence for analysis. 2009 is a cyclical trough, not a break in the data. Where income falls sharply, every share of GDP or of national income moves with the denominator as much as with the item itself: consumption in Finland, for example, rises from 84% to 93% of national income in 2009 although it grows by only 1.3%.[2] Compare levels before reading a change of a share as a change of the item.

[1] Eurostat, news release 22/2010, 12 February 2010, `https://ec.europa.eu/eurostat/documents/2995521/5047406/2-12022010-BP-FR.PDF/1d5f9762-e529-42cd-ac5c-cc4878fc584b`.

[2] Statistics Finland, Annual national accounts 2009, 15 July 2010, `https://stat.fi/til/vtp/2009/vtp_2009_2010-07-15_tie_001_en.html`.

E.1.2 The share of men in income, 2009

Pattern. The share of men in household primary income falls in 23 of the 26 countries in 2009, most in Lithuania (−6.7 points) and Latvia (−3.5). The fall is in employment income, and the share of men in unemployment benefits rises at the same time. The change unwinds slowly: the share of men in Lithuanian income is 49.5% in 2009 and 53.1% in 2014.

Explanation. Recessions reduce employment most in construction and manufacturing, where mostly men work.[3] The published employment indicators of Lithuania (Labour Force Survey) show it: the employment rate of men aged 20–64 falls by 9.4 points in 2009, that of women by 2.5.

Consequence for analysis. The profiles by sex of 2009 to 2013 in these countries show a recession and its slow recovery, not a lasting change in the distribution of income between men and women.

E.1.3 Interest rates, 2008–2011

Pattern. Property income received by households and the taxes on asset income fall in several countries in the years after the crisis. Interest paid by households falls even more, in 23 of 26 countries (median −39% in 2009). Because interest is paid at younger ages than property income is received, the age profile of net property income moves to younger ages: in Spain and Portugal the mean age of asset income falls by about three years in 2009. In 2008, before the cuts, households in Ireland and Romania paid more property income than they received.

Explanation. The European Central Bank cut its main refinancing rate from 4.25% in October 2008 to 1.00% in May 2009.[4] The negative net property income of Irish households in 2008 follows the growth of mortgage debt before the crisis.[5] For Romania no document was checked.

Consequence for analysis. The level and the age shape of property income, and the taxes on it, are not comparable across 2008 and 2009. The change reflects interest rates, not wealth or its distribution by age. Ireland and Romania in 2008 are the only country-years with negative net property income of households.

E.1.4 Ireland: GDP, 2015

Pattern. Irish nominal GDP rises by 34.2% in 2015, against an EU median of 3.5%, while national income rises by 8%. Nearly every Irish aggregate falls as a share of GDP in that year (social benefits from 11.1% to 8.3%), while the levels move as usual; only consumption of fixed capital rises as a share.

Explanation. The Irish statistical office attributes the level shift of GDP in 2015 to the relocation of entire balance sheets of companies to Ireland, dominated by intellectual property products; the

[3] H. Hoynes, D. L. Miller and J. Schaller, "Who Suffers during Recessions?," *Journal of Economic Perspectives* 26(3), 2012, pp. 27–48, for the United States. Construction employment is over 90% male in every EU member state (Eurostat, `lfsa_egan2`).

[4] European Central Bank, Key ECB interest rates, `https://www.ecb.europa.eu/stats/policy_and_exchange_rates/key_ecb_interest_rates/html/index.en.html`.

[5] P. Honohan, "The Irish Banking Crisis: Regulatory and Financial Stability Policy 2003–2008," report to the Minister for Finance, 31 May 2010.

capital assets that came with them also raised the estimates of depreciation (Central Statistics Office Ireland, 2016).

Consequence for analysis. Irish shares of GDP are not comparable across 2014 and 2015. Use national income, household income or values per capita. Independently of 2015, GDP and national income of Ireland in ENTA differ from Eurostat's figures in almost every year, in both directions and by up to about 1.6% of GDP (2014): ENTA build GDP as the sum of the sectors, and the Irish sector accounts do not add up to the GDP that Eurostat publishes.

E.1.5 National income as the reference: Cyprus, Luxembourg and the exchange rate

Pattern. Three cases in which national income, the denominator of many comparisons, moves on its own. In Cyprus it rises by 53% in 2008, the first year of the profiles, while GDP rises by 8%, because net property income from the rest of the world changes. In Luxembourg national income is 42% to 55% of GDP and changes by four to seven points in single years. In countries outside the euro area the values in euro move with the exchange rate: Swedish national income falls by 14% in 2009 and rises by 22% in 2010, against -4.5% and $+8.9\%$ in kronor.

Explanation. Luxembourg is a financial centre with many cross-border workers: wages paid to them and investment income paid abroad separate national income from GDP, and investment income is the volatile part.[6] Property income received from and paid to the rest of the world is five to six times GDP in Luxembourg and about as large as GDP in Cyprus, so small changes in either flow move national income. The reason for the Cypriot step of 2008 was not found. The exchange-rate effect follows from Eurostat's figures in national currency.

Consequence for analysis. For Cyprus in 2008 and for Luxembourg in general, a change of a share of national income may come from the denominator. For countries with their own currency, compare values in national currency or shares, not values in euro, across years with large exchange-rate changes.

E.1.6 Large swings in corporate income

Pattern. In four countries the primary income of the corporate sector changes by a third or more in one year, and corporate saving with it: Denmark in 2021–2023, the Netherlands in 2019–2021 (83.2, 58.0 and 128.8 billion euro), Hungary in 2011–2013, and Slovakia in 2021–2023, with negative corporate saving in 2022.

Explanation. The swings are in the sector accounts of Eurostat. What caused them was not established.

Consequence for analysis. The corporate age profiles and private and total saving follow the swing in these years. When corporate saving is close to zero or negative, its age profile changes

[6] Conseil national des finances publiques, *Note au sujet de l'évolution comparée du PIB et du RNB aux niveaux national et européen*, April 2026, `https://cnfp.public.lu/dam-assets/documents/informations/notes/note-pib-rnb.pdf`: the ratio of gross national income to GDP was about 90% in 1995–2001, fell to 64.4% in 2015 and has fluctuated around 70% since; the gap reflects the compensation of cross-border workers and cross-border flows of capital income.

shape, because it is a sum of profiles with different signs. These values are not errors, and they say nothing about households.

E.1.7 COVID-19, 2020–2021

Pattern. Household saving (SH) rises as a share of household primary income in 25 of the 26 countries in 2020 (median +5.8 points) and falls back in 2022; in Italy it is 1.8%, 9.0%, 6.7% and 2.2% of national income in 2019 to 2022. Social benefits rise in all 26 countries, government saving falls in all 26, and public health consumption rises as a share of GDP in all 26. Subsidies on production turn the taxes of some sectors negative: taxes less subsidies of Austrian corporations are 9.5 billion euro in 2019 and −4.4 billion in 2020. In Ireland unemployment benefits halve in 2022.

Explanation. The pandemic and the support measures of governments. Household saving rose because consumption fell: in the second quarter of 2020 the saving rate of households in the EU showed its largest increase on record, with a fall of household consumption by 17.3%.[7] Short-time work schemes and compensation for firms were recorded as other subsidies on production (D.39): in Austria subsidies paid by government rose by 14.1 billion euro in 2020, mainly for short-time work, the fixed-cost grant and the turnover compensation.[8] In Poland D.39 received by households rose from 4,057 to 12,573 million euro in 2020; unincorporated businesses belong to the household sector. The Irish Pandemic Unemployment Payment closed to new applicants in January 2022.[9]

Consequence for analysis. 2020 and 2021 are exceptional years for saving, benefits, government accounts and corporate taxes. ENTA add taxes less subsidies on production to employment and mixed income and removes them again as a tax, so the subsidies appear inside taxes on employment income: Polish taxes on employment income fall by 7.9% in 2020, but net of the subsidy they rise slightly. Breaks of single countries in these series in 2021 to 2023 are the end of the pandemic measures.

E.1.8 Energy prices and the cost of living, 2022–2023

Pattern. Miscellaneous current transfers (D.75) received by households rise in 2022 and fall back partly in 2023, most in Austria, Germany and Czechia; net other current transfers (TX) turn strongly negative there. Subsidies on products rise sharply in other countries (Bulgaria, Poland and Croatia in 2022, Slovakia in 2023). In these countries national income in ENTA, which is measured before taxes on products and after subsidies on them, rises by two to three points relative to Eurostat's net national income.

Explanation. Governments paid relief for high energy prices; since September 2021 about 650 billion euro were allocated in European countries to shield consumers.[10] Which national measures

[7] Eurostat, "Impact of COVID-19 on household consumption and savings," 10 November 2020, https://ec.europa.eu/eurostat/fr/web/products-eurostat-news/-/ddn-20201110-2. For the return in 2022: Istat, *Conti economici nazionali per settore istituzionale, anni 2020–2022*, https://www.istat.it/it/files/2023/10/Conti-economici-istituzionali-2022.pdf.

[8] Statistics Austria, as reported by ORF, "Rekorddefizit mit 8,9 Prozent des BIP," 1 April 2021, https://orf.at/stories/3207526/.

[9] Government of Ireland, COVID-19 Pandemic Unemployment Payment, https://www.gov.ie/PUP.

[10] Bruegel, "National fiscal policy responses to the energy crisis," https://www.bruegel.org/dataset/national-policies-shield-consumers-rising-energy-prices.

stand behind the figures of each country was not traced, and countries did not all record their relief under the same transaction.

Consequence for analysis. Differences between countries in 2022 and 2023 partly reflect how the relief was recorded, not its size. Other current transfers have the age shape of employment income plus pensions, so a payment tied to the number of children reaches children only through the transfers within households.

E.1.9 Interest rates, 2022–2023

Pattern. Property income received by households in 2023 is above the average of 2019–2021 in every country, most in Ireland, Czechia and Slovenia. Interest paid rises even more: in France from 16 to 62 billion euro in 2023. The age profile of net property income moves to older ages: its mean age rises by five to eleven years in Finland, France and Luxembourg in 2023. Taxes on asset income rise as well.

Explanation. The European Central Bank raised its main refinancing rate from 0.00% to 4.50% between July 2022 and September 2023 (see footnote 4). Interest is paid at younger ages than property income is received, so higher rates lower net property income at young ages and raise it at old ages.

Consequence for analysis. Property income and what is derived from it (taxes on asset income, disposable income, saving) are not comparable across 2022 without the change in interest rates in mind, in level and in age shape. This is the reverse of 2009.

E.2 National policy and law

E.2.1 Hungary and other countries: reforms of funded pensions

Pattern. Hungarian contributions to funded pensions fall from 2,271 million euro in 2010 to 1,102 in 2011 and 766 in 2012, and stay at about 690 afterwards. In other countries with a funded pillar, contributions and net benefits step to a new level within one year, often as mirror images (Poland 2011, Slovakia 2013). Two large steps have no known reason: Romanian contributions rise from 2.9 to 8.2 billion euro in 2023, and Cypriot benefits double in 2013 (490, 941 and 460 million euro in 2012 to 2014).

Explanation. Hungary ended the compulsory payments to the private pension funds with a law of December 2010 and transferred their assets to the state.[11] Poland cut the contribution rate to its open pension funds from 7.3% to 2.3% in 2011 and moved half of their assets to the public system in 2014; Lithuania cut its second-pillar rate from 5.5% to 2% in 2009; Slovakia from 9% to 4% in 2012–2013.[12] Portugal transferred the pension funds of banks, about 3.5% of GDP, to the public system in 2011.[13] For Romania 2023 no reform was found, and the step is not in the contributions

[11] M. Bocian and T. Dąborowski, "Hungary's risky economic strategy," Centre for Eastern Studies (OSW), 29 December 2010, `https://www.osw.waw.pl/en/publikacja/18113/pdf`.

[12] OECD, *Pensions at a Glance* 2011, 2013 and 2015, OECD Publishing, Paris.

[13] OECD, *OECD Economic Surveys: Portugal 2012*, OECD Publishing, Paris.

actually paid: the supervisor reports contributions to the second-pillar funds of 12.8 billion lei in 2023, 17% more than in 2022, which is about 2.6 billion euro.[14] The step is therefore in the recording of the sector accounts; its reason was not found. For Cyprus 2013 no reform was found; the step coincides with the banking crisis of that year.

Consequence for analysis. Funded pensions of these countries are two periods, before and after the reform, and the periods are not comparable. A high value before the reform is not an anomaly.

E.2.2 Estonia: withdrawal from the second pension pillar, 2021

Pattern. Estonian taxes on property income more than triple in 2021 (from 95.9 to 321.8 million euro in the series of DG TAXUD) and halve in 2022. Funded pensions received by households rise from 54 to 219 million euro in 2021, and contributions fall from 389 to 220 million.

Explanation. The funded second pillar became voluntary in 2021, and members could withdraw their savings before retirement age, taxed at 20%.[15] In September 2021 the pension registrar paid out more than 1.3 billion euro to the members who had applied to leave, about a quarter of a billion euro of it withheld as income tax.[16] A tax of 20% on that amount is about 260 million euro, close to the excess of the tax series in 2021.

Consequence for analysis. Estonian property income and the taxes on it in 2021 and 2022 include a one-off withdrawal of pension savings.

E.2.3 Poland: the family benefit Rodzina 500+, 2016

Pattern. Polish family benefits (`BSFAM`) rise from 1.84% to 3.45% of household primary income in 2016.

Explanation. The programme "Family 500+" has paid a monthly benefit of 500 zloty for every second and further child, and for the first child in families with low income, since April 2016.[17] 2016 holds nine months of it.

Consequence for analysis. Polish family benefits before and after 2016 are not comparable.

E.2.4 Income tax reforms: Austria 2016, Portugal 2013

Pattern. Taxes on income paid by Austrian households fall by 10% in 2016 while GDP grows by 4%; taxes on benefits and on pensions fall with them. In Portugal current taxes on income and wealth rise by 28% in 2013, from 14,995 to 19,254 million euro.

[14] Autoritatea de Supraveghere Financiară, "Evolution of the private pension system in 2023," press release, 2 April 2024, `https://asfromania.ro/en/a/2970/evolution-of-the-private-pension-system-in-2023`.

[15] Estonian Ministry of Finance, "Restructuring the second pillar of the funded pension," `https://www.fin.ee/en/restructuring-second-pillar-funded-pension`.

[16] Pensionikeskus, as reported by ERR News, "Pensionikeskus has transferred 99 percent of second pillar funds," 3 September 2021, `https://news.err.ee/1608327707/research-education`.

[17] Act of 11 February 2016 on State Assistance in Raising Children; A. Sowa, "'Family 500+': A new family income-supporting benefit in Poland," ESPN Flash Report 2016/45, European Commission, June 2016.

Explanation. The Austrian tax reform that came into force on 1 January 2016 raised the number of income tax brackets from four to seven and aimed at a noticeable reduction of personal income taxation.[18] The Portuguese budget of 2013 reduced the number of brackets, raised the average tax rate and added a surcharge of 3.5% on taxable income above the minimum wage.[19]

Consequence for analysis. Taxes on income of the two countries are on different schedules before and after these years.

E.2.5 Value added tax: Luxembourg 2015, Spain 2008–2010

Pattern. Taxes on products paid by the total economy of Luxembourg (Eurostat sector accounts `nasa_10_nf_tr`, item D.21 paid) fall by 16% in 2015, from 5,712 to 4,791 million euro, while GDP grows by 4.5%. In Spain taxes on products fall from 111 billion euro in 2007 to 73 billion in 2009 and rise to 92 billion in 2010; national income in ENTA, measured before taxes on products, moves from 88% to 92% of Eurostat's net national income and back to 90%.

Explanation. Since 1 January 2015 the value added tax on telecommunications, broadcasting and electronic services is charged in the member state of the consumer and no longer in that of the supplier (Council Directive 2008/8/EC); Luxembourg, the seat of many suppliers, lost this revenue. In Spain the collection of value added tax fell by 14% in 2008 and 30% in 2009, and the standard rate was raised from 16% to 18% in July 2010.[20]

Consequence for analysis. Consumption taxes of Luxembourg before 2015 include tax paid by consumers in other countries. For Spain, shares of national income in 2008 to 2010 move partly with the denominator.

E.2.6 Cyprus: General Healthcare System, 2019

Pattern. Taxes and contributions on employment income (`TYEM`) rise from 19.9% to 22.1% of national income in 2019 and stay there. Social contributions paid by households rise from 2,380 to 2,894 million euro.

Explanation. Contributions to the new General Healthcare System have been collected since 1 March 2019, at 1.85% of earnings for employers and 1.70% for employees, and at 2.90% and 2.65% from 1 March 2020.[21]

Consequence for analysis. Cypriot taxes on employment income before and after 2019 are not comparable.

[18] European Commission, "Personal Income Taxation in Austria: What do the Reform Measures Mean for the Budget, Labour Market Incentives and Income Distribution?," Economic Brief 030, `https://ideas.repec.org/p/euf/ecobri/030.html`.

[19] Council Implementing Decision 2013/64/EU of 20 December 2012 on granting Union financial assistance to Portugal, recital 4, `https://www.legislation.gov.uk/eudn/2013/64/introduction`.

[20] D. Romero-Jordán and J. F. Sanz-Sanz, "Impact of changes in Spain's VAT rates during the economic crisis: a comparative analysis," *Spanish Economic and Financial Outlook* 2(4), 2013, `https://www.funcas.es/wp-content/uploads/Migracion/Articulos/FUNCAS_SEFO/008art06.pdf`.

[21] Health Insurance Organisation of Cyprus, "Financing and Global Budget," `https://www.gesy.org.cy/en-us/hiofinancing`.

E.2.7 Lithuania: taxes on benefits, 2009 and 2019

Pattern. Taxes on pay-as-you-go pensions rise from 41 to 269 million euro in 2009, and the mean age of taxes on the income of the non-employed (TB) rises from 40 to 59. In 2019 taxes on other benefits rise from 93 to 218 million euro and the mean age of TB falls from 63 to 58.

Explanation. Both years bring a reform of the financing of social insurance. From 2009 health insurance contributions of 6% were separated from the personal income tax, whose rate was reduced to a flat 15%.[22] From 2019 most social contributions of employers were shifted to employees and gross wages were raised by the factor 1.289.[23] ENTA divide the taxes of the non-employed between pensions and other benefits with rates from EUROMOD, and these rates change in the two years. That the reforms are the reason is the reading of the timing; no document links them to the EUROMOD rates.

Consequence for analysis. The split of Lithuanian taxes on benefits, and the age profile of their total, are not comparable across 2008–2009 and across 2018–2019.

E.2.8 Bulgaria: no taxes on social benefits

Pattern. Taxes on social benefits are zero for Bulgaria in every year.

Explanation. DG TAXUD reports taxes on labour income of the non-employed as exactly zero for Bulgaria in every year from 1995 to 2024, while all other functions have positive values. This agrees with Bulgarian tax law, which exempts pensions, unemployment benefits and social assistance from income tax.[24]

Consequence for analysis. The zero profile is correct and is not a gap. Leave Bulgaria out of comparisons of taxes on benefits between countries.

E.2.9 Denmark: taxes on pension capital, 2009 and 2013–2015

Pattern. Danish taxes on funded pensions (TBPPEN) rise from 3,568 to 6,119 million euro in 2009 and return to 3,352 in 2010; they rise again to 12,866 in 2014 and fall to 3,564 in 2016. In DG TAXUD's national tax list the tax on payments from capital pensions accounts for the rise.

Explanation. Two measures fall into these years. In 2009 the special pension savings could be paid out between 1 June and 31 December, with a duty of 35% on the first 15,000 kroner and 50% on the rest.[25] In 2013 to 2015 capital pensions could be converted into the new old-age savings against a duty of 37.3% instead of 40%; by 1 October 2013 about 761,000 persons had converted and paid 15.7

[22] Lithuanian Free Market Institute, "Taxation in Europe 2009: a year of transition: Lithuania," 2010, https://old-en.llri.lt/news/economic-policy/tax-and-budget/taxation-in-europe-2009-a-year-of-transition-lithuania/lrinka.

[23] Grant Thornton Baltic, "Tax Reform in 2019," https://www.grantthornton.lt/globalassets/1.-member-firms/lithuania/pdf/tax-reform-2019.pdf.

[24] Bulgaria, Income Taxes on Natural Persons Act, Art. 13(1); summary at https://www.bulgaria-tax-law.bg/taxes-other-income.html.

[25] Danish Ministry of Taxation, "Mulighed for udbetaling af Den Særlige Pensionsopsparing," 4 March 2009, https://skm.dk/media/pqemdvdq/mulighed-for-udbetaling-af-den-saerlige-pensionsopsparing.pdf.

billion kroner.[26] That these measures stand behind the figures of DG TAXUD is the reading of the timing.

Consequence for analysis. Danish taxes on funded pensions in 2009 and 2013–2015 include one-off duties on pension capital and do not measure the tax on pensions received. The age profile overstates taxes at pension ages in these years.

E.2.10 Greece: taxes on pensions

Pattern. Greek taxes on pay-as-you-go pensions (`TBSPEN`) rise by 66% in 2012, fall by 73% in 2014 and by another 66% in 2019, from 744 to 252 million euro. From 2014 almost all Greek taxes of the non-employed in DG TAXUD's data are a levy on pensions: about 580 million euro in 2014, 710 million in 2018 and 230 million in 2019.

Explanation. The levy is the solidarity contribution of pensioners, introduced in 2010.[27] The fall of 2014 is a change in DG TAXUD's split of personal income tax. The reason for the fall of 2019 was not found; the contribution still exists.

Consequence for analysis. Greek taxes on pensions after 2013 measure essentially one levy and are not comparable with the years before.

E.2.11 Estonia: tax exemption of pensions, 2023

Pattern. Estonian taxes on pay-as-you-go pensions fall by 39% in 2023, from 174 to 106 million euro, while pensions rise by 13%.

Explanation. From 1 January 2023 the tax-free allowance of persons of retirement age equals the average old-age pension, 704 euro a month.[28]

Consequence for analysis. Estonian taxes on pensions before and after 2023 are on two different schedules.

E.2.12 Denmark: early-retirement pay (*efterløn*)

Pattern. Danish unemployment benefits (`BSU`) are high at ages 60 to 65 up to 2018 (7,000 to 12,000 euro per person at 63–65) and fall to almost zero at 66, where pensions rise by a similar amount. From 2019 the pattern largely disappears: in EU-SILC unemployment benefits per person at ages 60–69 fall from 3,152 to 1,156 euro, and old-age benefits rise from 12,043 to 13,558 euro.

[26] Danish Ministry of Taxation, "Omlægning af kapitalpensioner nedbringer den offentlige gæld," 5 October 2013, `https://skm.dk/media/Skatteministeriet/Dokumenter/PDF'er/2013-10-05-kapitalpension.pdf` (the duty of 37.3% for 2013, extended to 2014); the extension to 2015: Den Store Danske, "Kapitalpension," `https://lex.dk/kapitalpension`.

[27] *To Vima*, "Greece plans four measures for pensioners in 2027," `https://www.tovima.com/society/greece-plans-four-measures-for-pensioners-in-2027/`.

[28] ERR News, "Basic exemption for people of retirement age to rise to 704 euro on January 1," December 2022, `https://news.err.ee/1608774130/basic-exemption-for-people-of-retirement-age-to-rise-to-704-on-january-1`.

Explanation. At these ages the survey variable for unemployment benefits recorded mainly *efterløn*, the voluntary early-retirement pay, which is paid through the unemployment insurance funds (Goedemé & Zardo Trindade, 2020). Statistics Denmark moved it to old-age benefits with the revision of the EU-SILC income data in the survey of 2020, that is, from the income year 2019 (Statistics Denmark, 2021). The fall of 2019 is this reclassification, not a change of the scheme.

Consequence for analysis. Danish unemployment benefits at ages 60–65 up to 2018 measure early retirement and are comparable neither with other countries nor with Denmark from 2019. Use total benefits (`BS`). The treatment in ENTA is listed in Table 6.2.

E.3 Breaks caused by the data

The breaks below come from the way a series is measured, reported or built. Where ENTA use a different method for a country for this reason, Table 6.2 lists it. For the breaks in the source data whose reason is not known, the entry says so.

Aggregates

- **Ireland, 2011: the split of benefits breaks.** Public pensions fall from 16,313 to 12,017 million euro and family benefits double, while total benefits are unchanged: more than 4 billion euro move between functions in Eurostat's government expenditure by function. Total benefits are not affected; the split is not comparable across 2010 and 2011. See Section 6.5.3.

- **Greece, 2014: taxes on assets halve and taxes on production rise.** About 2.7 billion euro of household taxes move from other current taxes (D.59) to other taxes on production (D.29) with the new property tax of 2014. Taxes on assets (`TYA`) and taxes on employment income, which include the household's taxes on production, are not comparable across 2013 and 2014; total taxes are hardly affected.

- **Bulgaria, 2020–2021: property income of households falls.** Net property income (`YP`) falls from 9.1% of national income in 2019 to 6.1% in 2020 and 3.0% in 2021. In the sector accounts interest paid by households rises from 57 to 849 million euro in 2020 and interest received falls from 803 to 147. The reason was not found; the series are not comparable across 2019 and 2020.

- **Croatia, 2021, and Bulgaria, 2009–2010: mixed income.** Croatian mixed income (`YM`) doubles in 2021, from 5.0% to 9.0% of national income, and stays there. Bulgarian mixed income falls from 22.8% in 2008 to 15.3% in 2009 and 13.0% in 2010. Both changes are in the sector accounts (B.3g); the reasons were not found. Use employment income (`YEM`) across these years, not its split.

- **Romania, 2010–2013: private transfers between corporations and households.** Non-financial corporations pay 5.0 and 6.5 billion euro of miscellaneous current transfers (D.75) in 2011 and 2012, against less than 1 billion from 2013. Private transfers received by households

(BPB) are 3.0%, 5.8%, 7.2% and 1.9% of national income in 2010 to 2013, with the opposite sign for corporations. The reason was not found.

- **Romania, 2010, and Hungary, 2010: consumption by purpose.** Romanian private health consumption (CHH) doubles in 2010, from 2.2% to 4.3% of national income, in Eurostat's consumption by purpose. Hungarian public education consumption moves from primary to secondary education in 2010 (1,770 to 1,260 and 859 to 1,522 million euro) with an unchanged total, which raises its mean age by almost a year. The reasons were not found.

- **Czechia and Slovakia: taxes on benefits are high relative to the microsimulation.** The rate implied by the national accounts is about 10% against 0.2% for Czechia and 19% against 0.5% for Slovakia. The level is the concept of the national accounts, not a payment of households.

- **The split of taxes on benefits is absent in early years.** The split into taxes on pensions and on other benefits comes from EUROMOD, which starts between 2005 and 2007, and in 2011 for Croatia. See Section 6.5.2.

- **Germany and Ireland: the split of other current transfers begins late.** TXH and TXNPISH are separated only from 2012 for Germany and 2010 for Ireland; German TXH moves from −14,374 to +6,762 million euro in 2012 for this reason. Use TX.

- **Estonia, 2013–2019: funded pensions received are negative.** Eurostat reports negative benefits paid by financial corporations; ENTA keep the source values.

- **Romania, 2018: household operating surplus dips for one year.** YO falls by a third and returns in 2019; the dip is in the source (B.2g).

- **Bulgaria ends in 2022.** The sector accounts of 2023 were not published when the aggregates were built.

Age profiles

- **Germany, 2019: a new survey.** With the survey of 2020 (income year 2019) the German EU-SILC was integrated into the Microcensus and the old sample was discontinued.[29] The share of men in pensions is 56–58% in 2014–2018 and 52.3% from 2019; imputed rent and the operating surplus of households move to older ages (mean age of imputed rent 46.9 and 53.2). Do not compare profiles by sex, or the age shape of imputed rent, across 2018 and 2019.

- **Breaks that follow from a different method for a country** are described in Table 6.2: the step functions of the German profiles from 2022 and of the employment profiles of the Netherlands, Czechia, Ireland and Slovakia; Danish property income before 2019, built from rental income and imputed rent (with the change in 2019 the corporate profiles and taxes on assets move to older ages); the missing mortgage interest of Hungary 2018, France 2020 and

[29] Eurostat, EU-SILC national quality report for Germany, https://ec.europa.eu/eurostat/cache/metadata/en/ilc_simsilc_de.htm.

Germany 2008; Irish private pensions of 2018; and the flat private education consumption of Finland and Sweden.

- **Denmark, 2016–2018: benefits dip at 62 and rise at 63.** Early-retirement pay starts as a step at 62, which smoothing flattens by one age. Read ages 61–63 together.
- **Romania, 2014: other benefits and property income.** In the survey of 2015 (income year 2014) other social benefits have about a tenth of the receivers of the neighbouring years, and property income about ten times as many. Other benefits (`BSX`) are concentrated at old ages in 2014 only (39% at 65 and older, against about 10%). The reason was not found.
- **Greece, 2012–2013, and Cyprus, 2017: family benefits move to younger ages.** In Greece the share of family benefits at ages 65 and older falls from 21% in 2011 to 1% in 2013; the reform of family benefits of 2013 falls into these years, but no document was checked. In Cyprus the benefits move from ages 40–54 to ages 20–39 in 2017 and stay there; the reason was not found.
- **Austria, Spain, Belgium and Estonia: income data from registers.** These countries replaced survey answers by administrative data for most income variables: Austria and Spain around the survey of 2011, Estonia in 2013 and 2014, Belgium with the survey of 2019 (Goedemé & Zardo Trindade, 2020, 2022; Törmälehto, 2019). Levels do not break. Some age shapes change more than usual near these years: property income in Austria 2011 and Spain 2012–2013, pensions in Belgium 2018.
- **Croatia starts in 2009; Bulgaria, 2010.** The tax rates by age of Croatia for 2009 and 2010 are those of 2011, the first EUROMOD year. In the Bulgarian survey of 2010 the income components do not add up to the household total, so the pattern of some components by age and sex is less reliable in that year.
- **Asset income by sex, and at ages 18–19.** The share of men in net property income moves by up to 12 points from one year to the next, already in the survey, and corporate income inherits it; in Latvia it falls from 57% to 45% in 2009 and in Lithuania the share of men in the corporate profiles falls by nine points in 2012. Few receivers with large amounts decide these values. Net property income can be negative at 18–19. See Section 6.5.1.
- **Transfers between households (`TFB`).** The profile is a sum of positive and negative flows with different age shapes. Its age shape and its share of men change when the size of the flows changes, and are unstable when the total is close to zero (Luxembourg 2010, Hungary 2009 and 2014, Slovenia 2020, Romania 2012).
- **Unemployment benefits end at the pension age.** The profile keeps the fall at the age at which pensions start and is smoothed only up to four years before it, so single ages between 58 and 68 can stand out.
- **Netherlands, Ireland and Poland, 2008–2010: pre-school education.** Pre-primary enrolment is reported only in part in these years; Dutch public education consumption at age 3

goes from about 100 to 5,479 euro per person in 2011. Private education consumption at these ages is affected in the same way, because it follows the same enrolment data. See Section 6.5.6.

Population

A census usually finds fewer residents than the estimates of the years before it, mostly because emigrants were still counted. Where the earlier years are not revised, the population of Eurostat falls in one step: in Germany by 1.5 million between 1 January 2010 and 2011, in Poland by 0.7 million and in Czechia by 0.2 million between 2020 and 2021 (all three marked by Eurostat as a break in the series), and in Bulgaria by 0.2 million between 2001 and 2002. Romania loses 0.5 million between 2001 and 2002 and again between 2007 and 2009; the two steps coincide with the end of the visa requirement for travel to the EU in 2001 and with the accession to the EU in 2007 (Corsale, Iorio, & Salaris, 2025). The step is largest at ages 15 to 44 and small above 65. ENTA use the average of two consecutive 1 January figures, so each step appears in two years. Values per capita rise by about the same percentage in these years, at all ages; age shapes and shares of income are not affected.

In Romania the population aged 80 and older is uncertain between the censuses of 2002 and 2011: in the Eurostat data it does not change in 2003 and 2004, grows by 55% from 2004 to 2008 and does not change from 2009 to 2011. It is 16% above the population of the UN World Population Prospects in 2008 and 2009; from 2011 the two sources agree. Values per capita of flows concentrated at old ages, such as pensions and public health consumption, are therefore somewhat too low for Romania in 2008 to 2010. The population data are described in Appendix D.

References

Abio, G., Patxot, C., Souto, G., & Istenič, T. (2021). The role of gender, education and family in the welfare organization: Disaggregating national transfer accounts. *The Journal of the Economics of Ageing*, *20*, 100348.

Central Statistics Office Ireland. (2016). *National Income and Expenditure 2016: Frequently Asked Questions.* `https://www.cso.ie/en/media/csoie/methods/nationalincomeandexpenditureannualresults/NIE_2016_FAQS.pdf`.

Corsale, A., Iorio, M., & Salaris, L. (2025). Chi resterà? Il declino della popolazione in Romania [Who will remain? Population decline in Romania]. *Bollettino della Società Geografica Italiana*, *8*(1), 45–58. doi: 10.36253/bsgi-7538

Dramani, L., & Oga, I. J.-B. (2017). Understanding demographic dividends in Africa: The NTA approach. *Journal of Demographic Economics*, *83*(1), 85–101.

d'Albis, H., & Moosa, D. (2015). Generational economics and the national transfer accounts. *Journal of Demographic Economics*, *81*(4), 409–441.

European Commission. (2018). *The 2018 ageing report: Economic and budgetary projections for the EU member states (2016–2070)* (European Economy No. 8/2018). Brussels: European Commission, Directorate-General for Economic and Financial Affairs. Retrieved from `https://economy-finance.ec.europa.eu/publications/2018-ageing-report-economic-and-budgetary-projections-eu-member-states-2016-2070_en`

European Commission. (2026). *Data on Taxation.* `https://webgate.ec.europa.eu/taxation_customs/redisstat/databrowser/view/tax_ec_func/default/table`. Taxation and Customs Union.

EUROSTAT. (2013). *European system of accounts: ESA 2010.* Publications Office of the European Union. `https://ec.europa.eu/eurostat/documents/3859598/5925693/KS-02-13-269-EN.PDF`.

Goedemé, T., & Zardo Trindade, L. (2020). *MetaSILC 2015: A report on the contents and comparability of the EU-SILC income variables* (INET Oxford Working Paper 2020-01 and CSB Working Paper No. 20/01). Institute for New Economic Thinking, University of Oxford, and Herman Deleeck Centre for Social Policy, University of Antwerp.

Goedemé, T., & Zardo Trindade, L. (2022). Lessons and recommendations regarding the comparability of the EU-SILC income variables. In P. Lynn & L. Lyberg (Eds.), *Improving the measurement of poverty and social exclusion in Europe: reducing non-sampling errors* (pp. 249–261). Luxembourg: Publications Office of the European Union. doi: 10.2785/18803

Hammer, B. (2020). *National Transfer Accounts for Austria 2015 (OA edition).* AUSSDA. Retrieved from `https://doi.org/10.11587/4EOXZO` (Version 1) doi: 10.11587/4EOXZO

Hammer, B., Christl, M., & De Poli, S. (2023). Public redistribution in Europe: Between generations or income groups? *The Journal of the Economics of Ageing*, *24*, 100426.

Hammer, B., Spitzer, S., Vargha, L., & Istenič, T. (2020). The gender dimension of intergenerational

transfers in Europe. *The Journal of the Economics of Ageing*, *15*, 100234.

Istenič, T., Hammer, B., Šeme, A., Lotrič Dolinar, A., & Sambt, J. (2019). *European national transfer accounts*. Agenta. Retrieved from https://www.jstor.org/stable/pdf/26928909.pdf

Ladusingh, L., & Narayana, M. (2012). Demographic dividends for India: evidence and implications based on National Transfer Accounts. In *Aging, economic growth, and old-age security in Asia* (pp. 203–230). Edward Elgar Publishing.

Lee, R. D., & Mason, A. (2011). *Population aging and the generational economy: A global perspective*. Edward Elgar Publishing. https://idl-bnc-idrc.dspacedirect.org/server/api/core/bitstreams/168f97d0-da44-48d4-aa73-fb3eb4606b88/content.

Loichinger, E., Hammer, B., Prskawetz, A., Freiberger, M., & Sambt, J. (2017). Quantifying economic dependency. *European Journal of Population*, *33*(3), 351–380.

Mason, A., Lee, R., & members of the NTA network. (2022). Six ways population change will affect the global economy. *Population and development review*, *48*(1), 51–73.

Mérette, M., & Navaux, J. (2025). Did seniors benefit from consumption and public transfers at the expense of youth? National Transfer Accounts in Canada's provinces (1997–2017). *The Journal of the Economics of Ageing*, 100609.

Nelisse, R. (2021). *Non-Financial Corporations Split Into Subsectors.* https://www.cbs.nl/-/media/_pdf/2022/38/report-subsectoring-nfcs.pdf.

Oosthuizen, M. (2019). *Inequality and the generational economy: race-disaggregated national transfer accounts for South Africa, 2015* (No. 2019/24). Wider working paper.

Rohatgi, A. (n.d.). *WebPlotDigitizer.* https://automeris.io. (Version 5.2)

Shraberman, K., & Weinreb, A. A. (2024). The fiscal consequences of changing demographic composition: Aging and differential growth across Israel's three major subpopulations. *The Journal of the Economics of Ageing*, *27*, 100500.

Statistics Denmark. (2021). *Revision of SILC incomes – main elements. For Eurostat and users of SILC income statistics* (Tech. Rep.). Copenhagen: Statistics Denmark. (21 April 2021)

Törmälehto, V.-M. (2019). *Reconciliation of EU statistics on income and living conditions (EU-SILC) data and national accounts* (Eurostat Statistical Working Papers). Luxembourg: Publications Office of the European Union. doi: 10.2785/489634

United Nations. (2011). *System of National Accounts 2008.* https://unstats.un.org/unsd/nationalaccount/docs/sna2008.pdf.

United Nations. (2013). *National Transfer Accounts Manual: Measuring and Analysing the Generational Economy.* http://www.un.org/en/development/desa/population/publications/development/NTA_Manual.shtml.

Vanhuysse, P., Medgyesi, M., & Gál, R. I. (2023). Taxing reproduction: the full transfer cost of rearing children in Europe. *Royal Society Open Science*, *10*(10), 230759.